



Universität für Bodenkultur Wien
University of Natural Resources
and Life Sciences, Vienna

Master Thesis

Modelling nitrate-nitrogen transport in the vadose zone of an agricultural area in southern Styria using HYDRUS-1D

Submitted by

Stefanie MAIERHOFER, BSc

in the framework of the Master programme

Kulturtechnik und Wasserwirtschaft

in partial fulfilment of the requirements for the academic degree

Diplom-Ingenieurin

Vienna, April 2022

Supervisor:

Univ.Prof. Dipl.Geoökol. Dr.rer.nat. Karsten Schulz

Institute for Hydrology and Water Management (HyWa)

Department of Water, Atmosphere and Environment (WAU)

Supervisor team:

Dr. Bano Mehdi-Schulz

Univ.Prof. Dipl.Geoökol. Dr.rer.nat. Karsten Schulz

Student Number:

01231446

DANKSAGUNG

Ich möchte mich bei Prof. Karsten Schulz bedanken, dass er mich mit seiner eigenen Begeisterung für die Hydrologie während seinen Lehrveranstaltungen so sehr angesteckt hat, dass ich (für) immer mehr darüber lernen möchte. Und, dass er sich sofort bereiterklärt hat meine Masterarbeit zu betreuen und mich so gut unterstützt und sein Wissen mit mir geteilt hat.

Ein großer Dank gilt meiner wunderbaren Betreuerin Bano Mehdi-Schulz, die mich immer so freundlich und geduldig mit ihrem unglaublichen Wissensschatz beraten und mit mir über meine kleinen Hoppalas gelacht hat. Es gab keine Frage, die ich nicht stellen konnte, nur viel Verständnis und Hilfsbereitschaft.

Bedanken möchte ich mich auch bei Philipp Holler, dessen Messungen im Zuge seiner Masterarbeit die Grundlage für meine Masterarbeit gebildet haben. Ich bin sehr froh, dass ich die Daten verwenden durfte und bedanke mich auch für die Hilfsbereitschaft und die Beantwortung meiner Fragen.

Am wichtigsten ist es mir aber mich bei meiner Familie zu bedanken. Ihr stellt für mich Liebe, Unterstützung, Rückhalt, Verrücktheit, Spaß und den größten Wert in meinem Leben dar.

Papa, du hast es immer geschafft mir jeden Druck zu nehmen und mir die Zeit fürs Studieren zu geben, die ich brauche und mich gleichzeitig anzuspornen über mich hinauszuwachsen und das Lernen zu genießen. Und hast meine verzweifelte Anrufe wegen mathematischen Problemen bis spät in die Nacht entgegengenommen.

Mama, danke für deine scheinbar unendliche positive Energie mit der du mich so oft aufgebaut und immer wieder einmal aus dem wissenschaftlichen Sumpf zurückgeholt hast. Und für die unzähligen Rechtschreibkorrekturen, und Jausenpakete, um die unser Gepäck jedes Mal auf der Rückreise nach Wien glücklicherweise schwerer war.

Theresa, meine beste Freundin, und Schwester. Du hast mich am öftesten aufgefangen, in meinen schwierigen Zeiten, genau das Richtige gesagt und immer ein Glaserl Rosé bereitgehalten. Du steckst mich an mit deiner unendlichen Lebensfreude, Kreativität, Abenteuerlust und deinem Ehrgeiz. Ich werde immer die kleine Schwester bleiben die zu dir aufschaut und werde deine Hand nie loslassen.

Georg, du warst in den vergangenen Jahren ein wichtiger Fels in meiner sehr rauen Brandung und ein wunderbarer Freund. Danke, dass du mich immer mit deiner (oft tatkräftigen und muskelbepackten) Unterstützung, deinen Scherzen, und vor allem dem traumhaften Essen aufgebaut hast.

Und ich danke meinen vielen lieben FreundInnen und anderen wichtigen Menschen, die mich die letzten Jahre begleitet, unterstützt und mein Leben verschönert haben.

EIDESSTATTLICHE ERKLÄRUNG

Ich erkläre eidesstattlich, dass ich die Arbeit selbständig angefertigt habe. Es wurden keine anderen als die angegebenen Hilfsmittel benutzt. Die aus fremden Quellen direkt oder indirekt übernommenen Formulierungen und Gedanken sind als solche kenntlich gemacht. Diese schriftliche Arbeit wurde noch an keiner Stelle vorgelegt.

Ort, Datum

Vor- und Zuname

Unterschrift

ABSTRACT

Fertilization and intensive farming techniques are often inevitable for sustaining the yield nowadays, resulting in a soil degradation, the leaching of nutrients from the unsaturated zone and the pollution and eutrophication of groundwater and surface waters. The vadose zone affects many hydrological processes surrounding the soil surface, and the movement of water and dissolved nutrients in the unsaturated zone is not always easy to measure and calculate. As there are only a few ways to measure these fluxes, most of them being very time-consuming and expensive, simulating these fluxes with available measured data is a promising alternative. In the course of this thesis, a model was set up with the modelling environment HYDRUS-1D for an agriculturally used field site in Gabersdorf, in southern Styria, that is covered by different winter cover crop compositions to simulate the water flow and the nitrate-nitrogen (NO_3^- -N) transport in the vadose zone. At the field site, situated in the district Leibnitz that was affected by agriculturally induced groundwater pollution in the last three decades, soil samples were taken in the course of 9 months from August 2017 to May 2018 and were analysed for their water and mineralized nitrogen (N_{min}) content, prior to this study. The measured data was used for the calibration and validation of the HYDRUS-model that was set up for the time period from 3rd of August 2017 to the 16th of April 2018. The simulated water contents matched the measured ones quite well, with an overall coefficient of determination R^2 of 0.66 and a root-mean-square error (RMSE) of 2.68. The simulated NO_3^- -N concentrations also showed rather low errors with a R^2 of 0.62 and a RMSE of 2.87. The amount of NO_3^- -N leached during five months of the simulation period was estimated from the water flux and the NO_3^- -N concentration at the bottom of the soil profile. The HYDRUS-model was further evaluated by simulating the NO_3^- -N concentration during a heavy precipitation event that took place on the evening of the 16th of April 2018. A comparison of the simulated values with the measured water contents and NO_3^- -N concentrations on the 2nd of May 2018 revealed a good fit of the model, but also showed that the decrease of the NO_3^- -N concentration in all plots and all soil depths and thus the displacement capacity of the heavy precipitation was underestimated by the HYDRUS-model.

KURZFASSUNG

Für die Aufrechterhaltung der landwirtschaftlichen Ernteerträge kommen immer öfter starke Düngemittelgaben und intensive Anbaumethoden zum Einsatz, die jedoch auch zu einer Verschlechterung der Bodenqualität, der Auswaschung von Nährstoffen und der Verschmutzung von Oberflächen- und Grundwasser führen. Der ungesättigte Bereich im Boden bildet die Grundlage für zahlreiche hydrologische Prozesse, wie etwa Transportprozesse von Wasser und Nährstoffen zwischen Boden und Wurzel und an der Grenze zur Atmosphäre, die teils nur schwer durch Messungen und Berechnungen erfasst und abgebildet werden können. Die Messung dieser Ströme ist meist zeit- und kostenintensiv, wodurch alternative Methoden zur Bestimmung der Flusssichten, wie die Simulation mit physikalisch-basierten Modellen, an Bedeutung gewinnen. Das Ziel der vorliegenden Arbeit war das Erstellen eines Modells mit HYDRUS-1D für einen ackerbaulich genutzten Standort in der Südsteiermark mit unterschiedlichen Winterbegrünungen für die Simulation der Wasserbewegung und des Transports von Nitrat-Stickstoff (NO_3^- -N) im ungesättigten Bereich des Bodens. An dem Standort im Bezirk Leibnitz, der in den letzten Jahrzehnten mit Grundwasserverschmutzungen durch die starke ackerbauliche Nutzung zu kämpfen hatte, wurden im Laufe einer Masterarbeit im Zeitraum von 9 Monaten (2017 - 2018) Bodenproben genommen und auf deren Wasser- und mineralisierten Stickstoffgehalt (N_{min}) analysiert. Die Messwerte wurden für die Kalibrierung und die anschließende Validierung des HYDRUS-Modells verwendet, mit dem die Wassergehalte und die Nitrat-Stickstoff (NO_3^- -N) Konzentrationen über einen Zeitraum von 3. August 2017 bis 16. April 2018 simuliert wurden. Die simulierten Wassergehalte zeigen eine gute Übereinstimmung mit den gemessenen Werten, mit einem Bestimmtheitsmaß R^2 von 0.66 und einem root-mean-square error (RMSE) von 2.68. Die simulierten NO_3^- -N Konzentrationen weisen ebenfalls geringe Abweichungen zu den gemessenen auf, mit einem R^2 von 0.62 und einem RMSE von 2.87. Für eine Abschätzung der ausgewaschenen und in Richtung Grundwasser versickerten NO_3^- -N-Mengen während der Simulationsperiode, wurden die simulierten NO_3^- -N Konzentrationen am Fuße des Bodenprofils mit den Sickerwassermengen kombiniert. Die Bodenprobenentnahme nach einem Starkregenereignis am Abend des 16. April 2018 diente weiters für die Evaluierung des erstellten HYDRUS-Modells, wobei die NO_3^- -N-Flüsse im Zeitraum von 16. April bis 2. Mai 2018 simuliert wurden. Der Vergleich der simulierten mit den gemessenen NO_3^- -N Konzentrationen und Wassergehalten vom 2. Mai 2018 zeigte eine gute Vorhersagekraft des Modells mit geringen Abweichungen, aber offenbarte auch die ausnahmslose Überschätzung aller gemessenen NO_3^- -N Konzentrationen durch das HYDRUS-Modell. Die Abnahme der NO_3^- -N Konzentration im Bodenprofil nach dem Starkregenereignis wurde demnach unterschätzt.

CONTENTS

1. INTRODUCTION	1
2. MATERIAL AND METHODS	5
2.1 FIELD SITE DESCRIPTION	5
2.1.1 <i>Agriculture in Styria and Leibnitz</i>	5
2.1.2 <i>Legal basis for the groundwater protection in the Leibnitzer Feld</i>	6
2.1.3 <i>Geology, Pedology and Climate</i>	7
2.1.4 <i>Experimental setup, treatments and soil samples</i>	8
2.2 HYDRUS-1D DESCRIPTION	11
2.3 SOIL PHYSICAL BASICS IN HYDRUS	13
2.3.1 <i>Water transport</i>	13
2.3.2 <i>Root water uptake</i>	16
2.3.3 <i>Solute transport</i>	17
2.3.4 <i>Boundary Conditions</i>	18
2.4 HYDRUS-1D SETUP AND MODEL INPUT DATA	20
2.4.1 <i>Meteorological input data</i>	20
2.4.2 <i>Wet atmospheric nitrogen deposition</i>	21
2.4.3 <i>Soil parameters</i>	21
2.4.4 <i>HYDRUS-model setup</i>	21
2.5 SENSITIVITY ANALYSIS.....	26
2.6 HYDRUS-MODEL CALIBRATION AND VALIDATION	26
2.7 NITRATE-NITROGEN LEACHING ESTIMATION	28
3. RESULTS.....	29
3.1 RESULTS OF THE SENSITIVITY ANALYSIS	29
3.2 CALIBRATION RESULTS	33
3.3 VALIDATION RESULTS.....	38
3.4 RESULTS OF MODEL SIMULATIONS.....	41
3.4.1 <i>Evapotranspiration</i>	41
3.4.2 <i>Comparison of measured and simulated values of all plots</i>	42
3.4.3 <i>Actual root NO₃⁻-N uptake</i>	45
3.4.4 <i>Simulation of the heavy precipitation event</i>	46
3.4.5 <i>Bottom nitrate-nitrogen fluxes and leaching amounts</i>	49
4. DISCUSSION	51
4.1 SENSITIVITY ANALYSIS	51
4.2 CALIBRATION RESULTS	52
4.3 CALIBRATION RESULTS COMBINED WITH VALIDATION RESULTS	52
4.4 EVAPOTRANSPIRATION	54
4.5 SIMULATION OF THE HEAVY PRECIPITATION EVENT.....	55
4.6 HYDRUS-MODEL UNCERTAINTIES	55
5. CONCLUSION	56
BIBLIOGRAPHY	59
LIST OF FIGURES	64
LIST OF TABLES.....	67
APPENDIX.....	68

LIST OF ALL VARIABLES

A	horizontal surface area [L^2]
c	solution concentration [ML^{-3}]
c_0	concentration of the incoming fluid [ML^{-3}]
c_p	specific heat of moist air [$L^2 T^{-2} K^{-1}$] (i.e., $1013 \text{ kJ kg}^{-1} \text{ }^\circ\text{C}^{-1}$)
c_r	concentration of the sink term [ML^{-3}]
D_L	longitudinal dispersivity [L]
D_w	molecular diffusion coefficient in free water [$L^2 T^{-1}$]
D^w	solute dispersion coefficient for the liquid phase [$L^2 T^{-1}$]
E	maximum (potential) rate of infiltration or evaporation under the prevailing atmospheric conditions [$L T^{-1}$]
e_a	saturation vapour pressure at temperature T [$ML^{-1} T^{-2}$] (e.g., kPa)
e_d	vapour pressure at dew point [$ML^{-1} T^{-2}$] (e.g., kPa)
$(e_a - e_d)$	vapour pressure deficit [$ML^{-1} T^{-2}$] (e.g., kPa)
ET_0	evapotranspiration rate [$L T^{-1}$]
ET_{rad}	radiation term [$L T^{-1}$]
ET_{aero}	aerodynamic term [$L T^{-1}$]
G	soil heat flux [MT^{-3}] (e.g., $\text{MJ m}^{-2} \text{d}^{-1}$)
h	pressure head [L]
H	total hydraulic head [L]
h_A	minimum pressure head allowed at the soil surface [L]
h_S	maximum pressure head allowed at the soil surface [L]
K	unsaturated soil hydraulic conductivity [$L T^{-1}$]
K_r	relative soil hydraulic conductivity [-]
k_s	empirical constant relating the solution and adsorbed concentrations [$L^3 M^{-1}$]
K_s	saturated hydraulic conductivity [$L T^{-1}$]
l	pore-connectivity parameter [-]
L	water leaching flux [$L T^{-1}$]
n	exponent in the soil water retention function [-]
m	parameter in the soil water retention function [-]
M	mass of nitrate-nitrogen leaching below the profile depth [$ML^{-2} T^{-1}$]
q	volumetric flux density [$L T^{-1}$]
q_0	upward fluid flux [$L T^{-1}$]
r_a	root nutrient uptake term [$ML^{-3} T^{-1}$]
r_a	aerodynamic resistance (s m^{-1})
r_c	crop canopy resistance (s m^{-1})

R_n	net radiation at surface [MT^{-3}] (e.g., $MJ m^{-2} d^{-1}$)
S	sink term in the flow equation [T^{-1}]
s	adsorbed solute concentration [-]
S_e	effective saturation [-]
S_p	potential water uptake rate [T^{-1}]
t	time [T]
x	spatial coordinate [L] (positive upward)
α	parameter in the soil water retention function [L^{-1}]
α'	dimensionless water stress response function [-]
β	empirical constant in the adsorption isotherm [-]
γ	psychrometric constant [$ML^{-1} T^{-2} K^{-1}$] (e.g., $kPa \text{ } ^\circ C^{-1}$)
γ_w	zero-order rate constant for solutes in the liquid phase [$ML^{-3} T^{-1}$]
γ_s	zero-order rate constant for solutes adsorbed onto the solid phase [T^{-1}]
Δ	slope of the vapour pressure curve [$ML^{-1} T^{-2} K^{-1}$] (e.g., $kPa \text{ } ^\circ C^{-1}$)
η	empirical constant in the adsorption isotherm [$L^3 M^{-1}$]
θ	volumetric water content [$L^3 L^{-3}$]
θ_r	residual soil water content [$L^3 L^{-3}$]
θ_s	saturated soil water content [$L^3 L^{-3}$]
λ	latent heat of vaporisation [$L^2 T^{-2}$] (e.g., $MJ kg^{-1}$)
μ_w	first-order rate constant for solutes in the liquid phase [T^{-1}]
μ_w'	first-order rate constant for decay chain solutes in the liquid phase [T^{-1}]
μ_s	first-order rate constant for solutes adsorbed onto the solid phase [T^{-1}]
μ_s'	first-order rate constant for decay chain solutes adsorbed onto the solid phase [T^{-1}]
ρ	atmospheric density [ML^{-3}]
ρ	bulk density of porous medium [ML^{-3}]

1. Introduction

The vadose zone has long been recognized as a fundamental factor influencing hydrological processes, such as infiltration, surface runoff, erosion, root water uptake, evapotranspiration, soil moisture storage, seepage and groundwater recharge (Šimůnek et al. 2009). This zone can be defined as the geologic media, situated between the earth surface and the water table surface, that usually contains unsaturated soils and bedrock with pores that are partially filled with water, and a plant root zone and weathered soil horizons in the upper part (Stephens 1996; Wilson, Everett, and Cullen 1995). Although agricultural land use in Austria is slightly declining with ~1.4 million ha of cultivated area in 1995, compared to ~1.37 million ha in 2010 and ~1.32 million ha in 2020, it is still a major sector that strongly influences the soil quality and the environment. In 2020, approximately 13 % of the total area of Styria was used agriculturally (Statistik Austria 2020), with the greater share in the flatter parts in southern Styria.

Harvesting crops removes nitrogen, phosphorus and other nutrients from the soil, which have to be replaced by biological processes or manual addition to sustain agricultural production. Globally, fertilization is the primary way of nutrient addition, leading to quantities of new nitrogen and phosphorus in the terrestrial biosphere that have more than doubled since the invention of the Haber-Bosch process, which is a method for ammonia synthesis from hydrogen (H_2) and atmospheric nitrogen (N_2) under high pressure and temperature in 1908 (Vitousek et al. 2009; Humphreys, Lan, and Tao 2021). The use of fertilizers, pesticides and intensive farming techniques in general leads to a degradation of soils and a loss of biodiversity (Groenendijk et al. 2014). Environmental costs from nutrient pollution also include the degradation of downstream water quality, eutrophication of ecosystems and rising global concentrations of the greenhouse gas, nitrous oxide (Vitousek et al. 2009). Supplying the ever-growing population with enough food and simultaneously avoiding a surplus of nitrogen that is prone to leaching into groundwater is a global challenge (Groenendijk et al. 2014). Fertilizer applications represented the main anthropogenic cause of disturbances in the global nitrogen cycle in 2010 (Fowler et al. 2013). If nitrogen is excessively leached into groundwater or surface water, it may lead to accelerated eutrophication (Matteau et al. 2019). Nitrate (NO_3^-) represents, along with pesticides, a significant source of aquifer contamination in agricultural areas (Matteau et al. 2019). According to Jiang and Somers (2009), at a study area in Canada it would take several years to remove nitrate-nitrogen (NO_3^- -N) originating from intensive potato farming from the shallow aquifers, and even decades or longer for restoring water quality ($<10\text{mgNO}_3^- \cdot \text{N l}^{-1}$) in deeper portions of the aquifer (30–50 m depth).

Following the European Nitrate Directive (EU, 1991: Directive 91/676/EEC) approved by the European Commission (1991) the water pollution caused or induced by NO_3^- and phosphorus should be reduced to a minimum (Groenendijk et al. 2014). NO_3^- in drinking water in Austria is restricted to

$\leq 50 \text{ mg NO}_3^- \text{ l}^{-1}$ by the *Trinkwasserverordnung* (BGBl. II Nr. 304/2001 idgF) (TWV 2018) due to the risk of infant methemoglobinemia and may also increase the risk of birth defects, specific cancers, and thyroid diseases amongst others (Ward et al. 2018). However, nitrogen fertilization is the basis for modern agriculture as it is a limiting nutrient for crop growth (Matteau et al. 2019), and therefore the determination of nitrogen concentrations in the soil at all times is an important step to better adapt the fertilization amounts a plant needs and decrease the risk of NO_3^- leaching (Matteau et al. 2019; Gabriel, Muñoz-Carpena, and Quemada 2012).

Modelling the nonequilibrium flow and transport processes in soils for gaining insight into the hydrological, pedological and biological processes and agricultural practices affecting the risk of nitrate (NO_3^-) leaching, becomes more and more important (Groenendijk et al. 2014). The above mentioned processes are essential for the water supply of plants, plant growth and for maximising the yield and thus, modelling such processes with programmes such as HYDRUS-1D has often been the aim of soil water dynamic studies (e.g. Graham et al. 2018; Chen et al. 2020, Ursulino et al. 2019; Iqbal et al. 2020). For example in Switzerland, Richner et al. (2014) set up a basic model that simplifies processes surrounding nitrogen for a simpler estimation of the NO_3^- leaching amounts to compare different crops and treatments. Li et al. (2015) found the HYDRUS-1D software package to be an effective modelling tool for simulating the soil nitrogen and NO_3^- leaching. Tan et al. (2015) who also used HYDRUS-1D for assessing water and nitrogen transport at rice fields under varying water managements, found that alternate wetting and drying irrigation decreased the nitrogen leaching losses, but increased nitrogen losses of volatilization, nitrification and denitrification. Reducing NO_3^- losses by adjusting nitrogen and water application to crop requirements is a well-known technique (Gabriel, Muñoz-Carpena, and Quemada 2012). A greater target of modelling water flow and solute transport is often the reduction of NO_3^- leaching in future crop cycles at a farm scale through synchronising and thus optimizing nitrogen fertilization with crop uptake according to previous measurements and simulations.

According to Gabriel, Muñoz-Carpena, and Quemada (2012) NO_3^- leaching is difficult to measure without disturbing the soil, and often only little measured data is available to sustain the precise modelling of water and solute flow in the vadose zone. There exists no direct method for soil solution sampling under most soil conditions, but there are possibilities for the collection of water samples from the unsaturated zone, such as profile soil sampling, tile drains, drainage from watersheds, pan and wick lysimeters and monolith lysimeters as listed by Gabriel, Muñoz-Carpena, and Quemada (2012). Calculating water percolation amounts is one of the main factors determining NO_3^- leaching in indirect methods (Gabriel, Muñoz-Carpena, and Quemada 2012). With ceramic cup samplers the NO_3^- leaching can be quantified by measuring the NO_3^- concentration in the soil water and combining the measurements with a profound knowledge of the soil water dynamics. This indirect method allows for NO_3^- leaching estimations with a minimal soil disturbance (Gabriel, Muñoz-Carpena, and Quemada

2012). Veizaga, Rodríguez, and Ocampo (2016) for example investigated the NO_3^- dynamics in a fine-textured soil affected by feedlot effluents through measurements with suction cups amongst others and complemented field results interpretation with numerical modelling with HYDRUS-1D. Lysimeter data provides a solid foundation for setting up a model for the simulation of the water and solute flow in the soil. However, lysimeters require much effort and time for installation and service and are often not feasible for smaller institutions. Whichever measurements are available or can be conducted at a field site of interest, the modelling of the hydrological processes in the soil is an important completion for a NO_3^- leaching estimation. According to Veizaga, Rodríguez, and Ocampo (2016) there is a lack of combined field and numerical modelling investigations for the assessment of nitrogen, particularly NO_3^- displacement in natural surroundings. Especially, if there is little measured soil water and NO_3^- content data available, as it is often the case in smaller research settings. Further research in the modelling of vadose zone processes at field sites with little or poor information, such as soil hydraulic properties with low resolution, rare measurements and meteorological data that is not measured locally needs to be done. Groenendijk et al. (2014) compared the performances of different one-dimensional state-of-the-art models for NO_3^- leaching estimations, based on lysimeter data measured in Wagna, southern Styria. They concluded that models designed for NO_3^- leaching in high-input farming systems had problems to accurately predict the NO_3^- leaching amounts in low-input farming systems because the models tended to underestimate the influence of catch crops and NO_3^- fixation by legumes. The positive impact of cover crops on NO_3^- leaching reduction is also described by Gabriel, Muñoz-Carpena, and Quemada (2012).

In 2017, a field site was sampled for water contents, ammonium-nitrogen ($\text{NH}_4^+\text{-N}$) and nitrate-nitrogen ($\text{NO}_3^-\text{-N}$) concentrations from soil samples that were taken in a depth from 0-90 cm in August, September, October, November 2017, and April and May 2018 at an agricultural field in Gabersdorf, nearby Leibnitz in southern Styria. The soil samples were taken from three blocks with each six plots that were covered with different winter cover crop mixtures and fertilization amounts. The influence of the fertilization and the winter cover crop type on the amount of nitrogen in the soil layers and the solute uptake by the plants was investigated and quantified at the measurement dates in the course of a Master Thesis conducted by Philipp Holler (Holler 2019). However, there are only values available at the measurement dates, and no continuous course of neither the water contents, nor the $\text{NO}_3^-\text{-N}$ concentrations can be derived. This led to the necessity to analyse whether the whole measurement period from August 2017 until May 2018 can be covered by a simulation that fits to the measured water contents and $\text{NO}_3^-\text{-N}$ concentrations.

In the course of this thesis, a physically-based one-dimensional soil model HYDRUS-1D was established for the simulation of the water and $\text{NO}_3^-\text{-N}$ transport in the vadose zone over the measurement period of the study of Philipp Holler. Setting up a model in HYDRUS requires many soil hydraulic, root water and solute uptake and meteorological parameters that differ in its quantity

according to the chosen soil hydraulic property model, solute transport model and root water and root solute uptake model. In order to identify the most sensitive parameters of the HYDRUS-model for the subsequent calibration, a sensitivity analysis was carried out. The calibration and validation period was set according to the soil sampling dates, excluding the heavy precipitation event ($\sim 88 \text{ mm } 10 \text{ h}^{-1}$) that occurred on the evening of the 16th of April 2018 in the region surrounding Gabersdorf.

The main objectives of this thesis were the prediction of the water contents and NO_3^- -N concentrations at the soil sampling dates, the quantification of NO_3^- -N leaching amounts that reached the bottom of the profile in 1 m depth and consequently most likely leached into the groundwater during the simulation period, and the assessment of the influence of different winter cover crops and varying precipitation intensities. The simulation of the NO_3^- -N displacement during the heavy precipitation event was an additional objective of the study, where the course of the NO_3^- -N concentration in the soil profile was simulated for a period covering the heavy precipitation event. Thus, the HYDRUS-model could be evaluated by comparing the simulated to the measured NO_3^- -N concentrations from the soil sampling. The intention of this additional research question in the thesis was to find out whether the model can describe the changes in the soil after the heavy rainfall, although it was calibrated with meteorological data containing only much smaller precipitation amounts. The NO_3^- -N leaching into deeper soil layers was quantified from the HYDRUS-1D simulations, for the heavy precipitation event and for a longer time period from November 2017 until May 2018.

This study contributes to a better understanding of modelling soil processes at a field site and simulating the NO_3^- -N concentrations. Flow and solute transport modelling was used to complement the existing field studies at Gabersdorf for getting additional insights from process-based simulations. The validation revealed that the HYDRUS-model was able to describe the courses of the water and NO_3^- -N contents with acceptable deviations between the simulated and measured values. The NO_3^- -N amounts that were displaced over the winter months were quantified by combining the water percolation amounts and the NO_3^- -N concentrations at the bottom of the profile that were simulated by the HYDRUS-model. The simulation of the NO_3^- -N concentrations after the heavy precipitation event predicted lower values in all plots and soil depths compared to the measured values.

2. Material and Methods

2.1 Field site description

2.1.1 Agriculture in Styria and Leibnitz

The *Leibnitzer Feld* is a shallow aquifer in southern Styria, extending from *Wildoner Schlossberg* in the north to the Styrian-Slovenian border in the south, (Fig. 2) that provides the drinking water for ~100 000 inhabitants, according to Klammler et al. (2013). The district Leibnitz that fully covers the *Leibnitzer Feld* is also a very fertile area and is therefore intensely used for agricultural purposes (Fig. 1) (Klammler et al. 2013; Baumhackl 2013). Valleys are often used for agricultural purposes and at the same time for groundwater withdrawal for drinking water supply (Klammler et al. 2013), as it is the case in the district Leibnitz. In 2013, Klammler et al. (2013) named the NO_3^- leaching from the unsaturated zone as the most important threat to groundwater quality in Europe. Reszler and Fank (2016) state that the intensive agricultural cultivation in Leibnitz led to a massive NO_3^- problem in groundwater. In the 1990s the groundwater quality got endangered by high NO_3^- concentrations (Klammler et al. 2013). Since then, the NO_3^- concentration was decreased by agricultural action programmes (chapter 2.1.2).

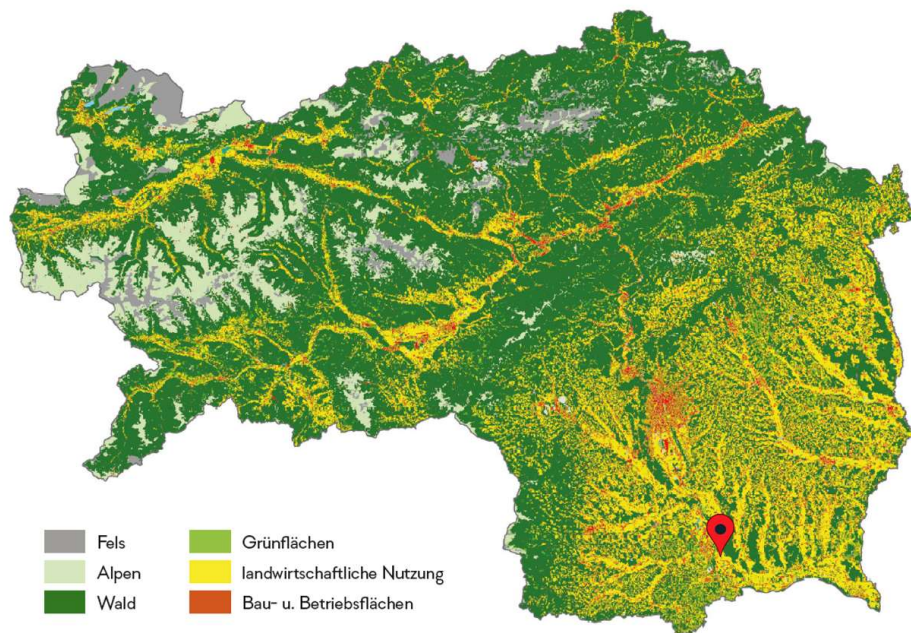


Fig. 1: Land use in Styria (status: October 2014) according to the *Wasserversorgungsplan* (Amt der Steiermärkischen Landesregierung 2015).

According to data from the *Statistik Austria*, in 2016 375 070 ha of the total area of Styria were agriculturally used and 136 216 ha were cultivated (Statistik Austria 2016). Preliminary results from the *Agrarstrukturhebung* in 2019 show a slight decrease in the agriculturally used and cultivated area in Styria to 365 029 ha and 126 368 ha, respectively (Statistik Austria 2019)

The main part of the cultivated crops was grain (~58%), followed by much smaller parts of green fodder (~18%), oil fruits (~11%), root crops (~5%), grain legumes (~2%) and other crops (~6%). In winter 2015 to 2016, nearly half of Styria's cultivated area (~44%) was not covered by any crops and ~14% of the cultivated area was planted with winter cover crops (Statistik Austria 2016). According to an investigation in 2005, almost half of the total area of the district Leibnitz (~43%) was agriculturally used (Land Steiermark 2009).

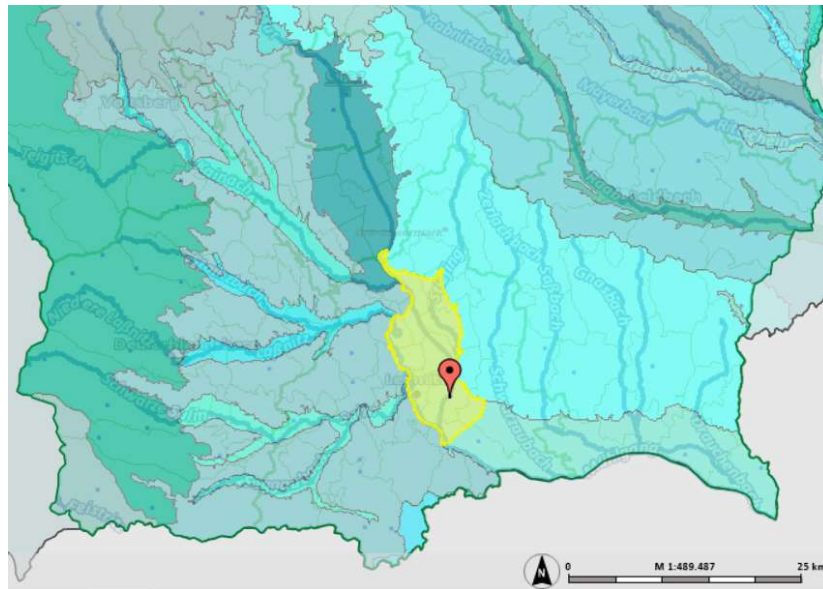


Fig. 2: Shallow groundwater aquifers in southern Styria with the position of the field site (red) where measurements used in this Master Thesis were conducted (Land Steiermark 2017).

2.1.2 Legal basis for the groundwater protection in the *Leibnitzer Feld*

The Austrian Water Law (*Wasserrechtsgesetz*) (WRG 1959) is the legal foundation for the protection of surface and groundwater in Austria. It builds the basis for the legally binding water quality assessment that may also activate other programmes for areas where threshold values are broadly exceeded. The *Trinkwasserverordnung* (BGBl. II Nr. 304/2001 idgF) regulates the requirements of the water quality for human use in Austria and specifies a threshold limit value of $50 \text{ mg NO}_3^- \text{ l}^{-1}$ (TWV 2018).

According to Klammler et al. (2013), the groundwater quality in the area around Leibnitz was endangered by high NO_3^- concentrations in the early 1990s. Since then, the agricultural action programmes that are described hereafter have been established with the aim to decrease the NO_3^- concentration in the groundwater to $\leq 50 \text{ mg NO}_3^- \text{ l}^{-1}$. The *Aktionsprogramm Nitrat* that was enacted by the Austrian ministry in response to legislation from the European Union for the protection of ground- and surface water against NO_3^- deposition from agricultural sources, contains temporal, quantitative and spatial restrictions for the use of nitrogenous fertilizers. The directive has been adapted several times, the most recent adaption in 2017 (BMLFUW 2017), where Gabersdorf is also listed as an area with increased actions for the protection of waterbodies against NO_3^- . Another

groundwater protecting decree is the *Schongebietsverordnung* (LGBI. Nr. 70/2020) (SGV 2018) for agricultural areas in the *Leibnitzer Feld* in the reach of groundwater wells for drinking water supply (Klammler et al. 2013). These efforts already payed off in the region around Leibnitz, Klammler et al. (2013) for example describes a NO_3^- concentration reduction from 60 mg l^{-1} in 1992 to approximately 30 mg l^{-1} in 2013 at a groundwater observation well in Wagna, Styria.

2.1.3 Geology, Pedology and Climate

Located at the transition between the Alps in the West and the Pannonian Basin in the East, Styria contains diverse rocks from active mountain building to sedimentary basin formation. Whereas the northern, western and southwestern parts of Styria are made up of rugged mountains, the south-eastern part is relatively flat and called Styrian Basin. In the district Leibnitz, Plio- and Pleistocene sediments are deposited (Fig. 3) (Gasser et al. 2009).

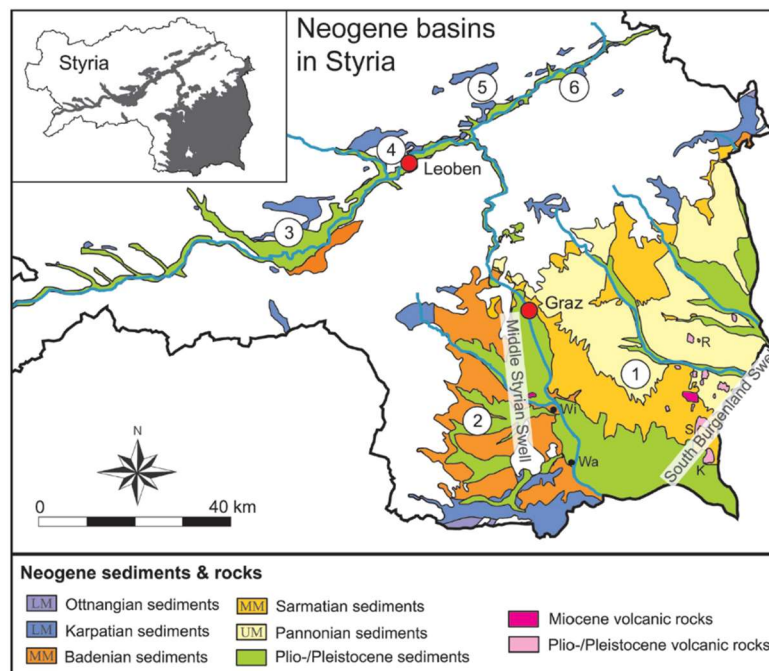


Fig. 3: Geological map of Neogene sediments and volcanic rocks in Styria (Gasser et al. 2009).

The field site in the southeast of Graz, where water and mineralized nitrogen (N_{min}) content measurements were conducted that were used in this study, is situated at the geological unit *Hochterrasse*, on the border to the *Niederterrasse* according to the geological map (GEOFAST 191) from the Geological Survey of Austria (GBA 2015). The soil consists of silica-based sediment-brown earth, originating from loamy-silty quaternary terrace material (Fig. 4). Hence the soil has a high water storage capacity and is very suitable for agricultural use (BFW 2004).

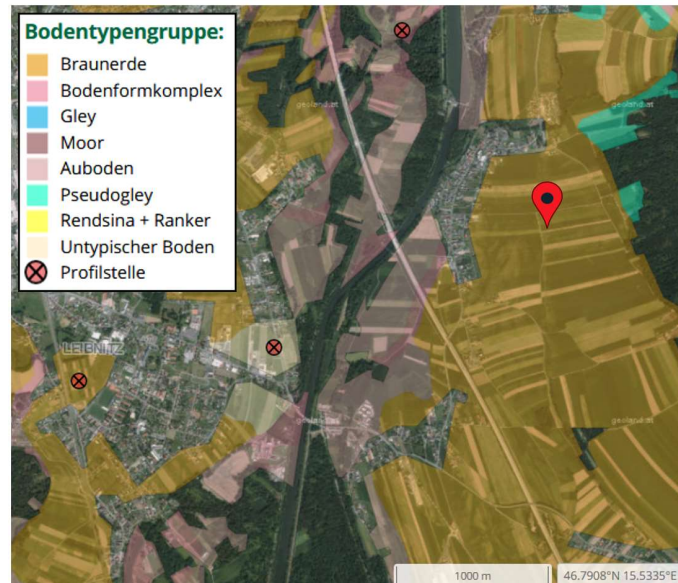


Fig. 4: Soil type (brown earth) at the field site (red position marker) according to the Austrian *Digitale Bodenkarte* (eBOD) (BFW 2004).

According to Klammler et al. (2013), the region receives sufficient precipitation and has a high sunshine duration, which leads to excellent conditions for agricultural use. The mean annual precipitation and the mean annual air temperature, measured for the period from 1999 - 2011 at the ZAMG measurement site in Wagna, are 908 mm and 10.2 °C, respectively (Klammler et al. 2013).

2.1.4 Experimental setup, treatments and soil samples

The field layout and the sampling at the field site as well as the analysis of the soil samples in the laboratory were conducted by Philipp Holler during his Master Thesis. The following information about the field site and the measurements is summarized from his thesis (Holler 2019).

At the field site in Gabersdorf, southern Styria, three blocks were established, each containing six plots (P1 – P6) with a different mixture of winter cover crops and fertilizer application. Each plot is 15x3 m, and therefore each plot covers an area of 45 m², each block an area of 270 m² and the total agricultural area used is 810 m².

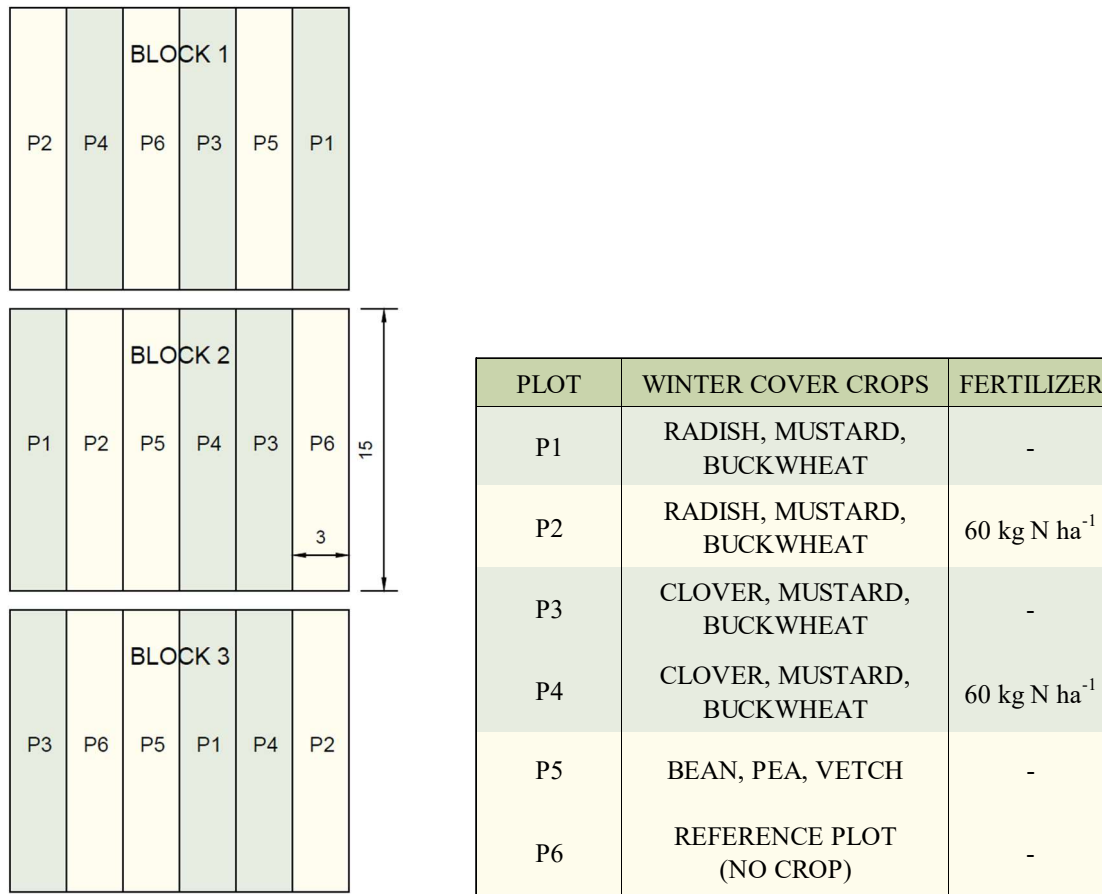


Fig. 5: Layout of the arrangement of the plots at the field site (left); Treatment of the different plots (right); light green representing the plots for the calibration and dark green the plots for the validation.

Three different compositions of winter cover crops were used for the five plots of each block. The sixth plot served as a reference plot, respectively. There was no cultivation of winter cover crops at the reference plot, it was just left as it was. The randomized complete block design was established. Since the plots were randomly arranged for every block (Fig. 5), the impact of the heterogeneity of the plots on the analysis is reduced. Winter cover crops for the use as fodder crops or green manuring are cultivated between two main growing seasons over the winter months. At the study site the winter cover crops listed in Fig. 5 were cultivated at the different plots. Only plot 2 and 4 were fertilized, which was done on the 3rd of August 2017. The fertilization amount was rather low, due to a restriction of fertilization in autumn in the study area.

The water and Nmin contents in the soil were determined by analysing approximately monthly soil samples, that were collected from August 2017 until May 2018 (Table 1). An auger was used for the extraction of the soil samples in the depths of 0–30 cm, 30–60 cm and 60–90 cm in every plot. For every depth layer, four soil samples were taken under disturbed conditions and mixed together for one representative sample, eventually resulting in 54 samples per sample taking for the analysis in the laboratory. According to Holler (2019) these samples were transported to the laboratory of the Plant Science Department of the University of Natural Resources and Life Sciences (BOKU) in Tulln in

coolers and then analysed for their N_{min} content following the ÖNORM L 1091 (1989). The cooling of the samples is necessary to avoid changes in the N_{min} contents due to microbial activity (Amelung et al. 2018). The samples were sieved with a mesh size of 5 mm and the extraction of N_{min} from 15 g of the soil samples was carried out with 60 ml of CaCl₂-solution (Amelung et al. 2018). The analysis of the mineralized nitrogen amounts was done with a spectrophotometer and is based on the absorption by parts of the N_{min} with different wavelength ranges. Ammonium (NH₄⁺) and nitrate (NO₃⁻) that make up the N_{min} are measured by the light absorption in a wavelength range of 660 nm and 540 nm, respectively (Holler 2019). The procedure in the laboratory is briefly summarized by Holler (2019).

Table 1: Schedule of actions taken at the field site.

DATE	ACTIVITY
3 rd August 2017	1 st soil sample taking
4 th August 2017	seeding of the winter cover crops
21 st September 2017	2 nd soil sample taking
30 th October 2017	3 rd soil sample taking
21 st November 2017	4 th soil sample taking
22 nd November 2017	ploughing of the fields
16 th April 2018	5 th soil sample taking
2 nd May 2018	6 th soil sample taking

The soil sampling in May 2018 took place after the heavy precipitation event on the evening of the 16th of April. The aim of this sample taking was to find out how this event has affected the soil properties and the NO₃⁻-N displacement into deeper soil layers. Originally it was intended to also use water and NO₃⁻-N content measurements from suction cups that were installed in every plot of block 2 for the calibration and validation. However, the suction cups failed to work consistently throughout the observation period. Additionally, the data would have required some conversion calculations containing inaccuracies, as the suction cups gathered seepage water over a time period that varied between days and weeks, before their content was emptied and analysed. Therefore, it was decided to only use the water and NO₃⁻-N content measurements from the soil samples.

2.2 HYDRUS-1D description

For the simulation of the water flow and the NO_3^- -N concentration in the vadose zone at the field site for the time period during which the soil samples were taken, a physically-based model was established with HYDRUS-1D. Since the water and solute flow in the plots at the field site can be simplified to a vertical movement, it was decided to use the one-dimensional HYDRUS software package. HYDRUS-1D is a modelling environment that allows simulating water, solute, and heat movement in one-dimensional variably saturated porous media. It has been developed by Jirka Šimůnek, Martinus Th. van Genuchten, and Miroslav Šejna (Šimůnek, van Genuchten, and Šejna 2008). The source code of the computational model of HYDRUS-1D is covered by the GNU General Public License. The numerical computer tool HYDRUS is composed of the computer program itself and the graphics-based user interface for data-preprocessing, discretisation of the soil profile and visualisation of the results. It contains both the pre-processing and post-processing units. In the pre-processing unit all necessary parameters to successfully run the HYDRUS FORTRAN codes are specified. The spatially distributed parameters (e.g. soil types in the soil column), initial conditions (e.g. water content and solute concentrations), root distribution, and the quantity and position of observation nodes are specified in a graphical environment (*Soil Profile – Graphical Editor*). With graphical tools the input as well as the output graphs, such as the distribution of pressure head, water content, solute fluxes, root water uptake and concentration over the depth of the soil column at preselected times or over time at specified depths (observation nodes) may be displayed using spatial or cross-sectional views and line graphs. The latter includes actual, potential, and cumulative fluxes also across defined upper or lower boundaries (Šimůnek et al. 2009). The programme is written in FORTRAN 77, whereas all modules are written in C++ (Vogel et al. 1996; Šimůnek et al. 2009). The graphs that are displayed in this thesis though are generated with the programming language Python3.8 (Van Rossum 1995) with the output files of HYDRUS-1D. The finite-element model HYDRUS numerically solves the Richards equation for the variably saturated water flow and advection-dispersion type equations for predicting solute transport in liquid phase as well as diffusion in the gaseous phase and if required also heat transport. However, in this work only the solute transport in the liquid phase was considered, as the observed solute (NO_3^- -N) is assumed to be not sorbed, and the solute transport in the gaseous phase and the heat transport were neglected. The water that is taken out of the soil by plants is accounted for by a sink term in the flow equation (Šimůnek et al. 2009). The model was run at the daily time step and was set up on the plots scale of 3 m wide. The soil profile was divided into 3 soil layers, each 30 cm deep and specified by different water retention parameters, bulk densities, initial values and solute specific parameters. The HYDRUS-model setup is further described in chapter 2.4.4. The structure of the HYDRUS-1D model implemented for the investigations in this thesis is schematically shown in the following diagram (Fig. 6).

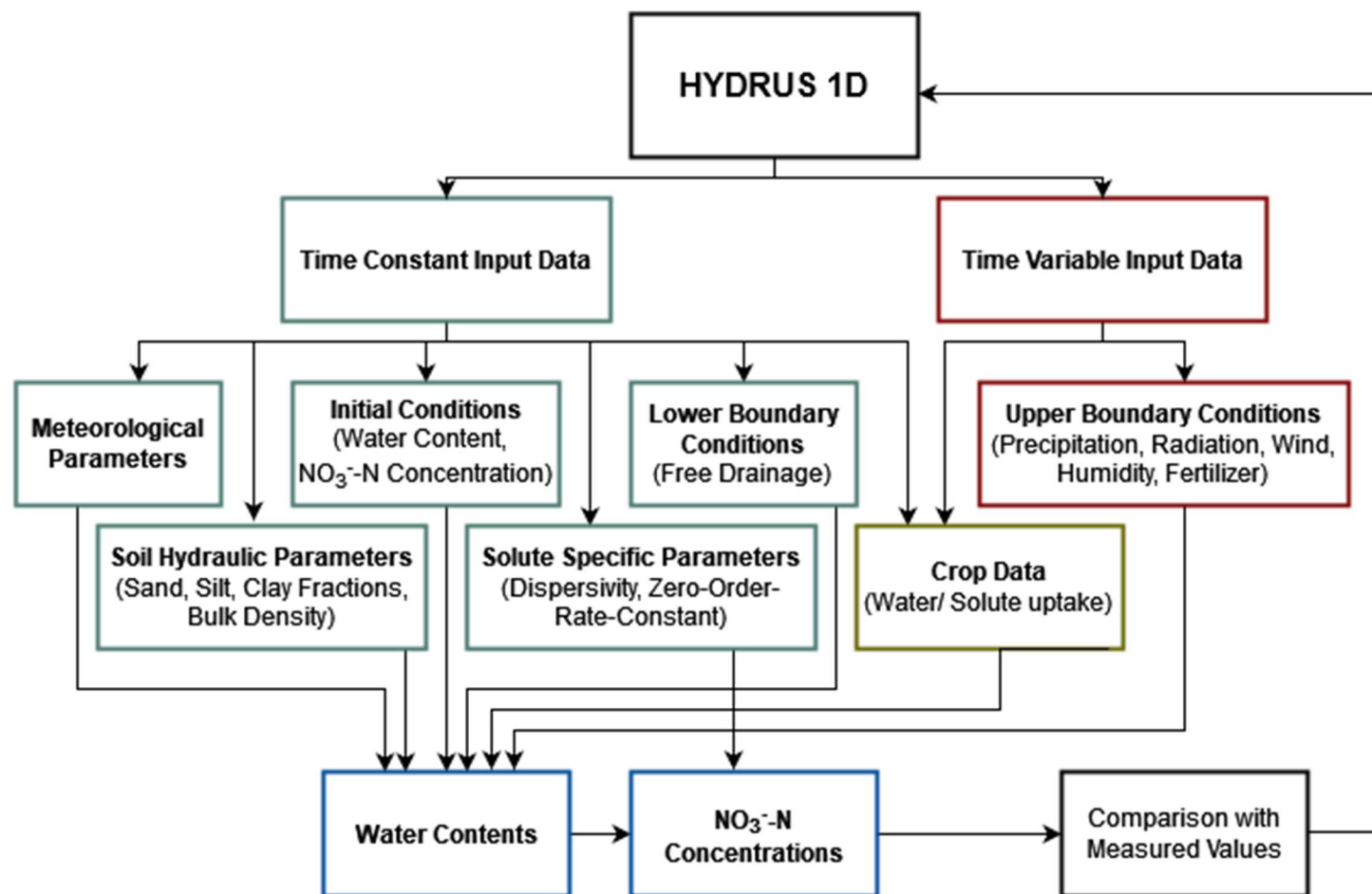


Fig. 6: Flow chart of the established model simulation process in HYDRUS-1D.

2.3 Soil physical basics in HYDRUS

To gain a better understanding of the parameters that needed to be characterized for setting up the HYDRUS-model at the field site, the underlying mathematical expressions that describe the flow and transport processes in the vadose zone are described in the following chapter (Stephens 1996). The information is based on the HYDRUS-1D Technical Manual (Šimůnek et al. 2009) and on textbooks by Amelung et al. (2018) and Taiz and Zeiger (2007). Especially the soil physical equations and the description of the parameters appearing in these equations are also described in the HYDRUS Manual, as these are equations underlying the calculations in HYDRUS-1D.

2.3.1 Water transport

Under natural conditions every soil contains water, which can be removed by oven-drying, which is normally done at 105 °C (Amelung et al. 2018). The amount of water that can be removed from the soil during that process in relation to the amount of soil is specified as the water content [%] (Amelung et al. 2018). The water content and the rate of water movement in the soil depend mainly on the pressure gradient through the soil and on the type of soil, its particle size distribution, and therefore its hydraulic conductivity. The addressed soil properties are described by the soil hydraulic parameters in the HYDRUS-model. When water that is coming from rain or other sources, hits the soil surface, then it percolates downward through the spaces between the soil particles due to gravity. It may then be adhered to the particle surface or fully fill the flow channels. Depending on whether the soil has a larger or a smaller pore size distribution, the water may just mainly flow through and drain out by gravity as for the first case, or it may be held against gravity due to adhesive and cohesive forces. The water that stays in a soil after the soil was fully saturated and then allowed to drain out the excess water is the moisture holding capacity, also referred to as the field capacity (Taiz et al. 2007). In soil science it is common to consider the potential, which is defined as the work needed for transporting a unit of water from one area to another. This is equivalent to the work that is needed to lift that unit water from an open water surface up into a pore (capillary) or to extract it from the soil. By quantifying all the movements with the potential concept, they can be compared and an overall movement in the soil can be derived. Water always moves from places with higher potential to those with lower potential, thus releasing energy, until every place holds the same total potential (Amelung et al. 2018). The water movement is also determined by the hydraulic conductivity of the soil. The simplest form of displaying the correlation between the water movement and the soil hydraulic conductivity is the Darcy equation [1856], named after the French engineer Henry Darcy, which is defined as:

$$q = -K \frac{dH}{dx} \quad (1.1)$$

In that equation q is defined as the volume of water that flows through a given cross section in a given time unit. K is the hydraulic conductivity of the soil [$L T^{-1}$] and describes rather the saturated than the

unsaturated flow in porous media, H is the driving total potential or total hydraulic head [L], and x is the spatial coordinate [L] (positive upward). A generalization of Darcy's Law is the Darcy-Buckingham-Law [1907], that describes both saturated and unsaturated water flow in soil:

$$q = -K(h) \left(\frac{\partial(h + x)}{\partial x} \right) \quad (1.2)$$

Buckingham found that in an unsaturated soil, the unsaturated hydraulic conductivity K is dependent on the water content. In this equation, K is the unsaturated hydraulic conductivity [$L T^{-1}$] and $H = h + x$. A combination with the continuity equation leads to a modified form of the Richards equation [1931], which is numerically solved by HYDRUS-1D:

$$\frac{\partial \theta(h)}{\partial t} = \frac{\partial}{\partial x} \left(K(h) \frac{\partial h}{\partial x} \right) + \frac{\partial K(h)}{\partial x} - S(h) \quad (1.3)$$

where θ is the volumetric water content [$L^3 L^{-3}$], h is the water pressure head [L], t is time [T], and S specifies a sink term [$L^3 L^{-3} T^{-1}$], accounting for the water uptake by plant roots. This water transport equation describes the change in the water content of a defined soil volume over time and is essential for the calculation of transport processes in the vadose zone (Hartge et al. 2014; Amelung et al. 2018). It is assumed that the soil matrix is homogeneous, rigid, inert and evenly distributed. K is again the unsaturated hydraulic conductivity given by

$$K(h, x) = K_s(x) K_r(h, x) \quad (1.4)$$

where K_r is the relative hydraulic conductivity [-] and K_s is the saturated hydraulic conductivity [$L T^{-1}$]. In the modified Richards equation only the water flow in the liquid phase is considered and the vapor movement as well as flow due to thermal gradients is neglected (Šimůnek et al. 2009).

The water in the soil is rarely in equilibrium, as processes like precipitation, evapotranspiration, root water uptake and percolation disturb the potential equilibrium most of the time. This is the case for saturated areas in the soil as well as for the vadose zone above the groundwater table (Amelung et al. 2018). If there is rainwater on the surface, the total potential gets less negative and the water infiltrates into the soil. The other way round, if there is evapotranspiration at the soil surface, the potential decreases and water may also travel upwards in the soil to compensate the gradient. These movements that aim at re-establishing the equilibrium concerning the potential, are called unsteady flow. Different from the steady flow is that the driving gradient changes over time (Amelung et al. 2018). The precipitation, as a soil water disequilibrating process, is defined in the time variable input data in the HYDRUS-model. The dependency of the water transport on the pore diameter and pore volume at different matric potentials leads to diverse water retention curves in different soils and different depth layers of soils. According to Amelung et al. (2018), the water retention curve for a soil is often derived by the van Genuchten-Mualem model through the pore size distribution curve and the saturated soil

hydraulic conductivity, where the pore connectivity is often neglected. In the HYDRUS-model three depth layers were defined and described with different water retention parameters. The unsaturated hydraulic functions are key input data in numerical models that describe the processes in the vadose zone. In HYDRUS-1D there are five analytical models for the hydraulic properties. In the established HYDRUS-model of this study, the unsaturated soil hydraulic properties are described with the van Genuchten-Mualem model (van Genuchten 1980) who used the statistical pore-size distribution model of Mualem (1976) to obtain a predictive equation for the unsaturated hydraulic conductivity function. They are given by:

$$\theta(h) = \begin{cases} \theta_r + \frac{\theta_s - \theta_r}{[1 + |\alpha h|^n]^m} & h < 0 \\ \theta_s & h \geq 0 \end{cases} \quad (1.5)$$

$$K(h) = K_s S_e^l [1 - (1 - S_e^{\frac{1}{m}})^m]^2 \quad (1.6)$$

where

$$m = 1 - \frac{1}{n} \quad n > 1 \quad (1.7)$$

and

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad (1.8)$$

θ_r is the residual water content, θ_s is the saturated water content, K_s is the saturated hydraulic conductivity, α is the inverse of the air-entry value (or bubbling pressure), n is a pore size distribution index and l is a pore connectivity parameter assumed to be 0.5 as an average for many soils. α , n and l are empirical parameters that affect the shape of the hydraulic functions. The above equations contain five independent parameters (θ_r , θ_s , α , n and K_s), which have to be specified in the *Water Flow Parameter* window in the HYDRUS-model. Alternatively, the water flow parameters can be estimated with pedotransfer functions, as the determination of the soil water retention curve is time-consuming and complex (Amelung et al. 2018; Šimůnek, van Genuchten, and Šejna 2008). The used programme for that estimation in HYDRUS-1D is the Rosetta (Schaap, Leij, and Van Genuchten 2001) that allows the prediction of the soil hydraulic parameters from more easily measured data. In the established HYDRUS-model of this study, the sand, silt and clay percentages and the bulk density that were estimated from nearby measurements were given, and the saturated hydraulic conductivity (K_s) and the water retention parameters according to Van Genuchten (1980) were predicted by the HYDRUS-model. There is also an integrated soil catalog in HYDRUS-1D with quite stable parameter suggestions taken from Carsel and Parrish (1988) and Rawls, Brakensiek, and Saxton (1982).

However, the catalog was only used for the estimation of possible parameter ranges for the sensitivity analysis and the calibration of the HYDRUS-model.

The potential evapotranspiration was calculated by HYDRUS-1D with the Penman-Monteith equation (Monteith 1981), specifically a combination equation recommended by the (Food and Agriculture Organization of the United Nations 1990) that combines the radiation term ET_{rad} [mm d⁻¹] and the aerodynamic term ET_{aero} [mm d⁻¹]:

$$ET_0 = ET_{rad} + ET_{aero} = \frac{1}{\lambda} \left[\frac{\Delta(R_n - G)}{\Delta + \gamma(1 + r_c/r_a)} + \frac{\rho c_p (e_a - e_d)/r_a}{\Delta + \gamma(1 + r_c/r_a)} \right] \quad (1.9)$$

ET_0 describes the evapotranspiration rate [mm d⁻¹], λ represents the latent heat of vaporization [MJ kg⁻¹], R_n is the net radiation at surface [MJ m⁻² d⁻¹], G is the soil heat flux [MJ m⁻² d⁻¹], ρ is the atmospheric density [kg m⁻³], c_p is the specific heat of moist air [i.e., 1013 kJ kg⁻¹ °C⁻¹], $(e_a - e_d)$ is the vapour pressure deficit [kPa], e_a is the saturation vapour pressure at temperature T [kPa], e_d is the actual vapour pressure [kPa], r_c is the crop canopy resistance [s m⁻¹], r_a is the aerodynamic resistance [s m⁻¹], Δ is the slope of the vapour pressure curve [kPa °C⁻¹], and γ is the psychrometric constant [kPa °C⁻¹]. More detailed information about the Penman-Monteith combination equation can be found in the HYDRUS technical manual (Šimůnek et al. 2009). Potential evaporation and potential transpiration are reduced to the actual values by the model based on the prevailing conditions in the soil and due to the specific properties of the vegetation (Šimůnek, van Genuchten, and Šejna 2012).

2.3.2 Root water uptake

In general, there is a high water vapour gradient between the atmosphere and the soil surrounding the roots. Hence, the plants emit water to the atmosphere with the intention to equilibrate that potential gradient, causing negative water pressure in the leaves. This in turn leads to water being taken out of the soil by the roots and vertical transport to the upper parts of the plants, where it is then transpired by the leaves (Taiz et al. 2007; Amelung et al. 2018). If water enters a cell, the positive hydrostatic pressure within the cell (turgor pressure) and therefore the water potential increases and the difference between inside and outside water potentials is reduced until it reaches equilibrium. The other way round if the water potential in the surrounding solution of the plant is lower, then the water may also move from the turgid cell to the soil. Hence, the water flow across membranes is a passive process (Taiz et al. 2007), driven by a gradient. In HYDRUS-1D, the sink term S that is already addressed in function (1.3), is defined as the volume of water that is removed by plants from a unit volume of soil in a unit time. The sink term S is given by:

$$S(h) = \alpha'(h) S_p \quad (1.10)$$

Herein, $\alpha'(h)$ is the water stress response function, a prescribed dimensionless function of the soil water pressure head ($0 \leq \alpha' \leq 1$), and S_p is the potential water uptake rate [T⁻¹]. In Fig. 7 the water

stress response function as assumed by Feddes, Kowalik, and Zaradny (1978) is shown. Near saturation the root water uptake ceases, as well as when the soil gets too dry ($h < h_4$). This pressure head h_4 is usually taken at the wilting point, where the water potential is so low that the plants cannot retain the turgor pressure, also if the water loss through transpiration ceases (Šimůnek et al. 2009; Taiz et al. 2007). In Fig. 7 it can also be seen that the root water uptake is at its highest between the pressure heads h_2 and h_3 , and that it decreases linearly between h_1 and h_2 as well as between h_3 and h_4 .

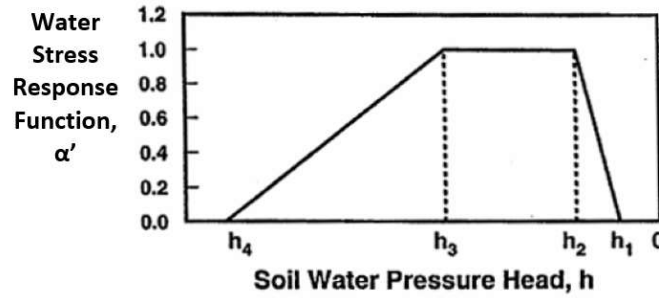


Fig. 7: Plant water stress response function, $\alpha'(h)$ by Feddes, Kowalik, and Zaradny (1978).

The cells are rarely fully hydrated because of transpirational water loss to the atmosphere, which may lead to such water deficits during droughts that the growth and the photosynthesis may be decelerated or also stopped. In relatively moist systems most of the water that is in a soil leaves it through root water uptake and subsequent plant transpiration rather than through evaporation from the soil surface. Only a very small part of the absorbed soil water remains in the plant to supply growth (~2%) or to be used in photosynthesis and other metabolic processes (~1%) (Taiz et al. 2007; Hupet et al. 2003).

2.3.3 Solute transport

With the water movement in the soil also other substances are transported, whether they are dissolved, suspended, or emulsified in the soil solution. This is important for many processes in the soil, such as reactions and weathering as well as exchange, leaching and accumulation processes (Amelung et al. 2018). The partial differential equation for one-dimensional nonequilibrium chemical transport of solutes with transient water flow in a variably saturated porous medium (Šimůnek and van Genuchten 1995) is as follows:

$$\begin{aligned} \frac{\partial \theta c_1}{\partial t} + \frac{\partial \rho s_1}{\partial t} = \frac{\partial}{\partial x} \left(\theta D_1^w \frac{\partial c_1}{\partial x} \right) - \frac{\partial q c_1}{\partial x} - r_{a,1} \\ - (\mu_{w,1} - \mu'_{w,1}) \theta c_1 - (\mu_{s,1} - \mu'_{s,1}) \rho s_1 + \gamma_{w,1} \theta + \gamma_{s,1} \rho \end{aligned} \quad (1.11)$$

where c and s are solute concentrations in the liquid [ML^{-3}] and solid [MM^{-1}] phase respectively, q is the volumetric flux density [L T^{-1}], μ_w and μ_s are first-order rate constants for solutes in the liquid and solid phase [T^{-1}], μ'_w and μ'_s are similar first-order rate constants providing connections between individual chain species [T^{-1}], γ_w and γ_s are zero-order rate constants for the liquid [$\text{ML}^{-3} \text{T}^{-1}$] and solid [T^{-1}] phase, ρ is the soil bulk density [ML^{-3}], r_a is the root nutrient uptake term [$\text{ML}^{-3} \text{T}^{-1}$] that for

passive root uptake is equal to the product of the sink term S in the water flow equation (1.3) and the concentration of the sink term c_r [ML⁻³], D^w is the dispersion coefficient for the liquid phase [L² T⁻¹]. The subscript w is for the liquid phase. In the simulations with HYDRUS-1D solutes are transported by advection and dispersion in the liquid phase, as well as by diffusion in the gaseous phase. Many other researchers who modelled nitrate transport assumed that the NO₃⁻-N is only present in the liquid phase and not sorbed (e.g. Chen et al. 2020; Veizaga, Rodríguez, and Ocampo 2016; Hanson, Šimůnek, and Hopmans 2006; Matteau et al. 2019; Neteler and Mitasova 2008; Li et al. 2015). Therefore, in this study the solid and gaseous concentration terms in (1.11) are also neglected. The adsorption isotherm that relates the solute concentrations in the solid s and the liquid c phase is described by:

$$s = \frac{k_s c^\beta}{1 + \eta c^\beta} \quad (1.12)$$

where k_s , η and β represent empirical factors. In this project $k_s=0$, leading to a solute transport without adsorption. This is also the reason why NO₃⁻ is displaced much faster with seepage water compared to the significantly retarded leaching of NH₄⁺ by the adsorption process (Li et al. 2015).

2.3.4 Boundary Conditions

Certain boundary conditions have to be specified in the HYDRUS-model. The model for the water flow part can cope with constant or time-varying prescribed head and flux boundaries, in particular also with atmospheric condition boundaries and free drainage. Solving the Richards equation (1.3) requires knowledge of the initial pressure head distribution, or if specified the initial water content distribution within the flow domain and for solving equation (1.11) the initial concentration within the flow region must be known.

$$h(x, t) = h_i(x) \quad t = t_0 \quad (1.13)$$

$$c(x, t) = c_i \quad t = t_0 \quad (1.14)$$

The chosen system-dependent upper boundary condition for the water flow, which cannot be defined beforehand is the Neumann boundary condition (Neuman, Feddes, and Bresler 1974) that involves the soil-air interface at the soil surface ($x=L$) and therefore the exposure to atmospheric conditions. The potential fluid flux across this boundary is driven by external conditions but depends also on the prevailing soil moisture conditions near the soil surface. The absolute surface flux is limited by two conditions:

$$\left| -K \frac{\partial h}{\partial x} - K \right| \leq E \quad \text{at } x = L \quad (1.15)$$

$$h_A \leq h \leq h_S \quad \text{at } x = L \quad (1.16)$$

where E is the maximum potential rate of infiltration or evaporation under the current atmospheric conditions [$L T^{-1}$]. In the second equation h_S represents the maximum pressure head at the soil surface and is normally set to zero. It can also be positive and thus allow ponding on top during heavy precipitation, prior to runoff. However, in this study every excess water is removed immediately, if h gets positive. h_A is the minimum pressure head determined from the equilibrium conditions between soil water and atmospheric water vapour (Šimůnek et al. 2009). For the lower boundary, a zero-gradient boundary condition is used to simulate a freely draining profile, where the water table is far below the bottom of the profile ($x=0$):

$$\frac{\partial h}{\partial x} = 0 \quad \text{at } x = 0 \quad (1.17)$$

For the solute transport, the HYDRUS-model again supports constant and time-varying prescribed concentration and concentration flux boundary conditions. To the upper and lower boundaries two different type of boundary conditions (Dirichlet and Cauchy type conditions) can be applied. The Cauchy type boundary condition is used for describing the concentration flux at the upper boundary:

$$-\theta D \frac{\partial c}{\partial x} + qc = q_0 c_0 \quad \text{at } x = L \quad (1.18)$$

In the equation q_0 is the upward fluid flux and c_0 is the concentration of the incoming fluid [ML^{-3}]. For the bottom of the profile, the water flow is directed out of the region. Therefore, another type of boundary condition (Neumann type) is applied, where (1.18) reduces to:

$$\theta D \frac{\partial c}{\partial x} = 0 \quad \text{at } x = 0 \quad (1.19)$$

2.4 HYDRUS-1D setup and model input data

2.4.1 Meteorological input data

Required meteorological input parameters for the HYDRUS-model were provided by the national meteorological service *Zentralanstalt für Meteorologie und Geodynamik* (ZAMG). The nearest ZAMG station is situated in Wagna, which is about 4km away from the field where the measurements were taken. The meteorological service provided hourly data for the period of June 2017 to May 2018, in particular precipitation [mm], air temperature [°C], solar radiation [J cm^{-2}], relative humidity [%] and wind speed [m s^{-1}] (Fig. 8).

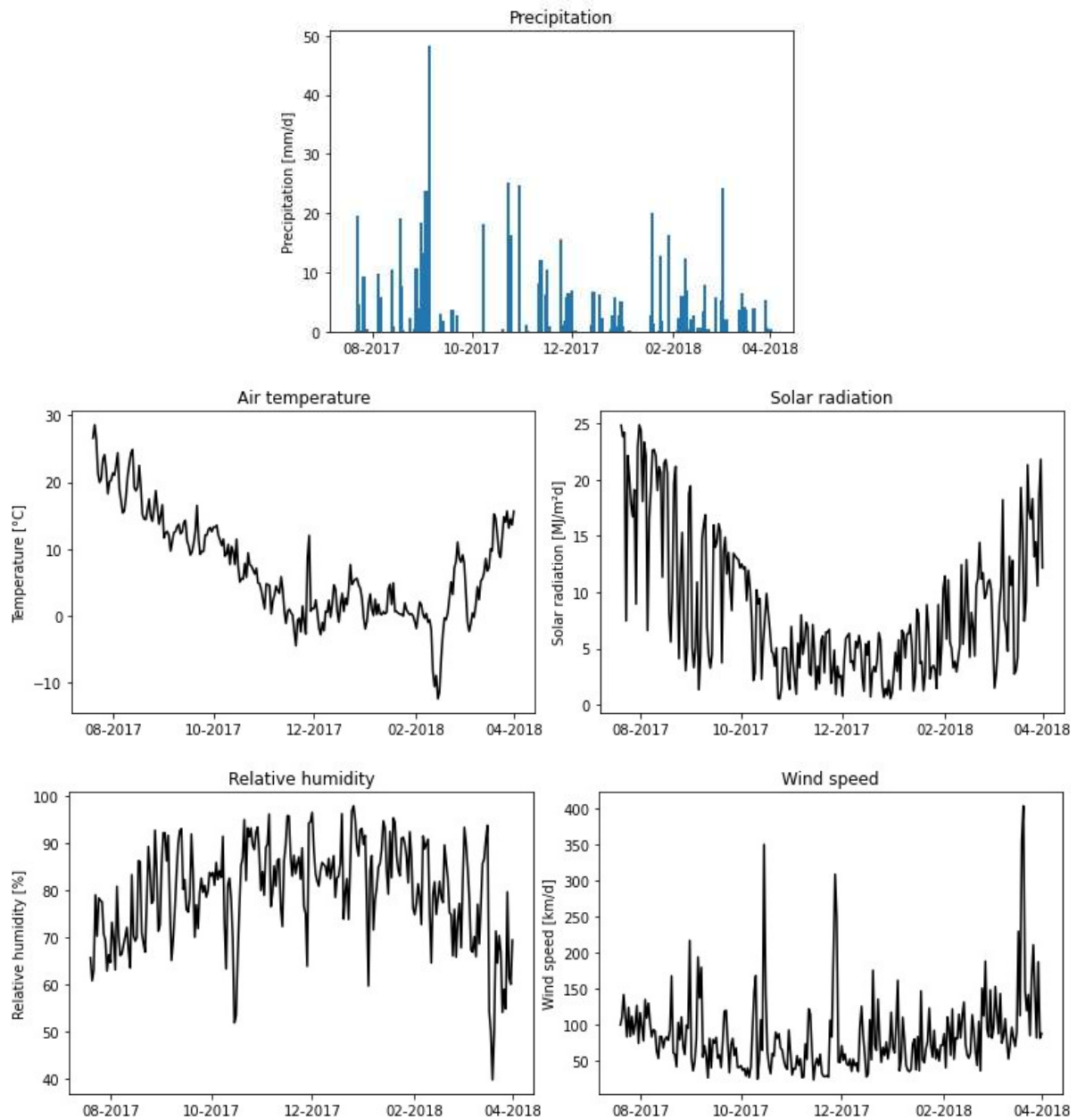


Fig. 8: Meteorological Data from the ZAMG station in Wagna (3rd August 2017 – 2nd May 2018) that serves as input for the HYDRUS-model; daily sums of precipitation (top) and solar radiation (middle right); daily mean values of air temperature (middle left), relative humidity (bottom left) and wind speed (bottom right).

2.4.2 Wet atmospheric nitrogen deposition

The wet atmospheric nitrogen (N) deposition is another required input for the HYDRUS-model. It was defined in the HYDRUS-model as a concentration input on the soil surface (c_{Top}) that is then multiplied by the amount of precipitation for the deposition flux. Concentration values were taken from the nearest measurement site (Arnfels) from the most recent reports of the wet deposition in Styria from the *Amt der Steiermärkischen Landesregierung* (Amt der Steiermärkischen Landesregierung 2018; 2019) in the years 2017 and 2018, and ranged from 0.13–0.65 mg l⁻¹. The measurements of the wet N-deposition show an annual variation with lower values in the winter months and a peak in spring (March to April).

2.4.3 Soil parameters

For setting up the model soil specific parameters are needed, such as the soil bulk density and textural distribution. Most of the information was collected from the Digital Soil Map from Austria (*Digitale Bodenkarte eBOD*) that is provided by the *Bundesforschungszentrum für Wald* (BFW 2004) and from the Digital Atlas of Styria (Land Steiermark 2017). The Styrian agricultural soil protection legislation (Stmk. LBSG 1987) and the soil conservation programme (Bodenschutzprogrammverordnung 1987) established a monitoring network of 1000 measuring points for pollutant, compaction and erosion assessment in agricultural areas in Styria. The sand, silt and clay fractions in the soil and the humus content are analysed. Two stations of the soil conservation programme LEI5 and LEI6 (Land Steiermark n.d.; n.d.) lie within a distance of 1 km of the field site and the information from these stations from the years 1986 - 1987 served as an initial guess for the soil parameters in the HYDRUS-model.

2.4.4 HYDRUS-model setup

For the simulation of the water flow in each plot with HYDRUS-1D, a soil column with a depth of 1 m and three different soil layers was defined. The first layer ranges from the soil surface to 30 cm, the second from 30 to 60 cm and the third from 60 cm to the bottom of the soil profile. In the graphical environment (*Soil Profile – Graphical Editor*) in HYDRUS-1D the material distribution was specified as the three soil layers. Mineralized nitrogen can be divided into NO₃⁻ and NH₄⁺. NH₄⁺ is sorbed and NO₃⁻ is not retained by soil particles and as described by Wang et al. (2010) readily mobile with water flow. For the simulation only the nitrate-nitrogen (NO₃⁻-N) was taken into account, as it represents the main part of the mineralized nitrogen in the soil and ammonium-nitrogen (NH₄⁺-N) is considered to be neglectable, like it was assumed by e. g. Gabriel, Muñoz-Carpena, and Quemada (2012). For the use in this study, the measured NO₃⁻-N concentrations [mg l⁻¹] by Philipp Holler were converted to mg kg⁻¹ by multiplying each measured value with the soil solution used for the analysis of the soil samples in the laboratory [0.06 l] and dividing each value by the mass of the respective analysed soil weights [~0.015 kg] times the dry matter [-]. This was followed by the conversion into mg cm⁻³ by multiplying the values with the corresponding bulk density, according to the soil depth layer (Table 2).

The water content measurements were already given in percentages and directly adopted for the use in the HYDRUS-model. The measured water contents [%] and the calculated NO_3^- -N concentrations [mg cm^{-3}] from the three blocks (with each six plots and three depths resulting in 54 values, respectively) were averaged over the three blocks, resulting in only one data set of mean values for each plot instead of three. In the *Graphical Editor* in HYDRUS-1D, the initial water content and NO_3^- -N concentration (the latter in “*Liquid Phase Concentrations [Mass of solute/Volume of water]*”) for the whole soil profile were defined (Fig. 9). These initial water content and NO_3^- -N concentration distributions are the values from the soil samples taken on the 3rd of August 2017. In the *Graphical Editor* also the number of nodes used to discretize the entire soil profile can be set, but is left as the default value of 101 nodes for the 1 m soil profile.

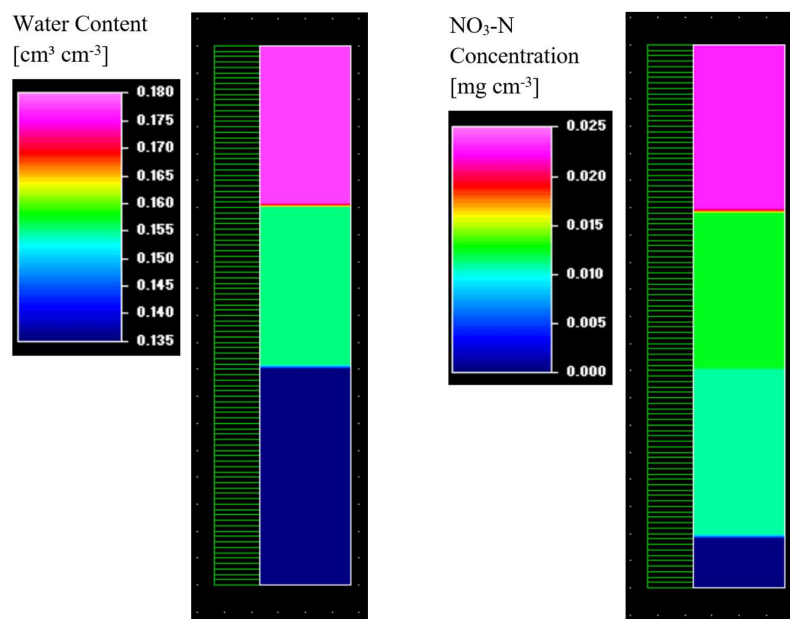


Fig. 9: Specifying the initial conditions (water content and solute concentrations) in the *Soil Profile – Graphical Editor* in HYDRUS, exemplary for plot 5.

The HYDRUS-model was set up for the period from 3rd of August 2017 until 16th of April 2018. In HYDRUS-1D the *Initial Time* and *Final Time* must be entered. As the Penman-Monteith equation is used for calculating the evapotranspiration, it is critical which day number is entered here. In the HYDRUS-model an initial time of e.g. 1 represents 1st of January and consequently 365 stands for the 31st of December in a common year. Thus, the initial and the final time were set with the numbers 214 – 470.

The time variable boundary conditions that are defined within the model are:

- Precipitation [cm d^{-1}]
- h_{CritA} [cm]
- Radiation [$\text{MJ m}^{-2} \text{d}^{-1}$]

- T_max [°C]
- T_min [°C]
- Humidity [%]
- Wind [km d⁻¹]

In the HYDRUS-1D model, the hCritA parameter indicates that as long as the pressure head of the observation node at the soil surface is below this value the actual evaporation is equal to the potential evaporation. When the pressure head falls beneath that value the actual evaporation decreases as the soil is too dry to deliver the potential rate. With a hCritA value of 21 000 cm the best results were achieved during calibration. The groundwater table at the study site is about 4m below the surface, according to the Austrian hydrological atlas (eHYD) *Bundesministerium für Land- und Forstwirtschaft, Umwelt und Wasserwirtschaft* (BMLFUW, n.d.). Therefore, capillary rise was neglected in the simulations. Concerning the iteration criteria, the maximum number of iterations for each time step was set to 10, as it is not likely that the model will converge if it does not within 10 iterations.

Measurements from the soil conservation programme (Land Steiermark 2017) served as a first guess for the estimation of the water retention parameters with the Rosetta programme, as well as information from the eBOD (BFW 2004) as already mentioned in chapter 2.4.3 . The final water retention and soil and solute specific parameter values are shown in Table 2. The zero-order rate constant for the liquid phase γ_w was set as $2 \times 10^{-8} \text{ mg cm}^{-3} \text{ d}^{-1}$ for the time from 3rd of August to the 21st of November and for the depth of 0–30 cm, assuming that all NH_4^+ -N produced by mineralization of organic nitrogen is transformed directly into NO_3^- -N by nitrification, similarly to the approach by Wang et al. (2010). On the 22nd of November the plots were ploughed and the winter cover crop residues remained on the field. To account for that input of organic matter that may be decomposed to NO_3^- -N in the soil, the zero-order rate constant γ_w was increased to $2.2 \times 10^{-4} \text{ mg cm}^{-3} \text{ d}^{-1}$ for the remaining time period. It was adjusted for the plots covered with winter crops and for the reference plot, where it was assumed that the grass that grew on plot 6 also served as NO_3^- -N input.

Table 2: Water retention parameters as described in the van Genuchten water retention function (1.5) and the soil and solute specific parameters of the three depth layers.

Soil layer	Residual water content θ_r [-]	Saturated water content θ_s [-]	Porosity distribution measure n [-]	Saturated hydraulic conductivity K_s [cm/d]	Air entry parameter α [1/cm]	Tortuosity parameter l [-]
0 - 30 cm	0.080	0.435	1.7	14.3	0.0052	0.5
30 - 60 cm	0.052	0.408	1.9	16.1	0.0054	0.5
60 - 100 cm	0.057	0.408	2.1	20.0	0.0056	0.5

Soil layer	Bulk density ρ [mg/cm ³]	Longitudinal dispersivity D_L [cm]	Diffusion coefficient in free water D_w [cm ² /d]	0-order rate constant for dissolved phase γ_w [mg/cm ³ d]	γ_w [mg/cm ³ d] for the 2 nd period from 22/11/2017 to 16/04/2018
0 - 30 cm	1380	12	1.64	2.00E-08	2.20E-04
30 - 60 cm	1430	12	1.64	0	0
60 - 100 cm	1440	12	1.64	0	0

The upper water flow boundary condition is defined as “*Atmospheric Boundary Condition with Surface Run Off*” and the lower as “*Free Drainage*”. The solute transport boundary conditions are set as “*Concentration Flux Boundary Condition*” for the upper and as “*Zero Concentration Gradient*” for the lower boundary. The equations according to these specifications are described in chapter 2.3.4 .

The cultivation of winter cover crops aims at the reduction of soil erosion, the increase of the humus content and improvement of soil quality in general, the suppression of weeds, the contribution and recycling of nutrients, the sequestration of carbon, the water protection and the preservation of natural shelter for animals (O’Connell et al. 2015; Holler 2019). Hence, a simulation of the water and solute flow in the soil has to include the processes in the soil root zone and the transport from the soil to the plants or reverse to cover that input or extraction of NO_3^- . The root water uptake was described by the Feddes approach. The water uptake by the roots is determined by the pressure head [cm] boundaries that are listed in Table 3. $P0$ describes the value of the pressure head below which the plant roots start to extract water and $POpt$ the pressure head below which the root water uptake rate is optimal. This extraction of water at the maximum possible rate, which is equivalent to a potential transpiration rate of $r2H$ [cm d⁻¹], lasts until a pressure head of $P2H$ is reached. Below that the roots cannot extract water at the maximum possible rate anymore. $P2L$ is similar to $P2H$, but the extraction of water decreases to a value that corresponds with a potential transpiration rate of $r2L$. Finally, if a pressure head of $P3$ is reached in the soil, the plant cannot extract any water anymore and the plant starts to wilt. These pressure heads correspond to the water stress response function (Fig. 7) by Feddes, Kowalik, and Zaradny (1978), so $P0$ is equivalent to h_1 , $POpt$ to h_2 , $P2H$ to h_3 and $P3$ to h_4 .

Table 3: Root water uptake parameters of the six plots, covered by different winter cover crop mixtures.

	P1	P2	P3	P4	P5	P6
P0 [cm]	0	0	0	0	0	-15
POpt [cm]	-1	-1	-1	-1	-1	-30
P2H [cm]	-500	-500	833.33	833.33	-750	-300
P2L [cm]	-900	-900	-1100	-1100	-2000	-1000
P3 [cm]	-16000	-16000	-16000	-16000	-16000	-16000
r2H [cm d ⁻¹]	0.4	0.4	0.4	0.4	0.4	0.4
r2L [cm d ⁻¹]	0.1	0.1	0.1	0.1	0.1	0.1

The pressure heads above are taken from an already existing database of suggested values for different plants in HYDRUS-1D that is based on studies by Wesseling et al. (1991) and Taylor and Ashcroft (1972). The winter cover crops of this study are not covered in the database, therefore the values were chosen on the basis of plants with similar root water uptake behaviour. Radish, mustard and buckwheat for example were described with the suggested parameters of wheat, and for bean, pea and vetch the parameters of beans were chosen. In plot 3 and 4, percentaged values of the root water uptake parameters of wheat (for mustard and buckwheat) and alfalfa (for clover) were used. The total root nutrient uptake is composed of the active nutrient uptake, where the plant takes up solutes against gradients and the passive nutrient uptake with the normal water movement (Šimůnek and van Genuchten 2008). Based on the assumptions by Matteau et al. (2019), Li et al. 2015 and Hanson, Šimůnek, and Hopmans (2006), only passive root solute uptake was adopted in the model and the part that is actively taken out by the plants was neglected. The crop data had to be specified in HYDRUS, whether there is no crop, or if the crop data are constant with time, entered in a table or given daily. In this study, the crop data was entered in a table with specific crop heights, albedos and root depths at certain dates and then the HYDRUS-model assumed a linear interpolation with time between the entered values. The values were defined in the beginning of the simulation in August 2017, shortly before the ploughing of the fields in November 2017 and once after 50% of the growing season. For plot 1 and 2 (radish, mustard, buckwheat), and plot 3 and 4 (clover, mustard, buckwheat) the maximum crop height was set as 70 cm, and the maximum root depth as 100 cm for plot 1 and 2, and 60 cm for plot 3 and 4. In plot 5 (bean, pea, vetch) the root depth and the crop height were limited by values of 60 cm and 100 cm, respectively. It was assumed that grass grew wildly on the reference plot (plot 6), and therefore the maximum root depth and crop height were set as 35 cm and 30 cm, respectively.

2.5 Sensitivity Analysis

Physically based numerical models are useful tools for the simulation of soil water dynamics, but the success of the predictions strongly depends on the parameter identification and the model's sensitivity to these parameters (Gabriel, Muñoz-Carpena, and Quemada 2012). Prior to the calibration, a sensitivity analysis was carried out to evaluate to which parameters the model is mostly sensitive to and which parameters do not really affect the simulated values. Saltelli (2002) described a sensitivity analysis as a “... *study of how the uncertainty in the output of a model (numerical or otherwise) can be apportioned to different sources of uncertainty in the model input.*” In the thesis the input parameters of the HYDRUS-model, such as soil and solute specific parameters, were varied one at a time and their influence on the output of the model was assessed. The chosen variation range for each parameter was varied dependent on the range of physically possible values of each parameter and on the values that were found in the literature. Some parameters differed already with the depth, as the soil was divided into three soil layers (0 – 30, 30 – 60, 60 – 90 cm). However, although the initial values from the three layers differed, the variation range for each parameter was the same. Alternatively, only the value of one depth layer was varied and the other two values were kept as they were.

2.6 HYDRUS-model calibration and validation

Model calibration is defined by Šimůnek, van Genuchten, and Šejna (2012) as a process where a model is adjusted for a particular problem, which is done by systematically altering the model parameters and maybe also initial and boundary conditions, until the simulated parameters match the observed values sufficiently (Šimůnek, van Genuchten, and Šejna 2012). The time period for the calibration and validation was set according to the dates where the soil samples were taken (Table 1). The starting point for the simulation was the 3rd of August 2017 and the end was set at the 16th of April 2018 (Fig. 10). The last measurements from May 2018 were not integrated due to a heavy precipitation event after the soil sampling on the 16th of April 2018. They were used for the evaluation of the HYDRUS-model as an additional research question. Data from plot 2, 5 and 6 served as input for the calibration. The initial water content and the NO_3^- -N concentration in the soil profile for starting the simulation were defined with the measured values from the plots, respectively. In HYDRUS-1D the model was set up for the conditions of each of these plots, like the root water uptake parameters appropriate for each winter cover crop and the initial conditions according to the measured water and NO_3^- -N contents of each plot. Soil layer depths, boundary conditions, observation nodes and discretisation, as well as the meteorological parameters and the time variable (meteorological) boundary conditions did not change for the different plots. The HYDRUS-model was fed with first estimations of all parameters and after running the model the output was compared to the measured values. According to the results and the deviations the parameters to which the model is sensitive to, that were identified by the sensitivity analysis beforehand, were adjusted. In the sensitivity analysis results (chapter 3.1), some of the calibrated parameters are presented, such as the saturated hydraulic

conductivity K_s , two shape parameters of the water retention function (α and n), the dispersivity D_L and the zero-order rate constant γ_w . The bulk density was also indirectly calibrated by varying the water retention parameters and K_s that too specify the soil properties. Furthermore, the residual and the saturated soil water content (θ_r and θ_s) and the diffusion coefficient were varied, but did not affect the outcomes as much as the other parameters. The interception constant was slightly varied in the beginning for assessing the impact on the model outcomes but was eventually set as the default value (0.25 mm d^{-1}). The root depths and crop heights of the various winter cover crops were only slightly varied in a physically reasonable range. It was decided to change one parameter at a time and keep the others fixed for one calibration round. For assessing the remaining deviations, the absolute errors between the measured and the simulated values from the corresponding day in the simulation were calculated. Overall, it was the aim to minimise the absolute errors between the measured and the simulated NO_3^- -N concentrations as well as for the water contents. Additionally, the root-mean-square error (RMSE) and the coefficient of determination for the simulated and measured water contents and NO_3^- -N concentrations, respectively were calculated.

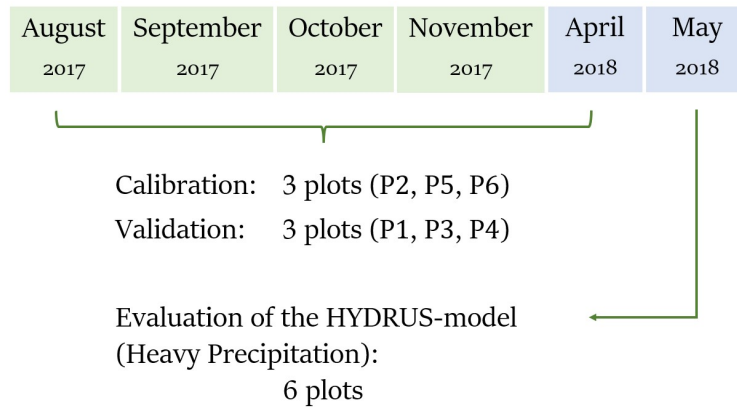


Fig. 10: Plots and time periods used for calibration, validation and the evaluation of the HYDRUS-model.

For the validation, the same time period as for the calibration was defined, but here the measurements of the remaining plots (1, 3 and 4) served as input. Plot 1 has the same winter cover crops as plot 2, and therefore the same model setup was used. However, as plot 2 was fertilized at the beginning of the simulation period, this NO_3^- -N input was removed for plot 1. Plots 3 and 4 feature the same winter cover crops that differ from all the other plots from the calibration, and therefore other root water uptake parameters had to be entered. Overall, the (time variable) boundary conditions, the meteorological parameters and the calibrated soil and solute specific parameters were not changed during validation. Solely the initial conditions (measured water and NO_3^- -N contents on the 3rd of August 2017) were adapted, respectively and as mentioned above the root water uptake parameters for plot 3 and 4.

2.7 Nitrate-nitrogen leaching estimation

The mass of nitrate-nitrogen leaching M [$\text{mgNO}_3\text{-N cm}^{-2}\text{d}^{-1}$] below the profile depth towards groundwater was assessed by multiplying the water leaching with the $\text{NO}_3\text{-N}$ concentration in the soil solution at the profile bottom:

$$M = \sum_{i=1}^n L_i C_i \Delta t_i A \quad (1.20)$$

This is a similar approach to the determination of nitrogen leaching below the root zone by Baram et al. (2016). In the above equation L is the water leaching flux below 1 m (bottom of the profile) [cm d^{-1}], Δt is a given time period [d], A is the given horizontal surface area [cm^2], and C is the average $\text{NO}_3\text{-N}$ concentration in the leaching water at the observed depth [mg cm^{-3}]. HYDRUS-1D performs that calculation internally and offers the $\text{NO}_3\text{-N}$ leaching mass as an output graph.

3. Results

3.1 Results of the sensitivity analysis

As a measure of the sensitivity of the HYDRUS-model to the varied parameter, the absolute errors between the measured and simulated water contents and NO_3^- -N concentrations, respectively were used. In the following graphs the behaviour of the HYDRUS-model with the alteration of sensitive parameters is shown. The variation of the absolute errors between the measured and simulated water contents (Fig. 11 to Fig. 13) with a variation of the saturated hydraulic conductivity K_s , the porosity distribution measure n and the air-entry parameter α , and the absolute errors of the measured and simulated NO_3^- -N concentrations (Fig. 14 to Fig. 16) with a variation of the saturated hydraulic conductivity, the longitudinal dispersivity D_L and the zero-order rate constant γ_w are displayed.

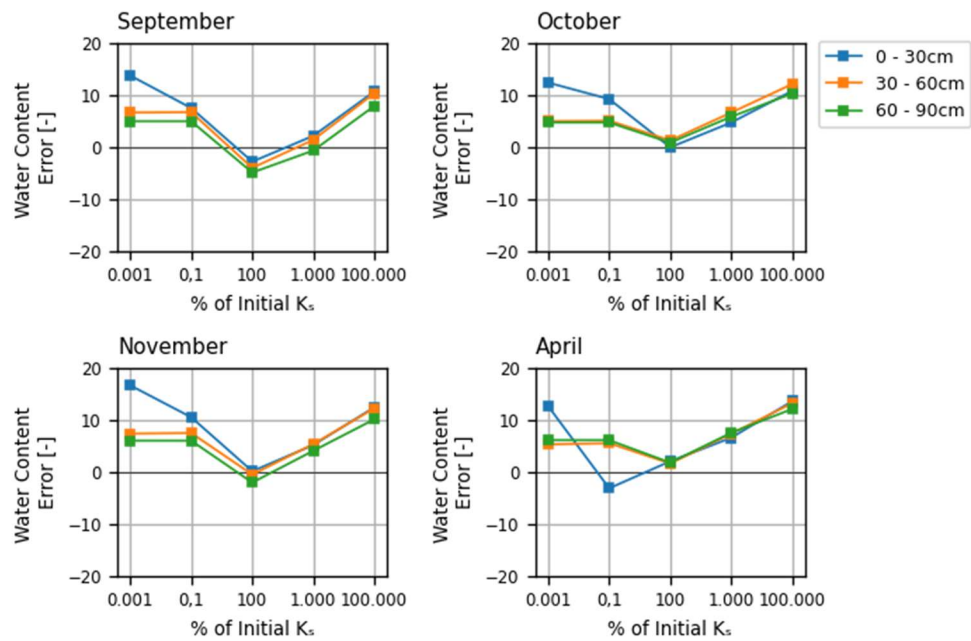


Fig. 11: Sensitivity analysis of the saturated hydraulic conductivity K_s concerning the absolute errors of the water contents in plot 6 (initial K_s of the three depth layers: ~ 14 , 16 and 20 cm d^{-1}).

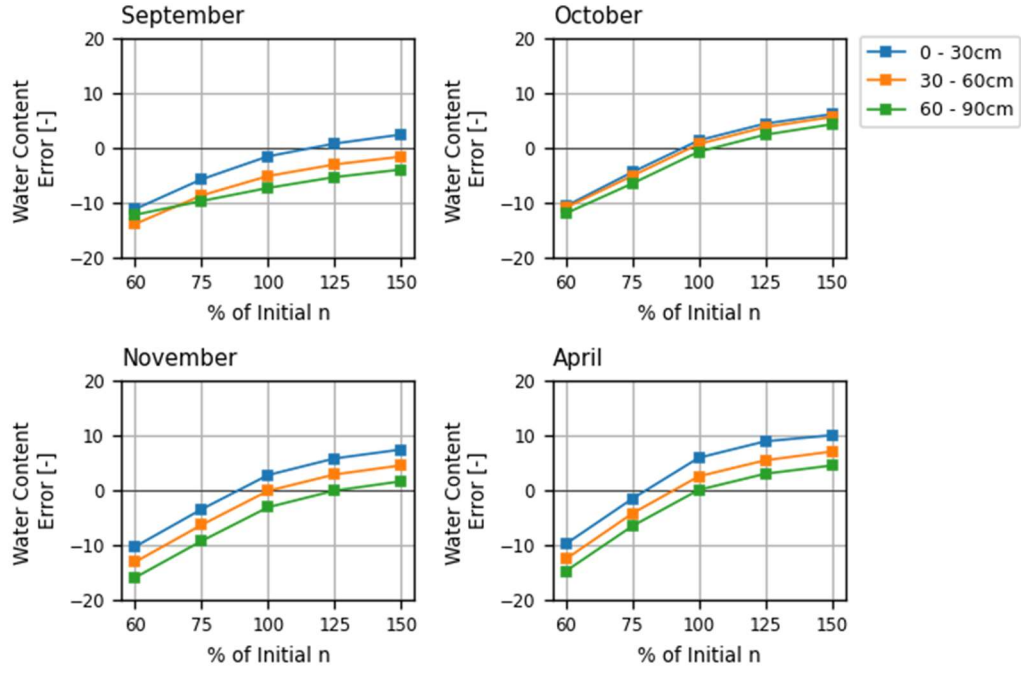


Fig. 12: Sensitivity analysis of the porosity distribution measure n concerning the absolute errors of the water contents in plot 2 (initial n : 2).

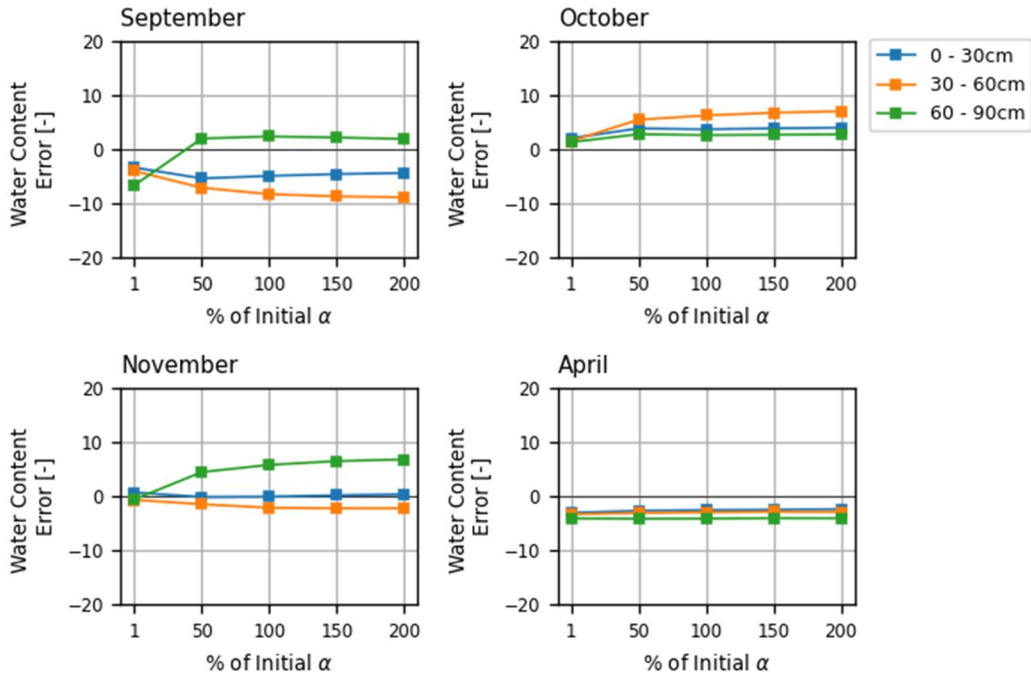


Fig. 13: Sensitivity analysis of the air-entry value α concerning the absolute errors of the water contents in plot 5 (initial α : 0.0775 cm⁻¹).

The variation of the saturated hydraulic conductivity K_s led to strong variations of the absolute errors between the observed and measured water contents. The absolute errors decrease with increasing K_s and show the smallest values at the initial K_s of the sensitivity analysis and then increase again with further increasing the K_s . Only the water content errors in the depth of 0–30cm behave a little differently, with higher water content errors at 0.001% of the initial K_s in all months and a much

lower, hence negative absolute error at 0.1 % in April 2018. Running the HYDRUS-model repeatedly with an increasing pore size distribution index n leads to increasing absolute errors from negative to positive values in all depths and all months. Contrarily, the water content errors of the three depth layers and of the different months behave differently with the variation of the air-entry value α . The water content errors in the depth layers from 0 – 60 cm slightly decrease with increasing α from 0 to 50 % of the initial value in the months September and November 2017, whereas the water content errors in the depth layer from 60 – 90 cm slightly increase with the same variation. Further increasing α does not lead to any more variation in the absolute errors. In the months October 2017 and April 2018, the water content errors even stay rather constant throughout the sensitivity analysis. Compared to the differences between the water content errors with varying K_s and n values, the water content errors do not change much with varying α values.

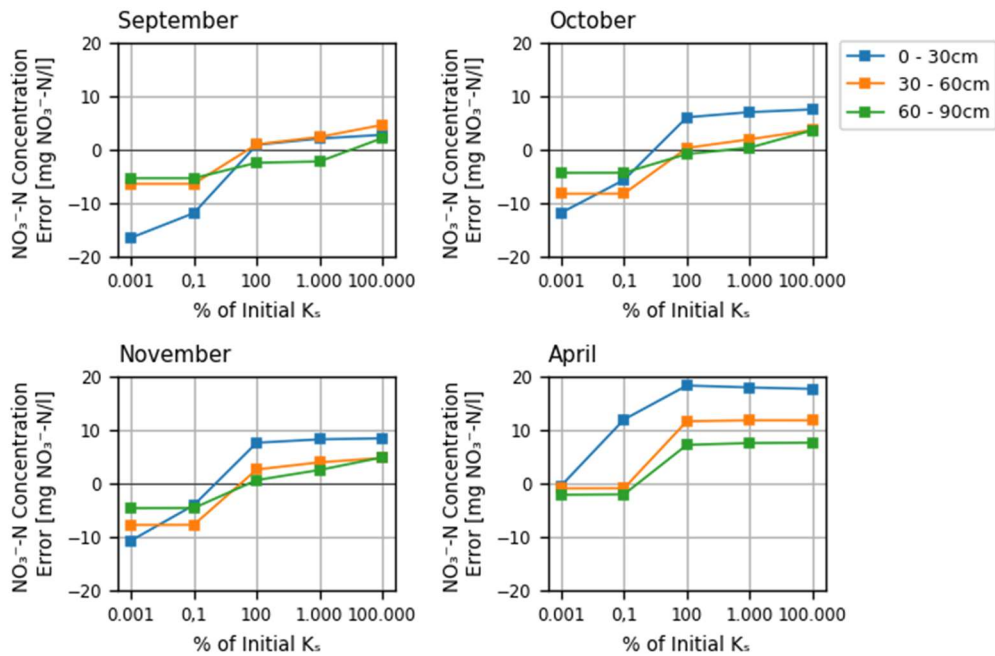


Fig. 14: Sensitivity analysis of the saturated hydraulic conductivity K_s concerning the absolute errors of the $\text{NO}_3\text{-N}$ concentration in plot 5 (initial K_s of the three depth layers: ~14, 16 and 20 cm d^{-1}).

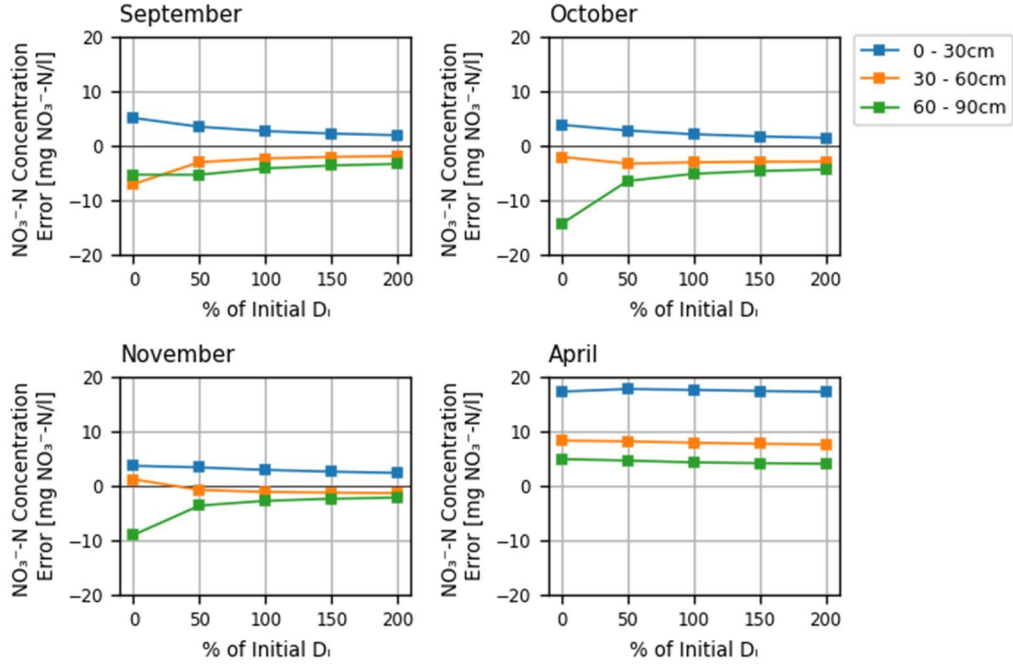


Fig. 15: Sensitivity analysis of the longitudinal dispersivity D_L concerning the absolute errors of the NO_3^- -N concentration in plot 2 (initial D_L : 15 cm).

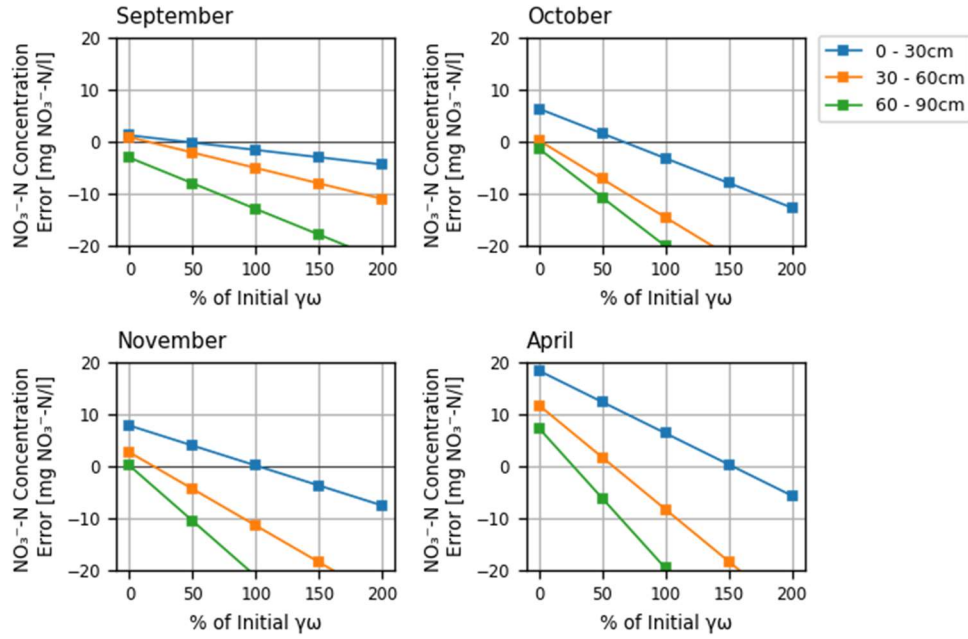


Fig. 16: Sensitivity analysis of the zero-order rate constant γ_w (0–30 cm) concerning the absolute errors of the NO_3^- -N concentration in plot 5 (initial γ_w : $0.00025 \text{ mg cm}^{-3} \text{ d}^{-1}$).

Similar to the water content errors, the absolute errors between the observed and measured NO_3^- -N concentrations also varied strongly with the variation of the K_s . Lower K_s values lead to negative absolute errors, which increase with increasing K_s . The difference between model runs with low and high K_s values is higher in the depth of 0 – 30 cm, compared to 30 – 90 cm. Additionally, the NO_3^- -N concentration errors from all three depth layers are much higher (i.e. more positive) in April than in the other three months. The same is the case for the NO_3^- -N concentration errors in April, if the

longitudinal dispersivity D_L and the zero-order rate constant γ_w are varied. Running the model with altered D_L input values does not lead to significant differences in the absolute errors of the NO_3^- -N concentrations. In October and November 2017, the absolute error in 60 – 90 cm depth decreased slightly from the model run with the smallest D_L to the other with higher D_L values. Concerning the zero-order rate constant γ_w , the NO_3^- -N concentration errors show a similar behaviour for all months and all depths. The errors get lower and eventually negative with higher γ_w values.

3.2 Calibration results

A comparison of the simulated HYDRUS-1D output curves of the three plots used for the calibration for the whole observation period from 3rd of August 2017 to the 16th of April 2018 with the punctual measured values of the water and NO_3^- -N contents in these plots can be seen in Fig. 17 to Fig. 22.

Plot 2:

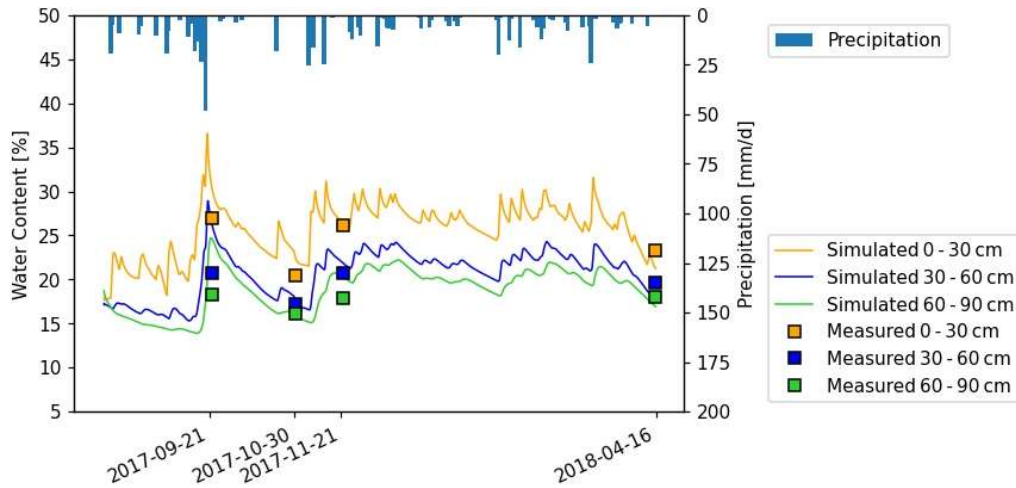


Fig. 17: Simulated water content curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 2 in comparison to the punctual measured water contents in plot 2.

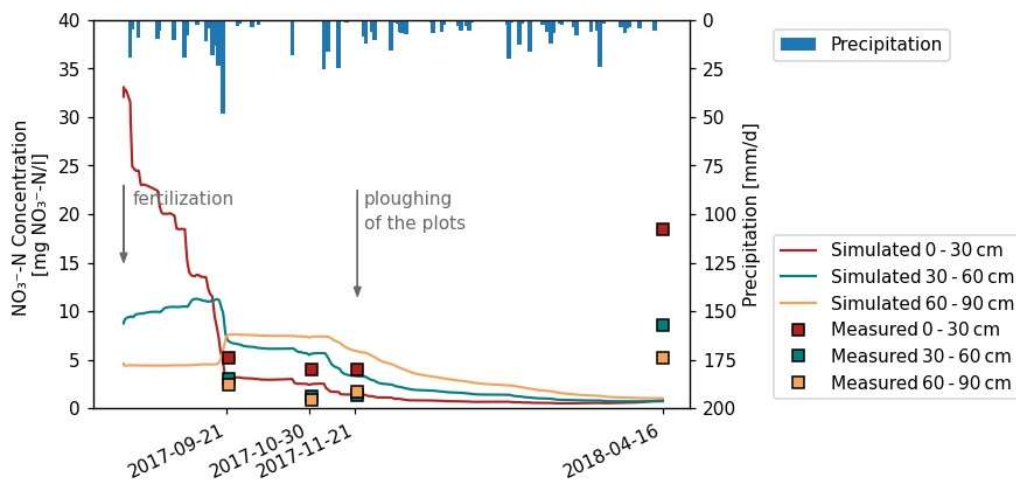


Fig. 18: Simulated NO_3^- -N concentration curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 2 in comparison to the punctual measured NO_3^- -N concentrations in plot 2.

The simulated water content curves of all depths behave similarly in response to the precipitation amounts, with the water content in the depth layer from 0–30 cm showing a bit more variation than in the other layers (Fig. 17, Fig. 19, Fig. 21). Apart from that, the water contents of the three soil depth layers are also on different levels throughout the simulation from 3rd of August 2017 to 16th of April 2018. The highest water content is in the soil layer near to the surface, followed by the middle soil layer and the lowest water content is in the deepest soil layer. The simulated water contents match the measured water contents well, only in September 2017 are the observed values to a small extent underestimated by the simulated values of the plots. The simulated NO₃⁻-N concentrations decrease rapidly until the second soil sampling on the 21st of September 2017 and then the curves flatten out until the end of the simulation in April 2018 (Fig. 18). The strong decline of the NO₃⁻-N concentration in the depth from 0–60 cm during the precipitation event on the 19th of September 2017 is followed by an NO₃⁻-N increase in the deepest soil layer. In September, October and November 2017, the observed NO₃⁻-N concentrations in 30–90 cm are slightly underestimated by the simulated NO₃⁻-N concentrations, whereas the simulated concentrations in 0–30 cm slightly overestimate the real NO₃⁻-N amount. The simulated NO₃⁻-N concentrations decrease towards April 2018 and highly underestimate the observed NO₃⁻-N concentrations in the soil from 0–90 cm.

Plot 5:

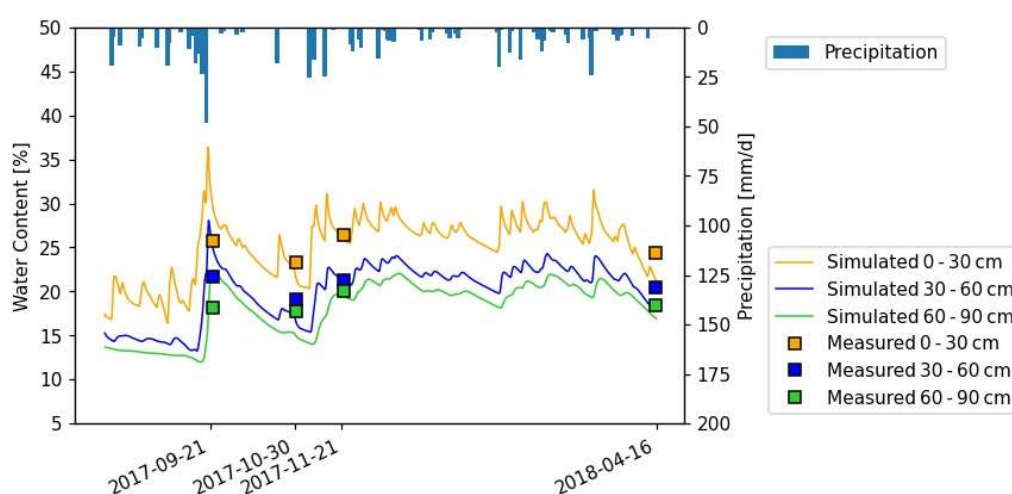


Fig. 19: Simulated water content curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 5 in comparison to the punctual measured water contents in plot 5.

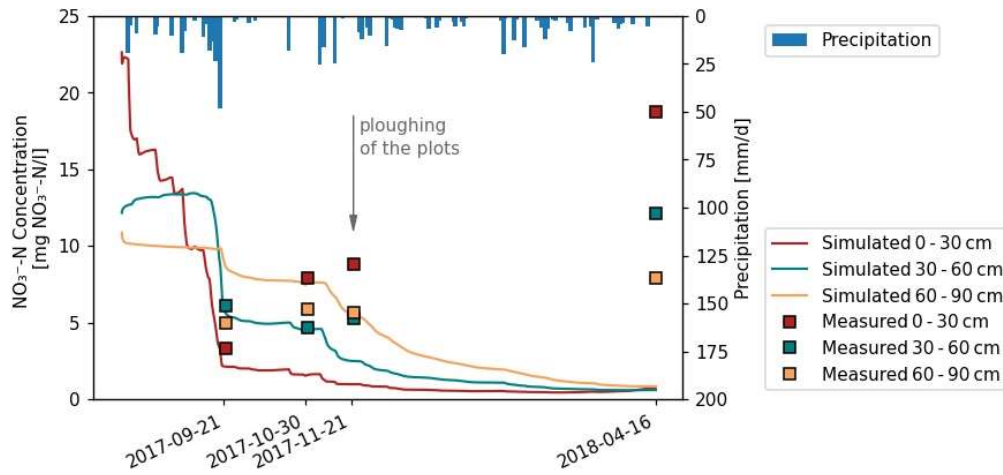


Fig. 20: Simulated NO_3^- -N concentration curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 5 in comparison to the punctual measured NO_3^- -N concentrations in plot 5.

The initial NO_3^- -N concentrations in the three depth layers of plot 5 are lower than the ones in plot 2, but the curves have the same decrease in the beginning of the simulation and flatten out until the 16th of April 2018 (Fig. 20). The simulated values on this date also underestimate the real NO_3^- -N levels in the whole soil profile. At the other measurement dates, the simulated NO_3^- -N concentrations achieve a better fit to the observed values, only the simulated NO_3^- -N concentration in 0 – 30 cm is much lower than the measured concentrations in October and November 2017.

Plot 6:

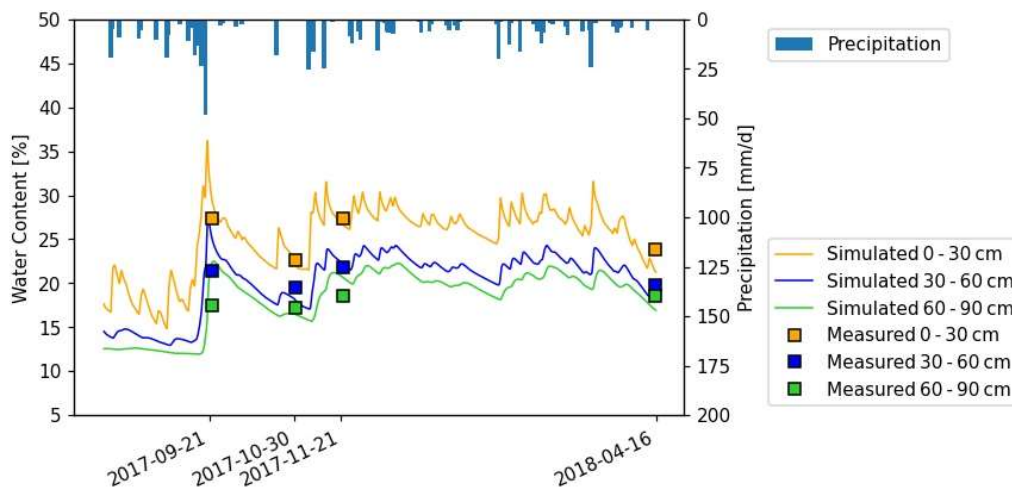


Fig. 21: Simulated water content curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 6 in comparison to the punctual measured water contents in plot 6.

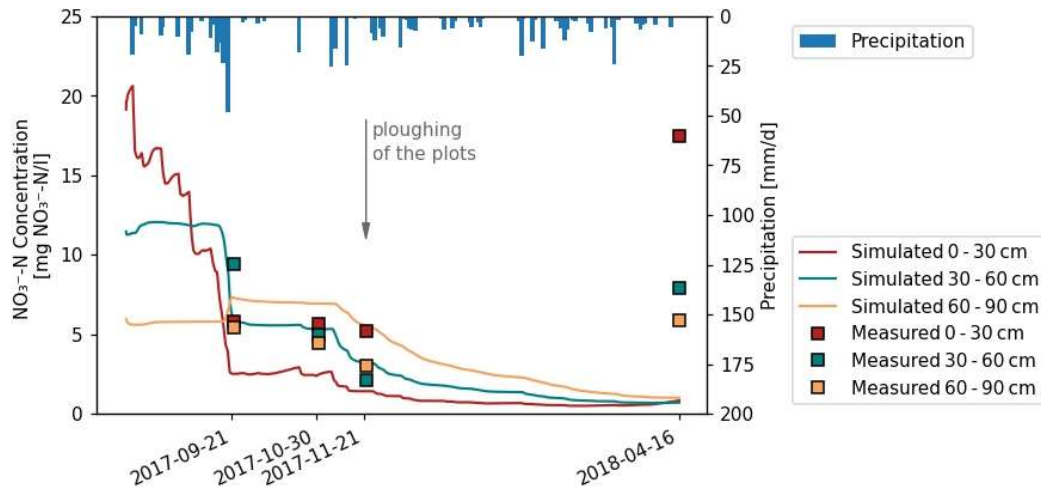


Fig. 22: Simulated NO_3^- -N concentration curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 6 in comparison to the punctual measured NO_3^- -N concentrations in plot 6.

The water content curves of plot 6 fit well to the punctual measurements in September, October, November 2017 and April 2018 (Fig. 21). There are more up and down movements in the water content curve of the soil layer from 0–30 cm than in the other two curves from deeper soil layers. Similar to plot 2 and plot 5, in the beginning the NO_3^- -N concentration in 0–30 cm decreases abruptly until the end of September 2017, when the precipitation amounts lessen (Fig. 22). In 30–60 cm depth the NO_3^- -N concentration only decreases with larger precipitation events, such as the one on the 19th of September 2017 with $\sim 48 \text{ mm d}^{-1}$. During the mentioned NO_3^- -N concentration drop in 0–60 cm, the NO_3^- -N concentration in 60–90 cm increases a few $\text{mg NO}_3^- \text{ N l}^{-1}$. From around November 2017 to April 2018, the NO_3^- -N concentration in the whole soil profile slowly stagnates and is much lower than the measured values on the 16th of April 2018.

Due to the large errors between the simulated and measured NO_3^- -N concentrations in April during the calibration, the zero-order rate constant γ_w for the dissolved phase was adapted. It was increased from 2×10^{-8} to $0.000375 \text{ mg cm}^{-3} \text{ d}^{-1}$ for the depth layer from 0–30 cm for the remaining simulated period after the ploughing in November 2017. Accordingly, the calibrated NO_3^- -N concentration curves from the plots changed to the curves that can be seen in Fig. 23 to Fig. 25.

Plot 2:

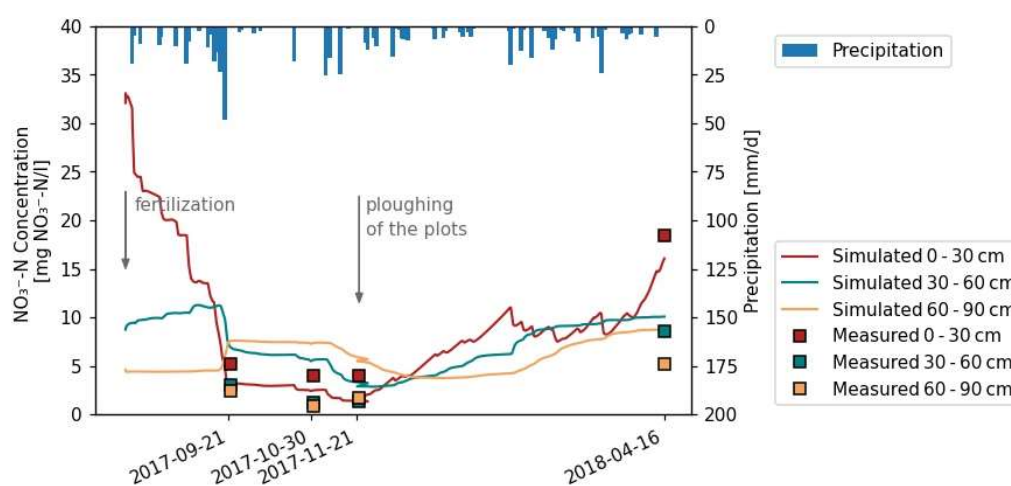


Fig. 23: Simulated NO_3^- -N concentration curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 2 with adapted zero-order rate constant for the time period from November to April in comparison to the punctual measured NO_3^- -N concentrations in plot 2.

Plot 5:

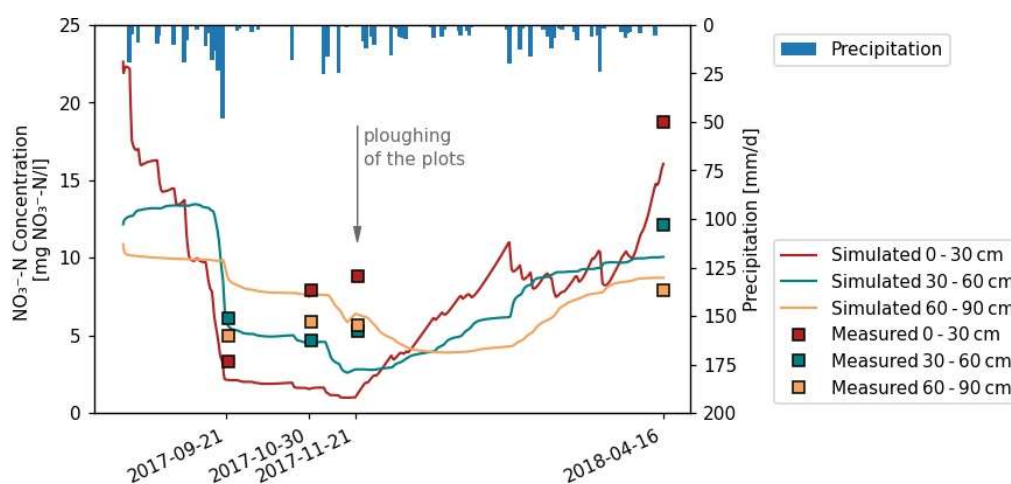


Fig. 24: Simulated NO_3^- -N concentration curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 5 with adapted zero-order rate constant for the time period from November to April in comparison to the punctual measured NO_3^- -N concentrations in plot 5.

Plot 6:

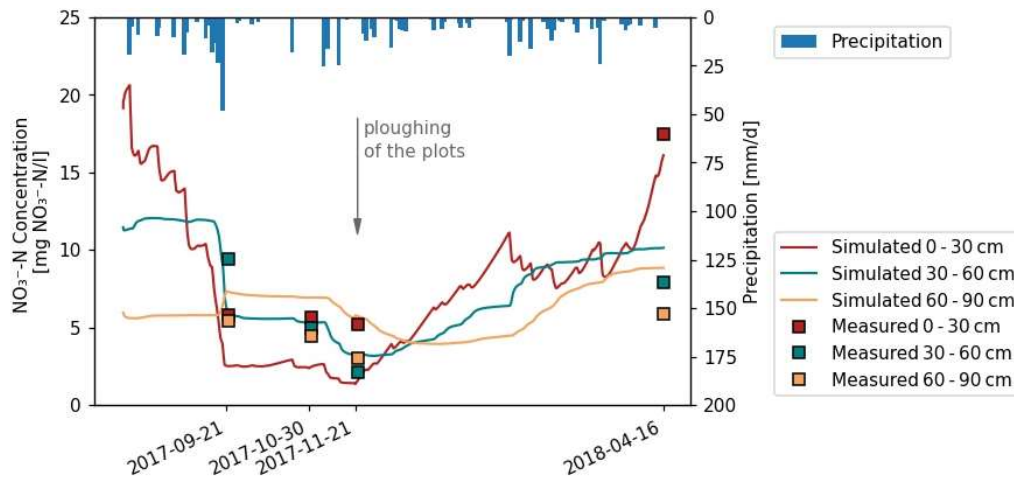


Fig. 25: Simulated NO_3^- -N concentration curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 6 with adapted zero-order rate constant for the time period from November to April in comparison to the punctual measured NO_3^- -N concentrations in plot 6.

As a consequence of the modified γ_w , the NO_3^- -N concentration rises in all plots and depths until the end of the simulation. Nearly consistently, the simulated NO_3^- -N concentrations in 0 – 30 cm in plots 2, 5 and 6 are a little lower than the measured values, and in 30 – 90 cm the simulated NO_3^- -N concentrations are slightly higher than the measured ones.

3.3 Validation results

For the validation of the model that was calibrated with plot 2, 5 and 6, it was implemented for the remaining three plots (P1, P3 and P4). The simulated water and NO_3^- -N curves of these three plots compared to the measurements are presented hereafter (Fig. 26 to Fig. 31).

Plot 1:

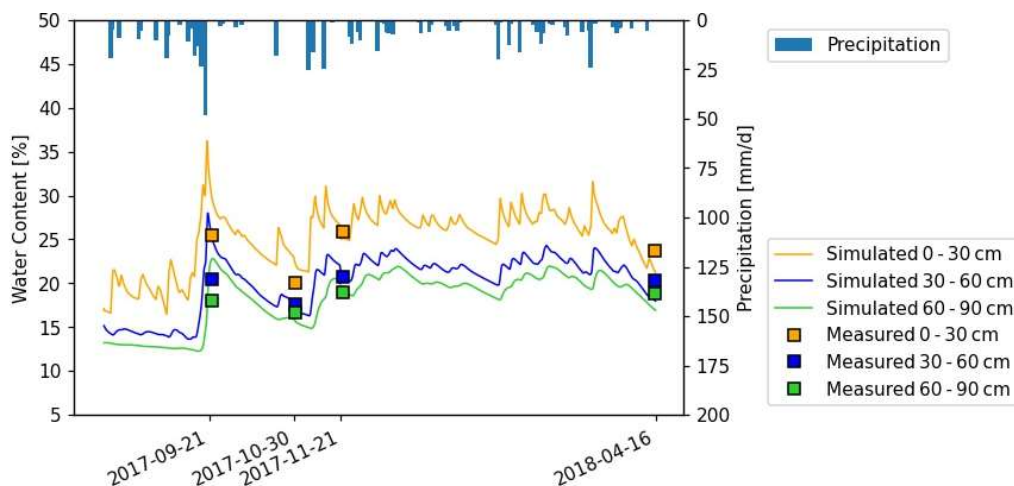


Fig. 26: Simulated water content curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 1 in comparison to the punctual measured water contents in plot 1.

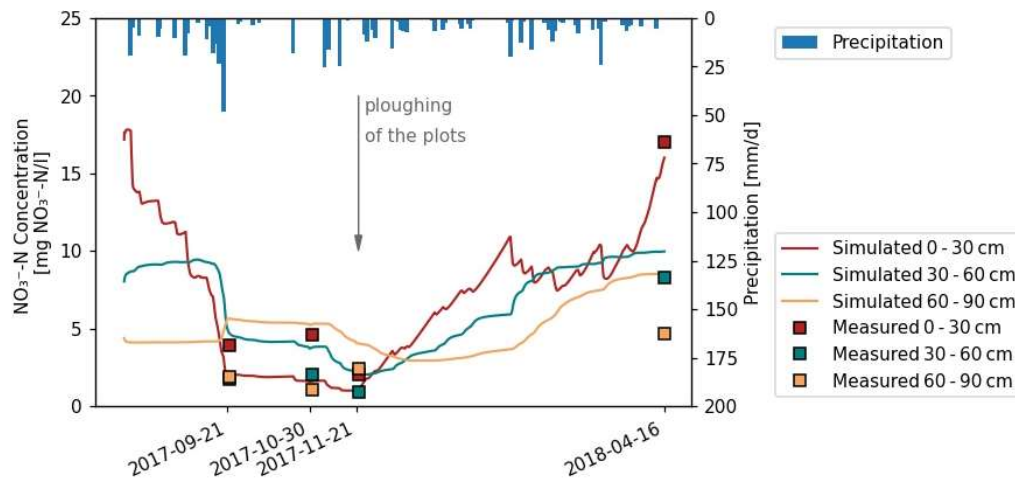


Fig. 27: Simulated $\text{NO}_3\text{-N}$ concentration curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 1 in comparison to the punctual measured $\text{NO}_3\text{-N}$ concentrations in plot 1.

Plot 3:

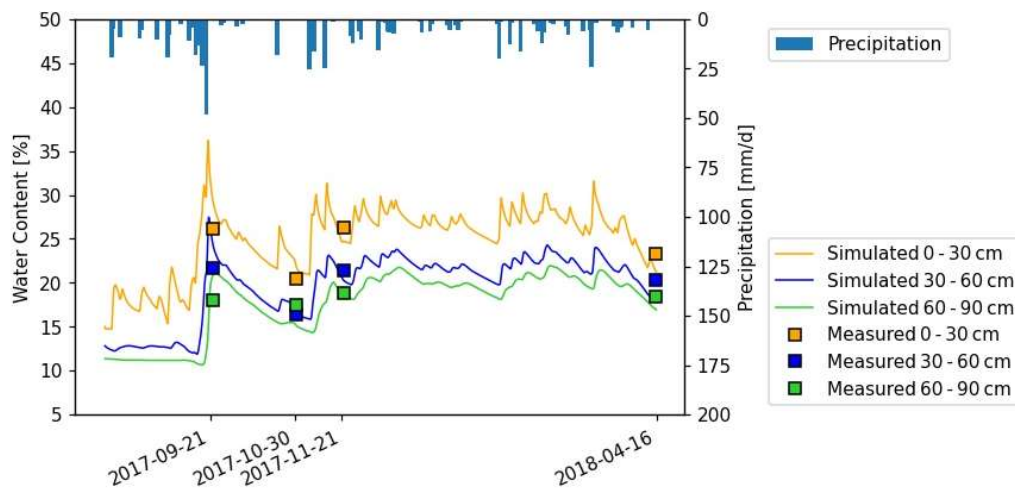


Fig. 28: Simulated water content curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 3 in comparison to the punctual measured water contents in plot 3.

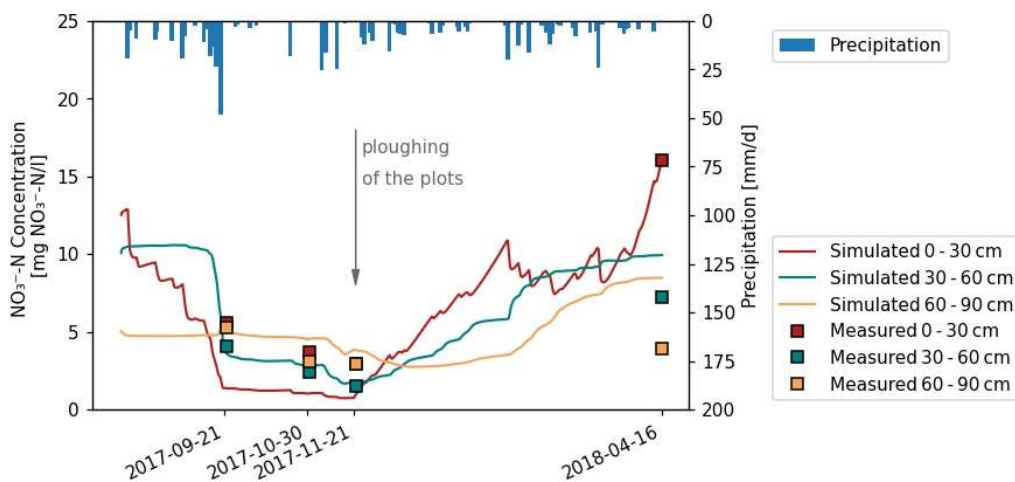


Fig. 29: Simulated $\text{NO}_3\text{-N}$ concentration curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 3 in comparison to the punctual measured $\text{NO}_3\text{-N}$ concentrations in plot 3.

Plot 4:

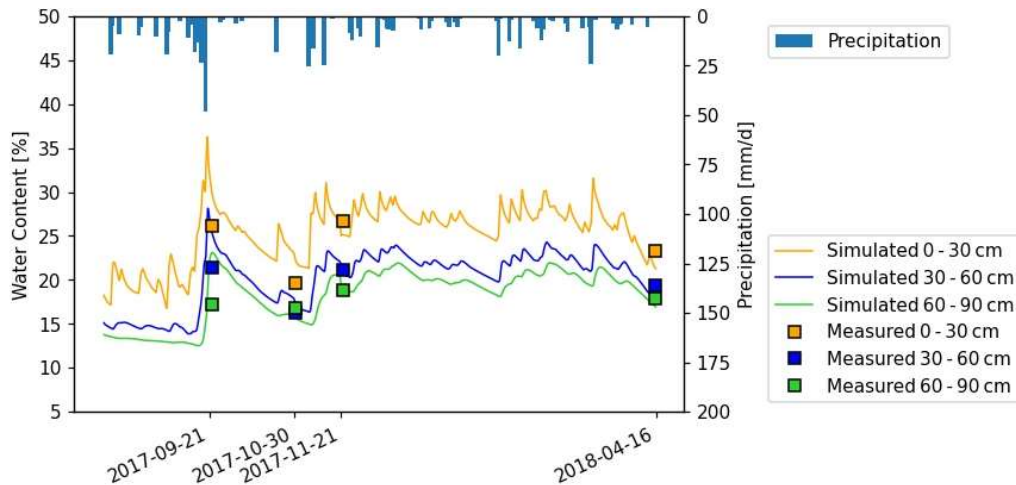


Fig. 30: Simulated water content curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 4 in comparison to the punctual measured water contents in plot 4.

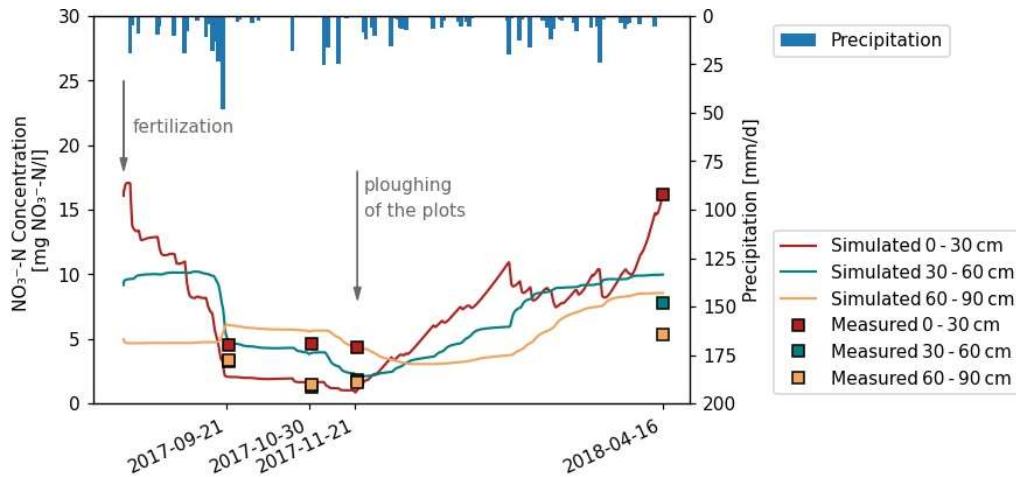


Fig. 31: Simulated NO₃-N concentration curves for the time period from 3rd of August 2017 to 16th of April 2018 of plot 4 in comparison to the punctual measured NO₃-N concentrations in plot 4.

The water contents of plot 1, 3 and 4 react similarly to the water contents in the calibrated plots, and also show more variation in 0-30 cm depth than in the other depths. The errors between the simulated and measured water contents in the three plots and three depths are quite low. The initial NO₃⁻-N concentrations that were measured on the 3rd of August 2017 in the depth of 0-30 cm in plot 1, 3 and 4 are lower than the initial NO₃⁻-N concentrations in the three calibrated plots. However, the decrease of the NO₃⁻-N concentration in the upper soil layer until the 21st of November and the increase after the ploughing due to the adjustment of the parameter γ_w in the HYDRUS-1D model is similar to the calibrated plots. Whereas the NO₃⁻-N concentrations decrease in the upper soil layers during the precipitation event in September 2017, in the deepest soil layer the NO₃⁻-N concentration temporarily increases. Apart from that, the NO₃⁻-N concentrations further decrease during the first half of the simulation period and increase again after the alteration of the γ_w in the HYDRUS-1D model.

3.4 Results of model simulations

3.4.1 Evapotranspiration

The actual evapotranspiration (ETa) [mm d^{-1}], derived from the potential evapotranspiration (ETp) that is calculated with the Penman-Monteith equation, for the whole time period for the calibrated plots (P2, P5 and P6) is displayed in Fig. 32. The potential evaporation and the potential transpiration are evaluated from the ETp using the leaf area index (LAI), and are displayed in Fig. 33 for plot 5. Both graphs illustrate the decline of the evapotranspiration during the months of winter and the peaks coinciding with the wind speed peaks.

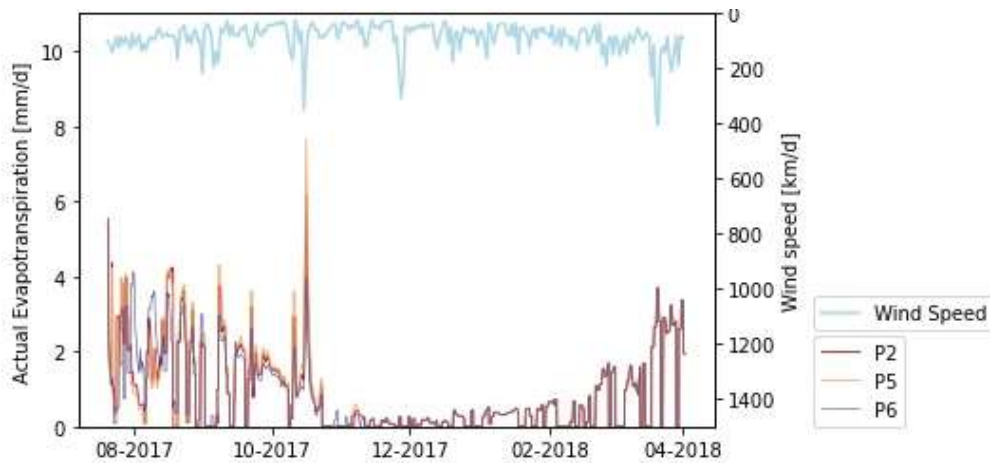


Fig. 32: Actual Evapotranspiration calculated by HYDRUS-1D for the calibrated plots (P2, P5 and P6), with the wind speed curve from the ZAMG measurement station.

Evaporation from the soil surface is driven by atmospheric conditions, surface and soil moisture. The potential fluid flux across the soil-air-interface is controlled only by external conditions, but the actual fluid flux depends also on the soil moisture conditions in the upper soil layer near the surface that are defined by specific pressure heads in HYDRUS-1D. The actual transpiration is calculated through the potential transpiration with a dimensionless water stress index (Šimůnek et al. 2009).

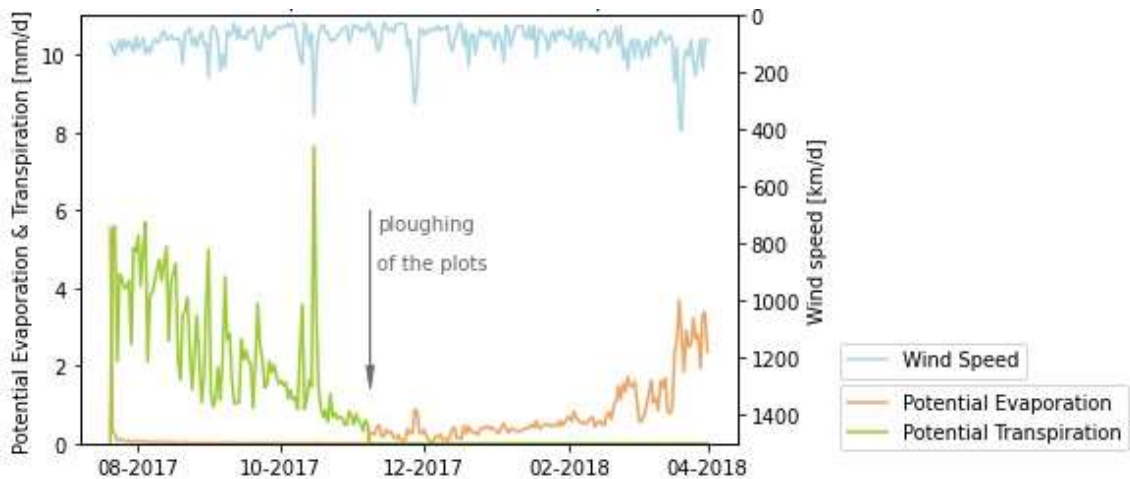


Fig. 33: Potential evaporation and potential transpiration calculated by HYDRUS-1D for plot 5.

3.4.2 Comparison of measured and simulated values of all plots

The results of the simulated water content and NO_3^- -N concentration simulations are shown in the calibration and validation results (chapter 3.2 and 3.3). For better comparability of the differences between the plots and especially the soil depths, the water content and NO_3^- -N concentration curves of all plots are plotted in one graph, respectively (Fig. 34, Fig. 35). Both graphs display the variation between the plots in the first half of the simulation, which can be assigned to the different winter cover crops. When the winter cover crops are removed and thereby their specific influence (e.g. root water and root solute uptake, and transpiration) on the simulated water contents and NO_3^- -N concentrations, the curves equalize. The only recognizable difference in the second half of the simulation is the difference between the three soil depth layers. The water contents and mostly also NO_3^- -N concentrations decrease with increasing soil depth.

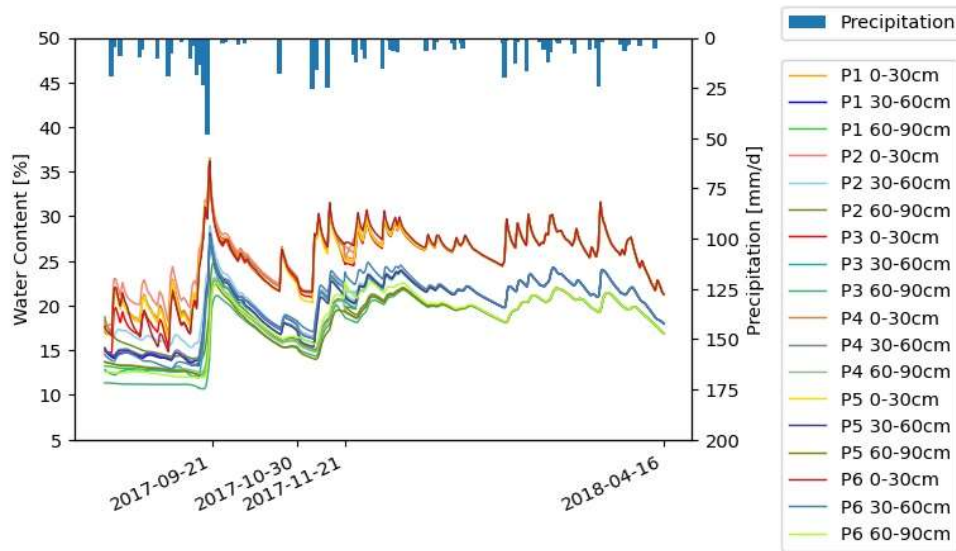


Fig. 34: Simulated water content curves for the time period from 3rd of August 2017 to 16th of April 2018 of all plots.

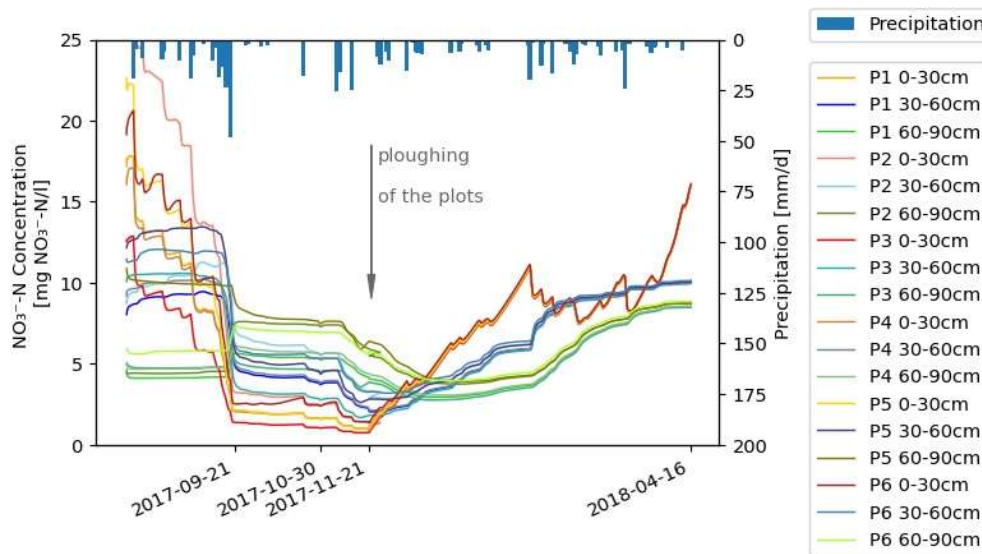


Fig. 35: Simulated NO_3^- -N concentration curves for the time period from 3rd of August 2017 to 16th of April 2018 of all plots.

The comparison of measured to simulated water contents of the three calibrated plots are displayed in Fig. 36, and the measured versus simulated NO_3^- -N concentrations in Fig. 37. The data points of the water contents aggregate around the identity line and roughly separate into three layers according to the three soil depth layers, with the measured and simulated water contents decreasing from shallow to deeper soil layers. Concerning the NO_3^- -N concentrations, the HYDRUS-model simulates mostly higher NO_3^- -N values in the soil layer from 60–90cm than the actual measured values, has the best fit for the NO_3^- -N concentrations in 30–60cm and somewhat underestimates the measured NO_3^- -N values in 0–30cm. The calibration process led to a minimisation of the absolute errors between the measured and simulated water contents [%] to values in the range of 0.08 to 6.43% with a mean value of 2.16%. The minimal absolute error of the measured to the simulated NO_3^- -N concentrations is 0.066mg l^{-1} , the maximal absolute error is 7.642mg l^{-1} and the mean of the absolute errors is 2.404mg l^{-1} . The coefficient of determination R^2 for the simulated and the measured water contents is 0.659 and the root-mean-square error (RMSE) is 2.678. The deviations of the simulated NO_3^- -N concentrations from the measured values are also rather low with a R^2 of 0.624 and a RMSE of 2.869.

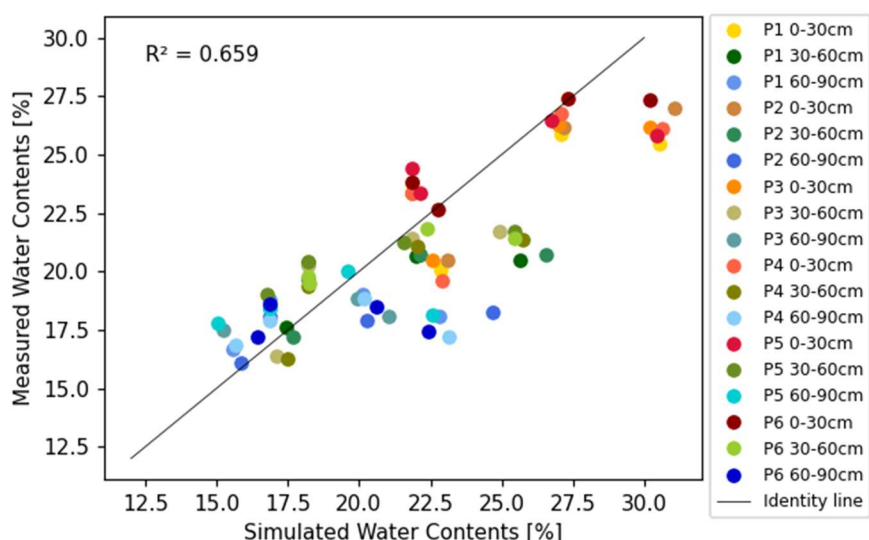


Fig. 36: Observed water contents (21st September, 30th October, 21st November, 16th April) plotted against the simulated water contents at that exact dates; with the colours dividing into different plots and plot depths, and an R^2 value of 0.659.

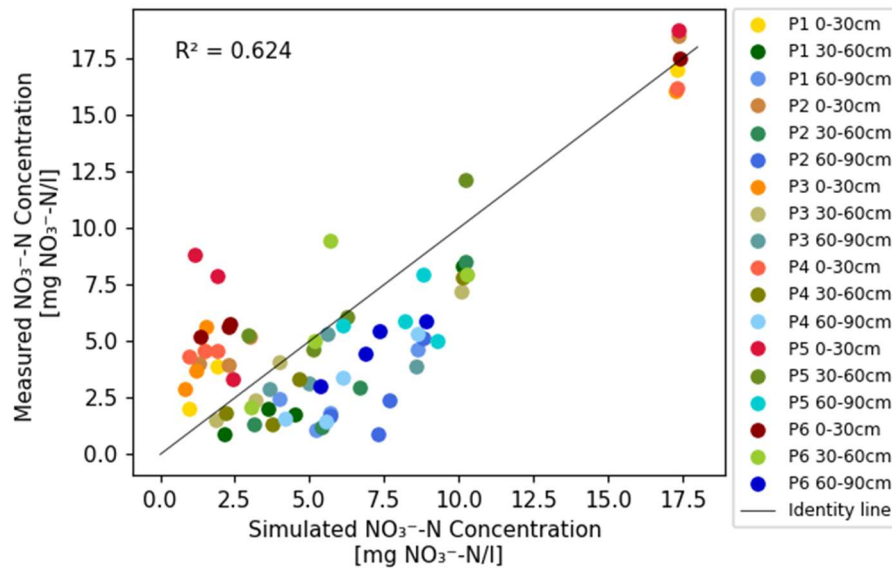


Fig. 37: Observed $\text{NO}_3\text{-N}$ concentrations (21st September, 30th October, 21st November, 16th April) plotted against the simulated $\text{NO}_3\text{-N}$ concentrations at that exact dates; with the colours dividing into different plots and plot depths, and an R^2 value of 0.624.

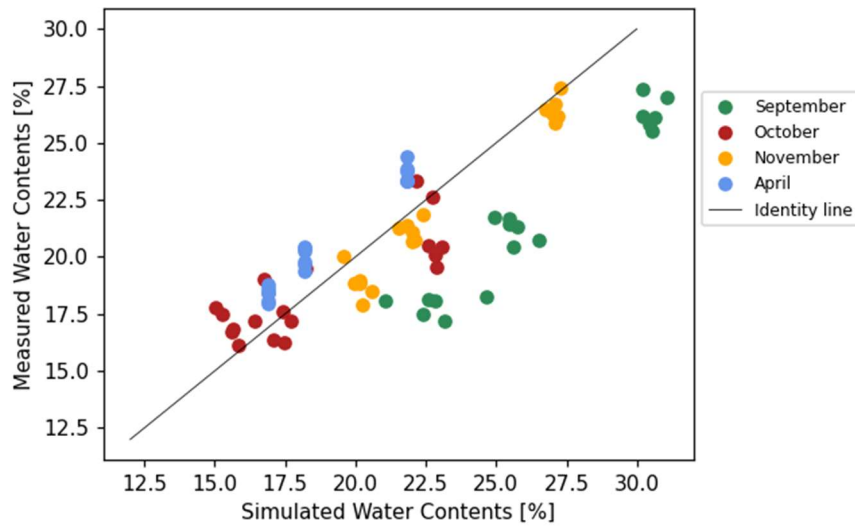


Fig. 38: Observed water contents (21st September, 30th October, 21st November, 16th April) plotted against the simulated water contents at that exact dates, with the colours dividing into the months when the measurements were conducted.

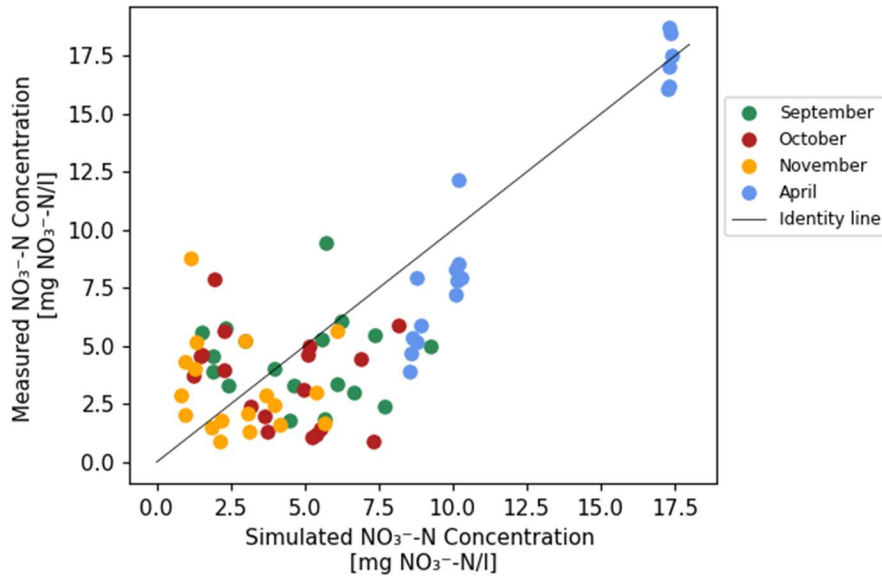


Fig. 39: Observed NO₃⁻-N concentrations (21st September, 30th October, 21st November, 16th April) plotted against the simulated NO₃⁻-N concentrations at that exact dates, with the colours dividing into the months when the measurements were conducted.

If the confronted measured and simulated water contents of all plots are divided by colours into the different months when the soil samples were taken, it can be observed that the simulated water contents are higher than the measured water contents in September 2017 (Fig. 38). The six NO₃⁻-N concentration points that are much higher than the other NO₃⁻-N concentrations are all measured/simulated in April 2018 and in the depth from 0–30 cm (Fig. 37, Fig. 39).

3.4.3 Actual root NO₃⁻-N uptake

The actual root NO₃⁻-N uptake of the winter cover crops of all plots was simulated by the HYDRUS-model and is displayed with the temperature curve (Fig. 40). In the beginning of the simulation in August 2017, the root NO₃⁻-N uptake reaches its highest amounts in plot 2 and 5, with a maximum value of $\sim 0.02 \text{ mg NO}_3\text{-N cm}^{-2} \text{ d}^{-1}$ on the 5th of August 2017 in plot 2. As the temperature declines towards the winter months, also the actual root NO₃⁻-N uptake decreases in all plots. After the ploughing of the plots on the 21st of November 2017, the simulated actual root solute uptake totally ceases due to the absence of winter cover crops.

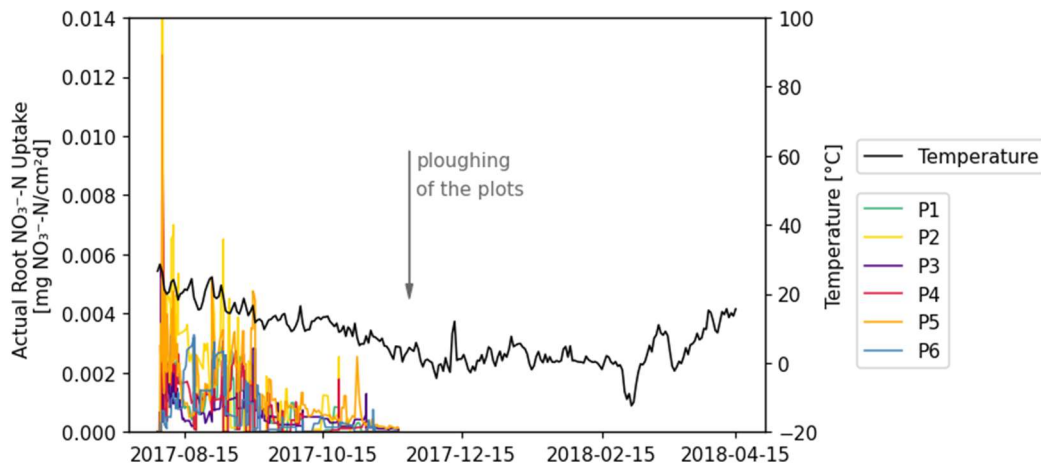


Fig. 40: Actual root NO_3^- -N uptake of the winter cover crops in all plots, with the temperature curve from the ZAMG measurement station.

3.4.4 Simulation of the heavy precipitation event

On the evening of the 16th of April 2018, a heavy precipitation event occurred in the region of the field site, and led to an inundation of the plots (Fig. 41). There exist different criteria that may define a precipitation event as a heavy precipitation event. Rimkus et al. (2011) for example, defined several indicators that describe heavy precipitation in the study area in Lithuania: e.g. the daily precipitation totally exceeding 10 mm and 20 mm and the 3-day precipitation totally exceeding 20 mm and 30 mm. The meteorological measurement station in Wagna reported nearly 88 mm in 10 hours. The course of the NO_3^- -N concentrations during the heavy precipitation event was simulated with the calibrated and validated HYDRUS-model as an additional research question.

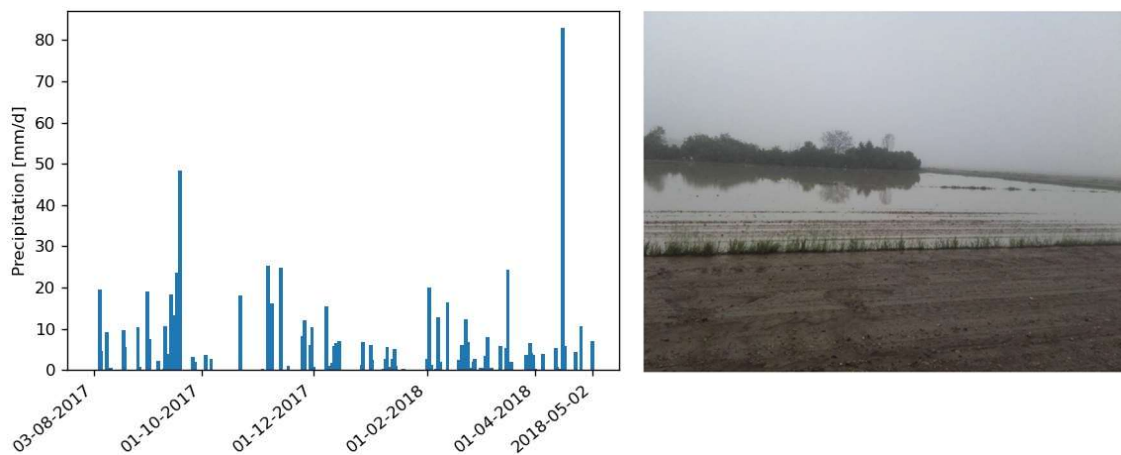


Fig. 41: Left: Precipitation curve from the ZAMG measurement station in Wagna, with the heavy precipitation event on the 16th of April 2018; Right: Inundated field site on the 17th of April 2018 (©Philipp Holler).

The water and NO_3^- -N content curves for the whole period covered by the soil sampling from 3rd of August 2017 to the 2nd of May 2018 of plot 5 are representatively displayed in Fig. 42 and Fig. 43. The simulation by the HYDRUS-model also includes the heavy precipitation event that has taken place on the 16th of April 2018 and is reflected in a strong increase of the water content, especially in the depth from 0–30 cm. During the heavy rainfall, the simulated NO_3^- -N concentration in the shallowest soil layer shows a recognizable drop in its course, whereas the NO_3^- -N concentrations in the soil layers beneath 30 cm mildly increase. On the 2nd of May 2018, the simulated NO_3^- -N concentrations in the three depth layers are higher than the measured values, especially in the depth from 60–90 cm there is a higher error.

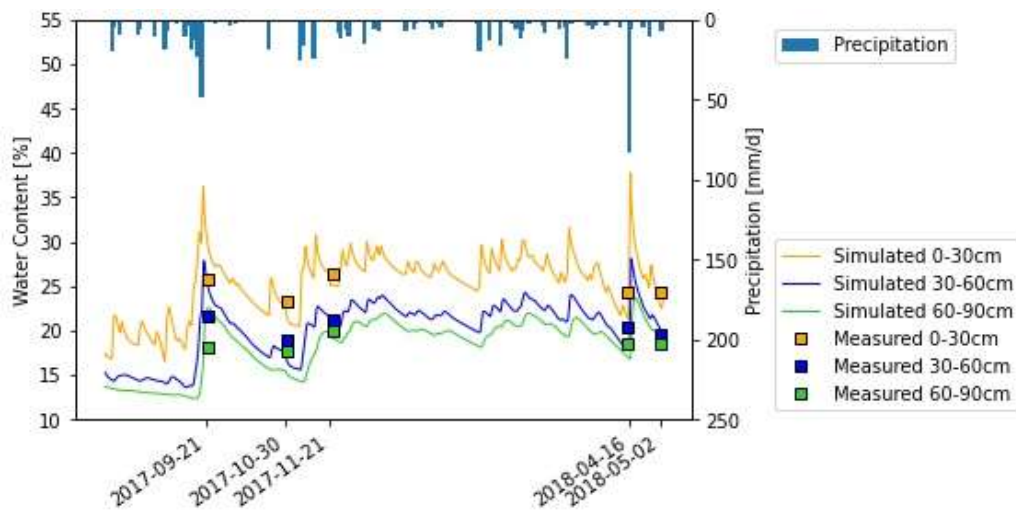


Fig. 42: Simulated water content curves for the time period from 3rd of August 2017 to 2nd of May 2018 of plot 5 in comparison to the punctual measured water contents in plot 5; with the heavy precipitation event on the 16th of April 2017 after the fourth soil sampling.

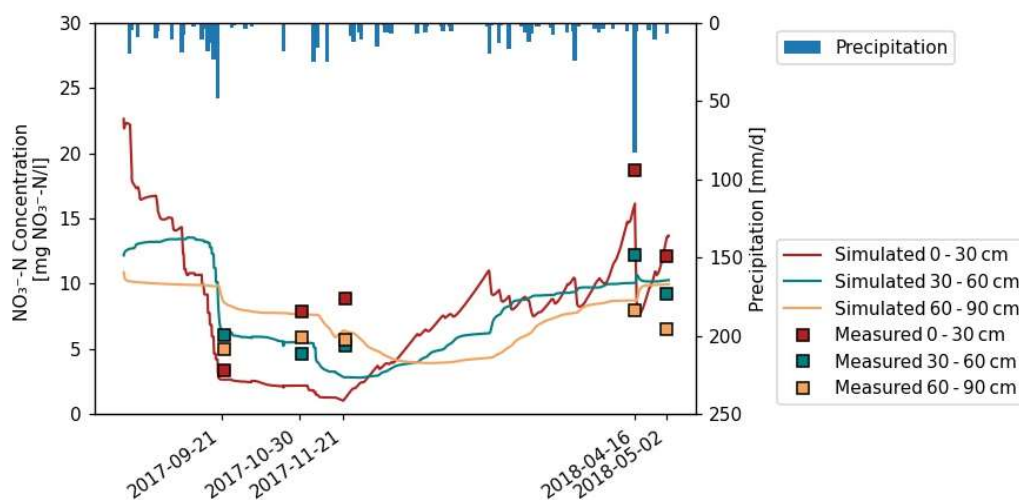


Fig. 43: Simulated NO_3^- -N concentration curves for the time period from 3rd of August 2017 to 2nd of May 2018 of plot 5 in comparison to the punctual measured NO_3^- -N concentrations in plot 5; with the heavy precipitation event on the 16th of April 2017 after the fourth soil sampling.

The comparison of all water content and NO_3^- -N concentration measurements to the simulated values from the 2nd of May 2018, respectively is shown in Fig. 44 and Fig. 45. Whereas the water contents assemble quite nicely around the identity line, the simulated versus the measured NO_3^- -N concentrations lie all on one side of the identity line, revealing that the model simulates higher NO_3^- -N concentrations in all plots and all depths on the 2nd of May 2018 than were measured on this date.

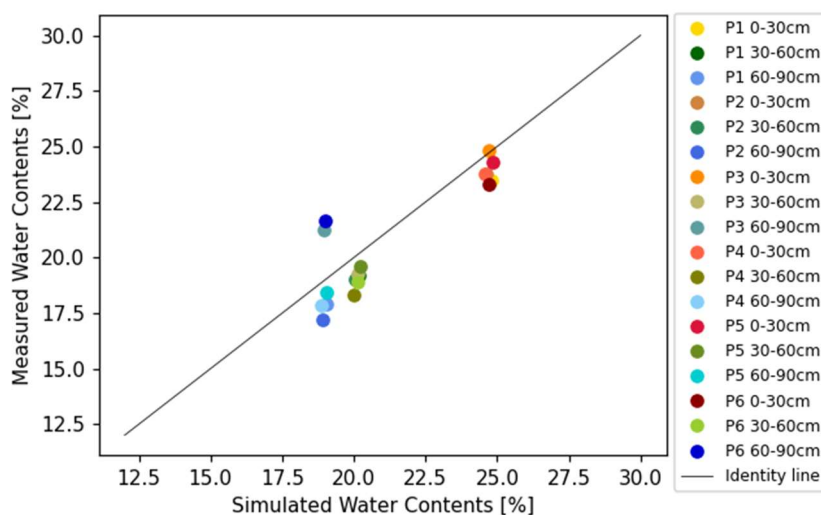


Fig. 44: Observed water contents (2nd May 2018) plotted against the simulated water contents at that exact date.

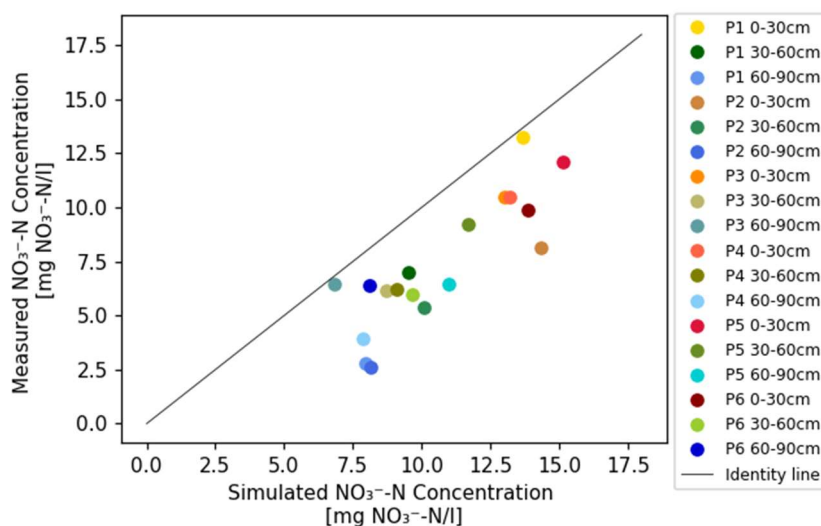


Fig. 45: Observed NO_3^- -N concentrations (2nd May 2018) plotted against the simulated NO_3^- -N concentrations at that exact date.

3.4.5 Bottom nitrate-nitrogen fluxes and leaching amounts

The simulated NO_3^- -N flux below the bottom of the profile (below 1 m) was simulated with the HYDRUS-model and the cumulative curve was calculated from the HYDRUS output files and plotted with Python (Fig. 46 and Fig. 47). The NO_3^- -N fluxes varied along with the water flux, in a relatively wide range from $\sim 0.08\text{--}0.27\text{ kg ha}^{-1}\text{ d}^{-1}$. The flux at 1 m depth is very low until around the 19th of September, when the accumulation of precipitation events leads to a higher amount of infiltrating water that carries the NO_3^- -N to deeper soil layers. A temporary strong increase in the bottom NO_3^- -N flux occurs in all plots, but to different extents. The highest simulated flux is in plot 2, followed by plot 5, 6, 4 and 1, and the lowest flux in plot 3. Towards the end of November 2017, there is another precipitation event that affects the NO_3^- -N flux beneath 1 m, but this time with the highest flux in plot 6. From the end of December 2017 until the 2nd of May 2018, the NO_3^- -N flux of the different plots align more and more with every day and eventually show the same course and amounts.

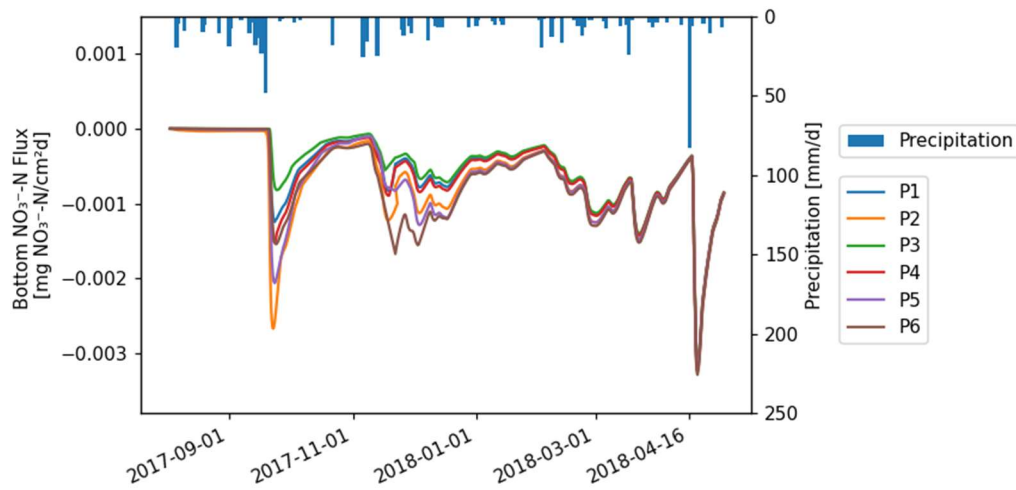


Fig. 46: Bottom NO_3^- -N flux [mg NO_3^- -N $\text{cm}^{-2}\text{ d}^{-1}$] in all plots calculated by HYDRUS-1D for the time period from 3rd of August 2017 to 2nd of May 2018.

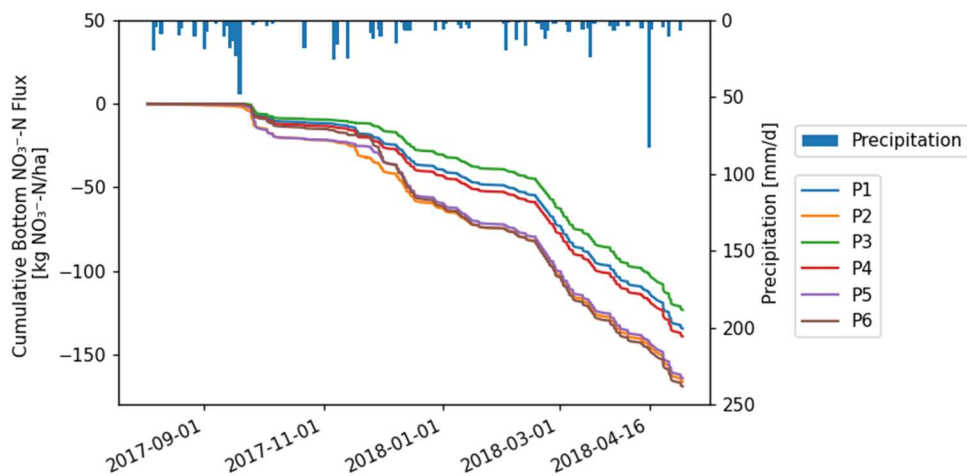


Fig. 47: Cumulative bottom NO_3^- -N flux [kg NO_3^- -N ha^{-1}] in all plots calculated by HYDRUS-1D for the time period from 3rd of August 2017 to 2nd of May 2018.

The NO_3^- -N leaching amount below 1 m was calculated by multiplying the NO_3^- -N concentration in the soil solution at the bottom of the soil profile with the seepage water of the calculated time interval (chapter 0). During the heavy precipitation event on the 16th of April 2018, an average amount of $0.787 \text{ kg NO}_3^- \text{ N ha}^{-1} 10 \text{ h}^{-1}$ was displaced into deeper soil layers according to the HYDRUS-1D simulation. The detailed amounts for every plot are listed in Table 4, as well as an estimation for the period from the ploughing in November 2017 until the beginning of May 2018, including the heavy precipitation event on the 16th of April 2018.

Table 4: Leached NO_3^- -N amounts [kg/ha] below 1 m, calculated exemplarily for two time periods.

	NO_3^- -N leaching amount [kg NO_3^- -N/ha]	
	16/04/18, 7pm - 17/04/18, 5am	22/11/17 - 02/05/18
P1	-0.77	-114.37
P2	-0.80	-145.80
P3	-0.77	-103.32
P4	-0.78	-118.99
P5	-0.80	-143.69
P6	-0.82	-148.14

4. Discussion

4.1 Sensitivity analysis

The sensitivity analysis confirmed that the saturated hydraulic conductivity K_s not only strongly influences the water content curve (Fig. 11), but also affects the course of the NO_3^- -N concentrations (Fig. 14). In general, parameters that affect the water content also influence the NO_3^- -N concentration, as the NO_3^- -N is dissolved in the water and therefore depends on the water content. Vice versa, parameters that may change the course of the NO_3^- -N concentrations do not necessarily affect the water content. The alteration of K_s leads to a similar response in the water content errors in the two deeper soil layers, whereas the soil layer from 0–30 cm behaves differently when the HYDRUS-model is run with lower K_s values. The differences between the results of model runs with varying K_s values are higher for the shallowest soil layer than for the others. Thus, the water content near the soil surface in plot 6 is more affected by the variation of the K_s than deeper in the soil. As the air-entry value α and the pore size distribution index n are empirical coefficients that affect the shape of the hydraulic functions and are part of the van Genuchten function, they certainly have an influence on the behaviour of the water displacement in the soil, although it is not as large as the influence of the K_s . The absolute errors between the measured and the simulated water contents decline with decreasing n values (Fig. 12), and show that behaviour at every measurement date and in all three depth layers. That means that a variation of the n affects the simulated water contents, but it is the same influence in every soil depth and month. For the sensitivity analysis the n values were varied over the spectrum that HYDRUS-1D suggests for all soil types. However, for the calibration the range of possible n values is delimited by the representativeness of these values for the considered soil type at the study site. Varying the α parameter does not affect the absolute errors of the water contents much, but it shows that the errors decrease with very low values (Fig. 13). Furthermore, the water content in the deeper soil layer from 60–90 cm behaves oppositely to the upper soil layers in September and November 2017 with a variation of α . It is therefore likely that the soil layers nearer to the surface have a slightly different soil water retention behaviour than deeper in the soil. The air-entry value α marks the transition between saturated and unsaturated soil mechanics (Soltani et al. 2021), and takes on different values for different soil properties as it is here the case for the different soil depths.

Beside the strong influence of the K_s , the NO_3^- -N concentrations were also affected by the longitudinal dispersivity D_L and the zero-order rate constant for liquid phase γ_w . The latter can strongly change the amount of NO_3^- -N in the soil water as it represents an input source for NO_3^- -N in the soil (Fig. 16). The dispersivity appears to have a higher influence in the deeper soil layers. The absolute errors of the NO_3^- -N concentrations decrease with higher dispersivity (Fig. 15), which is also why the values have finally been chosen slightly higher than the values from literature with comparable model set ups (Li et al. 2015; Hanson, Šimůnek, and Hopmans 2006; Sansoulet 2007).

4.2 Calibration results

Plot 2, 5 and 6 that were used for the calibration behave similarly concerning the simulated water and NO_3^- -N content curves, although plot 6 has different preconditions as it is not covered by winter crops. However, the weed and the grass that was growing wildly on the reference plot appears to have a similar influence or rather the winter cover crops do not have a significant influence on the simulation of the water and NO_3^- -N content curves by the HYDRUS-model. The simulated NO_3^- -N concentrations fit quite well at the dates of the measurements except for the simulated values in April. From August to November 2017 a smaller value for the zero-order rate constant γ_w was adopted, describing a rather low input of NO_3^- -N into the soil. However, on the 21st of November 2017 the fields were ploughed what led to an abrupt rise in the organic carbon and therefore nitrogen content that is then gradually converted to plant available forms of mineralized nitrogen in the soil. Increasing the γ_w reproduces that rise of NO_3^- -N over time. The NO_3^- -N concentration curves with the higher γ_w for the time period after the ploughing (Fig. 23 to Fig. 25) show an improved fit to the measured NO_3^- -N concentrations in April 2018. Nonetheless, the simulated values of April cannot accurately describe the variation of observed NO_3^- -N concentrations between the different soil depths as the upper soil layer (0–30 cm) has quite high NO_3^- -N concentrations in all plots compared to the deeper soil layers.

First, the calibration was done for only one plot (P5) and was then improved with the other two plots (P2 and P6). However, the other two plots did not present many challenges as they showed a quite similar behaviour and particularly similar water and NO_3^- -N content observations. The measured water contents and NO_3^- -N concentrations of all six plots are quite equally distributed around their mean values, with a maximum standard deviation of 2.98 %, 2.06 % and 4.84 % for the water contents of the three depth layers 0–30, 30–60, and 60–90 cm, respectively and 12.58 mg l⁻¹, 3.18 mg l⁻¹ and 3.76 mg l⁻¹ for the NO_3^- -N concentrations, respectively.

4.3 Calibration results combined with validation results

Fig. 36 and Fig. 37 display a comparison of each measured value with the simulated values. The water contents roughly group into three layers corresponding to the soil depth layers. Accordingly, the (simulated and measured) water contents tend to be higher in the upper soil layer and smaller in the deeper soil layer.

The best fit for the NO_3^- -N concentrations is achieved in the soil layer from 30 to 60 cm. The HYDRUS-model slightly overestimates the concentration in deeper soil layers and conversely underestimates the concentration in the soil layer nearer to the surface to a small amount (Fig. 37). April reveals quite high measured NO_3^- -N concentrations in the upper soil layer in all plots compared to the values in the depth of 30–90 cm, and especially compared to the values from the other months. This can be observed more easily in Fig. 39 where the concentration values are differentiated into

months by the different colours. The rise in the measured NO_3^- -N concentrations from November 2017 to April 2018 can be explained by the absence of cover crops on the plots over the winter months that would otherwise take out NO_3^- -N of the soil. Aside from that, the soil microorganisms continue with the conversion of organic matter into mineralized nitrogen that is not absorbed by plants anymore. Furthermore, the evaporation is much lower from November to April compared to the other months and the transpiration completely ceases with the removal of the winter cover crops, which leads to the NO_3^- -N remaining in the soil rather than partially evaporating with the water. Separating the water contents by the months of their measurements (Fig. 38) also reveals that the HYDRUS-model underestimates the September values in all plots slightly more than in the other months.

In the soil depth from 0–90 cm, the measured NO_3^- -N concentrations from 3rd of August 2017 to 16th of April 2018 varied in a range of $\sim 0.9 - 35.7 \text{ mgNO}_3^- \text{N l}^{-1}$, whereas the simulated NO_3^- -N concentrations varied between $\sim 0.7 - 33.1 \text{ mgNO}_3^- \text{N l}^{-1}$. The simulated maximum and minimum NO_3^- -N concentrations of each plot are all in the soil depth layer from 0–30 cm, indicating that the biggest variation in the concentrations is nearer to the soil surface. The simulated leachate NO_3^- -N concentration at the bottom of the profile in the HYDRUS-model varies between 1 - $10 \text{ mgNO}_3^- \text{N l}^{-1}$, thus being in the lower range of the values measured by Kuo, Huang, and Bembenek (2001) ($7.9 - 30.5 \text{ mg l}^{-1}$) and Gabriel, Muñoz-Carpena, and Quemada (2012) ($44 - 57 \text{ mg l}^{-1}$). These studies were conducted with partially different cover crops, setups and in areas with higher average annual temperature and lower average annual rainfall, what may describe deviations. On the contrary, the estimated total NO_3^- -N leached from November 2017 to May 2018 (Table 4) is rather high compared to the calculated amounts by Gabriel, Muñoz-Carpena, and Quemada (2012) and White et al. (2017), which may be due to higher rainfall amounts and increased seepage water at the field site in southern Styria. Disregarding the heavy precipitation event on the 16th of April 2018, the highest NO_3^- -N flux at the bottom of the profile (below 1 m) was simulated in September 2017, following a precipitation event ($\sim 48 \text{ mm d}^{-1}$).

Comparing the simulated NO_3^- -N concentrations of the plots amongst each other (Fig. 35 Fig. 34), the concentration curves seem to evolve quite similarly over time compared to much higher variance over time of the measured NO_3^- -N concentrations. This indicates that the prediction of the NO_3^- -N concentrations over time by the HYDRUS-model seems to be rather simplified and does not show much difference for diverse winter cover crops. Especially when the winter cover crops are removed on the 21st of November 2017, the water and NO_3^- -N content curves of the three soil depth layers begin to equilibrate, respectively. The HYDRUS-models of the six plots were set up with the same water retention parameters and soil and solute specific parameters and because of the limited spatial extent of the field site also the same time variable and time constant meteorological parameters were used. Therefore, besides the respective winter cover crops only the initial water contents and NO_3^- -N concentrations differed between the plots. As long as there are winter cover crops on the field, the

NO₃⁻-N concentration curves of the plots show some variation due to different root water uptake rates of the varying winter cover crops.

Using a model for the simulation of the complex soil processes leads to inevitable limitations and simplifications. Numerous processes taking place simultaneously, such as preferential flow and chemical reactions, and inhomogeneities in the soil can only be reproduced with a tremendous effort and the availability of numerous measured or otherwise determined soil specific parameters. Therefore, the simplifications had to be done during setting up the HYDRUS-model and are reflected in the deviations. Nonetheless, the simulated values matched the measured sufficiently good.

4.4 Evapotranspiration

The calculated actual evapotranspiration (Fig. 32) has a typical seasonal behaviour with a decline in the winter months due to the reduced solar radiation and a rise again in spring (March to April 2018). The major peaks coincide with the higher peaks in the wind speed curve. A comparison with calculated potential evapotranspiration values from measured meteorological data in Wagna by the *Amt der Steiermärkischen Landesregierung* over a time period from 2008 to 2014 reveals that the evapotranspiration generated by the HYDRUS-model is in general slightly higher (Table 5). This may be partially due to the fact that the evapotranspiration from the *Wasserversorgungsplan WVPL* (Amt der Steiermärkischen Landesregierung 2015) is calculated for a reference crop, which differs from the actual winter cover crops at the field site. Additionally, there are different observation periods covered. In comparison to mean evapotranspiration values for most cultivated plants in Central Europe of 2–3 mm d⁻¹, during main growing season of 3.5–4.5 mm d⁻¹ and on clear summer days of ≥ 8 mm d⁻¹, as suggested by Amelung et al. (2018), the simulated values stay within that range.

Table 5: Comparison of the monthly ETp sums of the HYDRUS-model and the *Wasserversorgungsplan Steiermark* (Amt der Steiermärkischen Landesregierung 2015)

Monthly ETp sums calculated by HYDRUS-1D		Mean monthly ETp sums over the years 2008 - 2014 (<i>WVPL</i>)	
August 2017	126.6	August	90.4
September 2017	67.7	September	52
October 2017	61.5	October	26.9
November 2017	16.6	November	11
December 2017	7.0	December	6.2
January 2018	11.2	January	8.6
February 2018	15.1	February	16.8
March 2018	33.9	March	40.6
April 2018	40.9	April	64.3

HYDRUS-1D divides the potential evapotranspiration into potential evaporation and potential transpiration, which is why the potential evaporation abruptly decreases when the winter cover crops start to grow, and the potential transpiration rapidly rises (Fig. 33). After the fields were ploughed, the potential transpiration goes to zero and the potential evaporation increases again.

4.5 Simulation of the heavy precipitation event

The precipitation curve from the 3rd of August 2017 to the 2nd of May 2018 (Fig. 41) displays the relatively large precipitation amount on the evening of the 16th of April 2018. Following the heavy precipitation event, the soil sampling on the 2nd of May 2018 revealed very low NO₃⁻-N concentrations, especially in comparison to the much higher values in the depth from 0–30 cm half a month earlier, on the 16th of April 2018 (Fig. 43). The absolute errors between the simulated and the measured NO₃⁻-N concentrations on the 2nd of May 2018 are within a relatively low range of 0.39 – 6.21 mgNO₃⁻-N l⁻¹, but all simulated concentrations of the six plots and three depth layers are higher than the observed NO₃⁻-N concentrations on the 2nd of May 2018 (Fig. 45). It can therefore be concluded that the HYDRUS-model that was calibrated with average precipitation events underestimates the NO₃⁻-N displacement capacity of such heavy precipitation.

4.6 HYDRUS-model uncertainties

The HYDRUS-model results inherited uncertainties by several sources, such as field measurements, model inputs, modelling approach and model structure. The HYDRUS-model setup was limited to basic models (e.g. soil hydraulic property model) in order to reduce the required parameters to run the model. Because of the limited availability of measured soil hydraulic, plant physiological and meteorological parameters, some parameters (e.g. bulk density, sand, silt and clay percentages, dispersivity, diffusion coefficient, root water uptake parameters and meteorological parameters) had to be estimated with available data from literature, nearby measurements and through the estimation by Rosetta and other included parameter databases in HYDRUS. Furthermore, the meteorological measurements and parameters were observed in Wagna, which is in approximately 4 km air-line distance from the field site, and thus were not fully representative. Especially the soil hydraulic parameters that were based on measurements in Wagna and large-scale information from the Digital Soil Map from Austria (eBOD) most likely caused deviations of the predicted values to the observed ones. Moreover, the concentration of the incoming water is defined through measured wet deposition values from Arnfels, located about 20 km linear distance away. Additionally, for each month only one wet deposition concentration was defined that stayed the same throughout the whole month, respectively. The estimation of the total net nitrogen mineralization during residue decomposition in the field is still a challenge (O'Connell et al. 2015). Not all processes along the nitrogen chain could be considered in this study, resulting in some workarounds where e.g. the conversion from organic matter into mineralized nitrogen skipped the ammonium and directly resulted in an increase of nitrate. The input of NO₃⁻-N coming from organic residues was reproduced with a constant parameter, the zero-order rate constant. This simplification of the HYDRUS-model most certainly led to a less precise description of the real state.

5. Conclusion

HYDRUS-1D was used for setting up a model to simulate the water contents, NO_3^- -N concentrations and NO_3^- -N leaching in the soil at a field site in southern Styria, thus testing the application of a model to such a field site and complement the existing field studies with water flow and solute transport simulations in the vadose zone. The HYDRUS-model was calibrated and validated with previously measured soil water contents and NO_3^- -N concentrations. HYDRUS-1D allowed for most parts of the water flow, solute transport, and root water uptake calculation a simplification of the process resulting in a setup with basic assumptions and models and few required parameters that nonetheless led to tolerably good results. Although the results of the established HYDRUS-model are site specific, the findings are relevant to advance the understanding of NO_3^- -N dynamics in the soil of agriculturally cultivated fields with winter cover crops and the process of applying a model to such a field site.

The use of the van Genuchten – Mualem model as soil hydraulic model and the equilibrium model for the solute transport enable the user to simulate the water and solute transport in the soil without a massive need for soil and solute specific parameters that are difficult to measure and is seldomly done. The neural network prediction, where water retention parameters and the saturated hydraulic conductivity are predicted from the soil textural distribution and the bulk density, was used during setting up the HYDRUS-model. Prior to the calibration and validation of the HYDRUS-model with the measured water contents and NO_3^- -N concentrations, a sensitivity analysis was conducted to identify the parameters to which the HYDRUS-model was sensitive to. The saturated hydraulic conductivity turned out to be a very sensitive parameter, which led to the conclusion that precise measurements of this parameter should have a high priority in similar future modelling approaches. The simulations revealed that especially the soil layer next to the soil surface is strongly influenced by parameter variations and external impacts, such as meteorological fluctuations. The maximum and minimum NO_3^- -N concentrations are all simulated and mostly also measured in the soil depth layer from 0–30 cm, leading to the conclusion that there is more variation of the NO_3^- -N concentrations nearer to the soil surface.

Without defining an input in the HYDRUS-model the NO_3^- -N concentration in the soil declined due to the displacement in the soil with seepage water and was much lower than the measured values. By assuming an external source of NO_3^- -N with the zero-order rate constant being higher than zero, the actual NO_3^- -N input coming from mineralization could be better reproduced and the deviations between the simulated and measured NO_3^- -N concentrations in April 2018 very much decreased. The simulated water contents were in really good accordance with the measured water contents, with a maximum absolute error of 6.43 %, a R^2 of 0.659 and a RMSE of 2.678. The errors between the simulated and measured NO_3^- -N concentrations are within an acceptable range, with a maximum absolute error of 7.642 mg l⁻¹, a R^2 of 0.624 and a RMSE of 2.869.

An objective of the study was to find out whether the simulation of a period covering a heavy precipitation event with the established HYDRUS-model delivers water contents and NO_3^- -N concentrations that are within an acceptable range of the measured contents. The errors between the simulated and measured NO_3^- -N concentrations are low, but the simulation revealed that the HYDRUS-model underestimates the NO_3^- -N displacement during the precipitation event that exceeds the boundaries of the setup chosen for the calibration and validation. It could certainly be improved by integrating more diverse precipitation events, but that would also require a wide range of observed data, as heavy precipitation events are scarce. However, the simulation revealed that especially during precipitation events with higher precipitation intensities (e.g. 19th September 2017 and 16th April 2018), increased water fluxes led to a notable NO_3^- -N displacement into deeper soil layers, that often showed an increase first, followed by a gradual decrease and NO_3^- -N leaching towards groundwater.

Future improvements would mainly cover the acquisition of input parameters of the HYDRUS-model, besides the modelling process itself. The simplifications due to the lack of measured soil specific parameters, such as bulk density and sand, silt and clay percentages at the study site led to a less complex model which allows simulating for the short time period but results also in a less accurate description of the actual state. There should be a special emphasis on the precise measuring of the saturated hydraulic conductivity, and soil property parameters determining the water retention parameters. For a more stable prediction of water and especially NO_3^- -N contents at a field site, there should be more observations available for the calibration and validation. As the standard deviation between all measured water contents and all NO_3^- -N concentrations were quite low, which indicates that the plots behave quite similarly, it would also be recommended to apply the model for more diverse plots and an expanded spatial extent. The next step could be testing the same procedure also for other sites with differing soil types.

The modelling environment HYDRUS is able to reproduce the processes surrounding plants, such as the root water uptake, solute uptake and transpiration. However, it is actually more of an approximation to the actual processes, as HYDRUS is rather designed for displaying the processes in the soil, such as water, solute and heat transport and the behaviour of numerous solutes in the soil. It would be interesting to test how the model behaves in comparison to a more plant processes-based programme (e.g. SWAT). The three different winter cover crop compositions that were cultivated on plot 1–plot 5 were specified with root water uptake parameters that were suggested by an integrated database of HYDRUS, which could be improved by actually measuring the required pressure heads that describe the root water uptake. Aside from that, the active root solute uptake by the different winter cover crops that was neglected in this study, could be specified and a more detailed NO_3^- -N uptake could be determined. The option to allow multiple seasons for the root growth is not yet integrated in HYDRUS, so the root growth function can only be used once. If root growth is simulated

for multiple years, there must be constant root depth and the surface fluxes must be manipulated by dividing evapotranspiration into potential evaporation and transpiration. Therefore, no second root growth phase in spring 2018 was established. However, as there was no seeding of crops after the ploughing, it can be assumed that it has not influenced the outcomes much.

Furthermore, the unsaturated zone may be quite diverse with different drainage paths according to the prevailing pore system (Reszler and Fank 2016). It could be attempted to describe this complexity better with a dual system consisting of a delayed matrix flow and a rapidly draining flow through preferential flow paths as proposed by Gerke and Van Genuchten (1993). Additionally, if the model can be assumed to be generating fairly good predictions, it could be used for calculating the present amount of nitrogen in the soil root zone and the required amounts of nitrogen fertilizer at all times of a cropping period. For the conducted research in the thesis, the basic model setup was sufficient and produced some interesting insights in the soil physics of an agricultural field.

BIBLIOGRAPHY

- Amelung, Wulf, Hans-Peter Blume, Heiner Fleige, Rainer Horn, Ellen Kandeler, Ingrid Kögel-Knabner, Ruben Kretzschmar, Karl Stahr, and Berndt-Michael Wilke. 2018. *Scheffer/Schachtschabel Lehrbuch Der Bodenkunde*. Springer-Verlag.
- Amt der Steiermärkischen Landesregierung. 2015. *Wasserversorgungsplan Steiermark*. Abteilung 14 – Wasserwirtschaft, Ressourcen und Nachhaltigkeit.
- . 2018. “Nasse Deposition Im Land Steiermark Oktober 16 - September 17.”
- . 2019. “Nasse Deposition Im Land Steiermark Oktober 2017 - September 2018.”
- Baram, S., V. Couvreur, T. Harter, M. Read, P.H. Brown, M. Kandelous, D.R. Smart, and J.W. Hopmans. 2016. “Estimating Nitrate Leaching to Groundwater from Orchards: Comparing Crop Nitrogen Excess, Deep Vadose Zone Data-Driven Estimates, and HYDRUS Modeling.” *Vadose Zone Journal* 15 (11): 1–13. <https://doi.org/10.2136/vzj2016.07.0061>.
- Baumhackl, Thomas. 2013. *Regionalentwicklung Im Bezirk Leibnitz*.
- BFW. 2004. “Digitale Bodenkarte von Österreich, 1km-Raster.” Bundesforschungs- Und Ausbildungszentrum Für Wald, Naturgefahren Und Landschaft (BFW). 2004.
- BMLFUW. n.d. “EHYD - Hydrographische Archivdaten Österreichs.” BMLFUW/Abteilung IV/4 - Wasserhaushalt.
- . 2017. “Aktionsprogramm Nitrat BGBl. II Nr. 385/2017.” *BGBl. II Nr. 385/2017*, no. 391: 1–33.
- Bodenschutzprogrammverordnung. 1987. *Bodenschutzprogrammverordnung*.
- Carsel, Robert F, and Rudolph S Parrish. 1988. “Developing Joint Probability Distributions of Soil Water Retention Characteristics.” *Water Resources Research* 24 (5): 755–69.
- Chen, Ning, Xianye Li, Jiri Šimůnek, Haibin Shi, Qi Hu, and Yuehong Zhang. 2020. “Evaluating Soil Nitrate Dynamics in an Intercropping Dripped Ecosystem Using HYDRUS-2D.” *Science of the Total Environment* 718. <https://doi.org/10.1016/j.scitotenv.2020.137314>.
- European Commission. 1991. *Directive 91/676/EEC. Council Directive of 12 December 1991 Concerning the Protection of Waters against Pollution Caused by Nitrates from Agricultural Sources. Official Journal of European Community L*. Vol. 375.
- Feddes, R A, P J Kowalik, and H Zaradny. 1978. “Simulation of Field Water Use and Crop Yield.” *New York, NY John Wiley*.
- Food and Agriculture Organization of the United Nations. 1990. “Expert Consultation on Revision of FAO Methodologies for Crop Water Requirements, ANNEX V, FAO Penman-Monteith Formula.” In *Rome (Italy). 28-31 May 1990*.
- Fowler, David, Mhairi Coyle, Ute Skiba, Mark A. Sutton, J. Neil Cape, Stefan Reis, Lucy J. Sheppard, et al. 2013. “The Global Nitrogen Cycle in the Twentyfirst Century.” *Philosophical Transactions of the Royal Society B: Biological Sciences* 368 (1621). <https://doi.org/10.1098/rstb.2013.0164>.
- Gabriel, J. L., R. Muñoz-Carpena, and M. Quemada. 2012. “The Role of Cover Crops in Irrigated Systems: Water Balance, Nitrate Leaching and Soil Mineral Nitrogen Accumulation.” *Agriculture, Ecosystems and Environment* 155 (3): 50–61. <https://doi.org/10.1016/j.agee.2012.03.021>.
- Gasser, Deta, Jürgen Gusterhuber, Oliver Krische, Barbara Pühr, Lorenz Scheucher, Thomas Wagner, and Kurt

- Stüwe. 2009. "Geology of Styria: An Overview." *Mitteilungen Des Naturwissenschaftlichen Vereines Für Steiermark* 139: 5–36.
- GBA. 2015. "Geofast - 191 Kirchbach i. Stmk." Zusammenstellung ausgewählter Archivunterlagen der Geologischen Bundesanstalt 1:50.000 - 191 Kirchbach i. Stmk.: Stand 2015, Ausgabe 2016/04; Geologische Bundesanstalt (GBA), Michael Moser.
- Genuchten, M.Th. van. 1980. "A Closed-Form Equation for Predicting the Hydraulic Properties of Unsaturated Soils." *Soil Science Society of America Journal* 44: 892–98.
- Gerke, H H, and M Th Van Genuchten. 1993. "A Dual-porosity Model for Simulating the Preferential Movement of Water and Solutes in Structured Porous Media." *Water Resources Research* 29 (2): 305–19.
- Graham, Scott L., M.S. Srinivasan, Nathalie Faulkner, and Sam Carrick. 2018. "Soil Hydraulic Modeling Outcomes with Four Parameterization Methods: Comparing Soil Description and Inverse Estimation Approaches." *Vadose Zone Journal* 17 (1): 170002. <https://doi.org/10.2136/vzj2017.01.0002>.
- Groenendijk, Piet, Marius Heinen, Gernot Klammler, Johann Fank, Hans Kupfersberger, Vassilios Pisinaras, Alexandra Gemitzi, et al. 2014. "Performance Assessment of Nitrate Leaching Models for Highly Vulnerable Soils Used in Low-Input Farming Based on Lysimeter Data." *Science of the Total Environment* 499: 463–80. <https://doi.org/10.1016/j.scitotenv.2014.07.002>.
- Hanson, Blaine R., Jirka Šimůnek, and Jan W. Hopmans. 2006. "Evaluation of Urea-Ammonium-Nitrate Fertigation with Drip Irrigation Using Numerical Modeling." *Agricultural Water Management* 86 (1–2): 102–13. <https://doi.org/10.1016/j.agwat.2006.06.013>.
- Hartge, Karl Heinrich, Rainer Horn, Jörg Bachmann, and Stephan Peth. 2014. "Einführung in Die Bodenphysik."
- Holler, Philipp. 2019. "Einfluss Der Düngung von Begrünungsvarianten Und Deren Wirkung Auf Nitrat Im Bodenwasser." BOKU - Universität für Bodenkultur.
- Humphreys, John, Rong Lan, and Shanwen Tao. 2021. "Development and Recent Progress on Ammonia Synthesis Catalysts for Haber–Bosch Process." *Advanced Energy and Sustainability Research* 2 (1): 2000043. <https://doi.org/10.1002/aesr.202000043>.
- Hupet, F., S. Lambot, R. A. Feddes, J. C. Van Dam, and M. Vanclooster. 2003. "Estimation of Root Water Uptake Parameters by Inverse Modeling with Soil Water Content Data." *Water Resources Research* 39 (11). <https://doi.org/10.1029/2003WR002046>.
- Iqbal, Mazhar, Md Rowshon Kamal, Mohd Fazly M., Hasfalina Che Man, and Aimrun Wayayok. 2020. "HYDRUS-1D Simulation of Soil Water Dynamics for Sweet Corn under Tropical Rainfed Condition." *Applied Sciences (Switzerland)* 10 (4): 1–17. <https://doi.org/10.3390/app10041219>.
- Jiang, Yefang, and George Somers. 2009. "Modeling Effects of Nitrate from Non-Point Sources on Groundwater Quality in an Agricultural Watershed in Prince Edward Island, Canada." *Hydrogeology Journal* 17 (3): 707–24. <https://doi.org/10.1007/s10040-008-0390-2>.
- Klammler, Gernot, Hans Kupfersberger, Gerhard Rock, and Johann Fank. 2013. "Modeling Coupled Unsaturated and Saturated Nitrate Distribution of the Aquifer Westliches Leibnitzer Feld, Austria." *Environmental Earth Sciences* 69 (2): 663–78. <https://doi.org/10.1007/s12665-013-2302-6>.
- Kuo, S., B. Huang, and R. Bembenek. 2001. "Effect of Winter Cover Crops on Soil Nitrogen Availability, Corn Yield, and Nitrate Leaching." *TheScientificWorldJournal* 1 Suppl 2: 22–29. <https://doi.org/10.1100/tsw.2001.267>.

- Land Steiermark. n.d. "Bodenschutz Analysen LEI5." Accessed April 12, 2021a.
https://sterz.stmk.gv.at/at.gv.stmk.capp/cms/cagris/process.do?app=AGRIS_BSP_Standort_Ana&embed=frame&output=1&ip01=NN&ip01def=LEI5&showBackButton=true&backUrl=app%3DAGRIS_BSP_Standort_Beu%26embed%3Dframe%26output%3D1%26ip01%3DNN%26ip01def%3DLEI5.
- . n.d. "Bodenschutz Analysen LEI6." Accessed April 12, 2021b.
https://sterz.stmk.gv.at/at.gv.stmk.capp/cms/cagris/process.do?app=AGRIS_BSP_Standort_Ana&embed=frame&output=1&ip01=NN&ip01def=LEI6&showBackButton=true&backUrl=app%3DAGRIS_BSP_Standort_Beu%26embed%3Dframe%26output%3D1%26ip01%3DNN%26ip01def%3DLEI6.
- . 2009. "Bericht Über Die Lage Der Land- Und Forstwirtschaft in Der Steiermark." *Fachabteilung 10A – Agrarrecht Und Ländliche Entwicklung*, 38.
- . 2017. "Digitaler Atlas Der Steiermark."
- Li, Yong, Jirka Šimůnek, Zhentin Zhang, Longfei Jing, and Lixiao Ni. 2015. "Evaluation of Nitrogen Balance in a Direct-Seeded-Rice Field Experiment Using Hydrus-1D." *Agricultural Water Management* 148: 213–22. <https://doi.org/10.1016/j.agwat.2014.10.010>.
- Matteau, J. P., S. J. Gumiere, J. Gallichand, G. Létourneau, L. Khiari, M. O. Gasser, and A. Michaud. 2019. "Coupling of a Nitrate Production Model with HYDRUS to Predict Nitrate Leaching." *Agricultural Water Management* 213 (November): 616–26. <https://doi.org/10.1016/j.agwat.2018.10.013>.
- Monteith, J. L. 1981. "Evaporation and Surface Temperature." *Quarterly Journal of the Royal Meteorological Society* 107 (451): 1–27. <https://doi.org/10.1256/smsqj.45101>.
- Mualem, Yechezkel. 1976. "Hysteretical Models for Prediction of the Hydraulic Conductivity of Unsaturated Porous Media." *Water Resources Research* 12 (6): 1248–54. <https://doi.org/10.1029/WR012i006p01248>.
- Neteler, Markus, and Helena Mitasova. 2008. "Open Source GIS." *North* 2. <https://doi.org/10.1007/978-0-387-68574-8>.
- Neuman, S. P., R. A. Feddes, and E. Bresler. 1974. "Finite Element Simulation of Flow in Saturated-Unsaturated Soils Considering Water Uptake by Plants."
- O'Connell, S., W. Shi, J. M. Grossman, G. D. Hoyt, K. L. Fager, and N. G. Creamer. 2015. "Short-Term Nitrogen Mineralization from Warm-Season Cover Crops in Organic Farming Systems." *Plant and Soil* 396 (1–2): 353–67. <https://doi.org/10.1007/s11104-015-2594-2>.
- ÖNORM L 1091. 1989. "1091 Chemische Bodenuntersuchung, N Min-Methode." *Österreichisches Normungsinstitut Wien*.
- Rawls, Walter J, Donald L Brakensiek, and K E Saxton. 1982. "Estimation of Soil Water Properties." *Transactions of the ASAE* 25 (5): 1316–20.
- Reszler, Christian, and Johann Fank. 2016. "Unsaturated Zone Flow and Solute Transport Modelling with MIKE SHE: Model Test and Parameter Sensitivity Analysis Using Lysimeter Data." *Environmental Earth Sciences* 75 (3): 1–13. <https://doi.org/10.1007/s12665-015-4881-x>.
- Richner, W, H R Oberholzer, Freiermuth Knuchel, Huguenin R. O., S Ott, and U Walther. 2014. *Modell Zur Beurteilung Der Nitratauswaschung in Ökobilanzen – SALCA-NO3. To Be Published at: Www.Agroscope.Ch*.
- Rimkus, Egidijus, Justas Kažys, Arunas Bukantis, and Aleksandras Krotovas. 2011. "Temporal Variation of Extreme Precipitation Events in Lithuania." *Oceanologia* 53 (1-TI): 259–77. <https://doi.org/10.5697/oc.53-1-TI.259>.

- Rossum, Guido Van. 1995. *Python Tutorial*. Amsterdam, the Netherlands: Technical Report CS-R9526, Centrum voor Wiskunde en Informatica Amsterdam.
- Saltelli, Andrea. 2002. "Sensitivity_analysis_for_importance_asse.Pdf." Society for Risk Analysis.
- Sansoulet, Julie. 2007. "Transferts d'eau et Des Ions Potassium et Nitrate Dans Un Sol à Capacité d'échange Anionique Sous Un Couvert Redistributeur de La Pluie."
- Schaap, Marcel G., Feike J. Leij, and Martinus Th Van Genuchten. 2001. "Rosetta: A Computer Program for Estimating Soil Hydraulic Parameters with Hierarchical Pedotransfer Functions." *Journal of Hydrology* 251 (3–4): 163–76. [https://doi.org/10.1016/S0022-1694\(01\)00466-8](https://doi.org/10.1016/S0022-1694(01)00466-8).
- SGV. 2018. *Schongebietsverordnung LGBl. Nr. 24/2018 IdF LGBl. Nr. 70/2020 (Steiermark)*. Regionalprogramm zum Schutz der Grundwasserkörper Grazer Feld, Leibnitzer Feld und Unteres Murtal (Grundwasserschutzprogramm Graz bis Bad Radkersburg 2018).
- Šimůnek, Jirka, and Martinus van Genuchten. 1995. "Numerical Model for Simulating Multiple Solute Transport in Variably- Saturated Soils." *Water Pollution III: Modelling, Measuring and Prediction*, 21–30.
- . 2008. "Modeling Nonequilibrium Flow and Transport Processes Using HYDRUS." *Vadose Zone Journal* 7 (2): 782–97. <https://doi.org/10.2136/vzj2007.0074>.
- Šimůnek, Jirka, Martinus van Genuchten, and Miroslav Šejna. 2008. "Development and Applications of the HYDRUS and STANMOD Software Packages and Related Codes." *Vadose Zone Journal* 7 (2): 587–600.
- . 2012. "HYDRUS: Model Use, Calibration, and Validation" 55 (1987): 1261–74.
- Šimůnek, Jirka, Miroslav Sejna, Hirotaka Saito, M. Sakai, and Martinus Van Genuchten. 2009. *The HYDRUS-1D Software Package for Simulating the Movement of Water, Heat, and Multiple Solutes in Variably Saturated Media. Technical Manual*. Department of Environmental Sciences, University of California Riverside, Riverside, California, USA.
- Soltani, Amin, Mahdich Azimi, Arezoo Boroomandnia, and Brendan C. O'Kelly. 2021. "An Objective Framework for Determination of the Air-Entry Value from the Soil–Water Characteristic Curve." *Results in Engineering* 12: 100298. <https://doi.org/10.1016/j.rineng.2021.100298>.
- Statistik Austria. 2016. "Agrarstrukturerhebung 2016. Betriebsstruktur. Schnellbericht 1.17.," 195.
- . 2019. "Landesentwicklung Steiermark; Land- Und Forstwirtschaft; Kulturarten Und Sonstige Flächen 2003 - 2019." *Land Steiermark*.
- . 2020. "Anbau Auf Dem Ackerland 2020."
- Stephens, Daniel B. 1996. *Vadose Zone Hydrology*. Boca Raton: CRC press - Lewis Publishers.
- Stmk. LBSG. 1987. *Steiermärkisches Landwirtschaftliches Bodenschutzgesetz*.
- Taiz, L, E Zeiger, I. M Møller, and A. Murphy. 2007. *Plant Physiology: Das Original Mit Übersetzungshilfen*. Transl. by Birgit Jarosch, Spektrum Akademischer Verlag.
- Tan, Xuezhai, Dongguo Shao, Wenquan Gu, and Huanhuan Liu. 2015. "Field Analysis of Water and Nitrogen Fate in Lowland Paddy Fields under Different Water Managements Using HYDRUS-1D." *Agricultural Water Management* 150: 67–80. <https://doi.org/10.1016/j.agwat.2014.12.005>.
- Taylor, Sterling A, and Gaylen L Ashcroft. 1972. *Physical Edaphology. The Physics of Irrigated and Nonirrigated Soils*. San Francisco, California: Freeman and Co.
- TWV. 2018. *Trinkwasserverordnung (BGBl. II Nr. 304/2001 IdgF)*.
- Ursulino, Bruno Silva, Suzana Maria Gico Lima Montenegro, Artur Paiva Coutinho, Victor Hugo Rabelo Coelho, Diego Cezar dos Santos Araújo, Ana Cláudia Villar Gusmão, Severino Martins dos Santos Neto,

- Laurent Lassabatere, and Rafael Angulo-Jaramillo. 2019. "Modelling Soil Water Dynamics from Soil Hydraulic Parameters Estimated by an Alternative Method in a Tropical Experimental Basin." *Water (Switzerland)* 11 (5): 4–6. <https://doi.org/10.3390/w11051007>.
- Veizaga, E. A., L. Rodríguez, and C. J. Ocampo. 2016. "Investigating Nitrate Dynamics in a Fine-Textured Soil Affected by Feedlot Effluents." *Journal of Contaminant Hydrology* 193: 21–34. <https://doi.org/10.1016/j.jconhyd.2016.08.005>.
- Vitousek, P. M., R. Naylor, T. Crews, M. B. David, L. E. Drinkwater, E. Holland, P. J. Johnes, et al. 2009. "Nutrient Imbalances in Agricultural Development." *Science* 324 (5934): 1519–20. <https://doi.org/10.1126/science.1170261>.
- Vogel, T, K Huang, R Zhang, and M Th Van Genuchten. 1996. "The HYDRUS Code for Simulating One-Dimensional Water Flow, Solute Transport, and Heat Movement in Variably-Saturated Media." *US Salinity Lab, Riverside, CA*.
- Wang, Huanyuan, Xiaotang Ju, Yongping Wei, Baoguo Li, Lulu Zhao, and Kelin Hu. 2010. "Simulation of Bromide and Nitrate Leaching under Heavy Rainfall and High-Intensity Irrigation Rates in North China Plain." *Agricultural Water Management* 97 (10): 1646–54. <https://doi.org/10.1016/j.agwat.2010.05.022>.
- Ward, Mary H, Rena R Jones Id, Jean D Brender, Theo M De Kok, Peter J Weyer, Bernard T Nolan, Cristina M Villanueva, and Simone G Van Breda. 2018. "Drinking Water Nitrate and Human Health : An Updated Review," 1–31. <https://doi.org/10.3390/ijerph15071557>.
- Wesseling, J G, J A Elbers, P Kabat, and B J Van den Broek. 1991. "SWATRE: Instructions for Input." *Internal Note, Winand Staring Centre, Wageningen, the Netherlands* 1991.
- White, Charles M., S. Tianna DuPont, Mena Hautau, Dave Hartman, Denise M. Finney, Brosi Bradley, James C. LaChance, and Jason P. Kaye. 2017. "Managing the Trade off between Nitrogen Supply and Retention with Cover Crop Mixtures." *Agriculture, Ecosystems and Environment* 237: 121–33. <https://doi.org/10.1016/j.agee.2016.12.016>.
- Wilson, L Gray, Lorne G Everett, and Stephen J Cullen. 1995. *Handbook of Vadose Zone Characterization & Monitoring*. Boca Raton: CRC press.
- WRG. 1959. *Wasserrechtsgesetz BGBl. Nr. 215/1959, Zuletzt Geändert Durch Das Bundesgesetz BGBl. I Nr. 58/2017*.

LIST OF FIGURES

FIG. 1: LAND USE IN STYRIA (STATUS: OCTOBER 2014) ACCORDING TO THE <i>WASSERVERSORGUNGSPLAN</i> (AMT DER STEIERMÄRKISCHEN LANDESREGIERUNG 2015).....	5
FIG. 2: SHALLOW GROUNDWATER AQUIFERS IN SOUTHERN STYRIA WITH THE POSITION OF THE FIELD SITE (RED) WHERE MEASUREMENTS USED IN THIS MASTER THESIS WERE CONDUCTED (LAND STEIERMARK 2017).....	6
FIG. 3: GEOLOGICAL MAP OF NEOGENE SEDIMENTS AND VOLCANIC ROCKS IN STYRIA (GASSER ET AL. 2009).....	7
FIG. 4: SOIL TYPE (BROWN EARTH) AT THE FIELD SITE (RED POSITION MARKER) ACCORDING TO THE AUSTRIAN <i>DIGITALE BODENKARTE</i> (EBOD) (BFW 2004).	8
FIG. 5: LAYOUT OF THE ARRANGEMENT OF THE PLOTS AT THE FIELD SITE (LEFT); TREATMENT OF THE DIFFERENT PLOTS (RIGHT); LIGHT GREEN REPRESENTING THE PLOTS FOR THE CALIBRATION AND DARK GREEN THE PLOTS FOR THE VALIDATION.....	9
FIG. 6: FLOW CHART OF THE ESTABLISHED MODEL SIMULATION PROCESS IN HYDRUS-1D.....	12
FIG. 7: PLANT WATER STRESS RESPONSE FUNCTION, $A'(h)$ BY FEDDES, KOWALIK, AND ZARADNY (1978).	17
FIG. 8: METEOROLOGICAL DATA FROM THE ZAMG STATION IN WAGNA (3 RD AUGUST 2017 – 2 ND MAY 2018) THAT SERVES AS INPUT FOR THE HYDRUS-MODEL; DAILY SUMS OF PRECIPITATION (TOP) AND SOLAR RADIATION (MIDDLE RIGHT); DAILY MEAN VALUES OF AIR TEMPERATURE (MIDDLE LEFT), RELATIVE HUMIDITY (BOTTOM LEFT) AND WIND SPEED (BOTTOM RIGHT).	20
FIG. 9: SPECIFYING THE INITIAL CONDITIONS (WATER CONTENT AND SOLUTE CONCENTRATIONS) IN THE <i>SOIL PROFILE – GRAPHICAL EDITOR</i> IN HYDRUS, EXEMPLARY FOR PLOT 5.....	22
FIG. 10: PLOTS AND TIME PERIODS USED FOR CALIBRATION, VALIDATION AND THE EVALUATION OF THE HYDRUS-MODEL.....	27
FIG. 11: SENSITIVITY ANALYSIS OF THE SATURATED HYDRAULIC CONDUCTIVITY K_s CONCERNING THE ABSOLUTE ERRORS OF THE WATER CONTENTS IN PLOT 6 (INITIAL K_s OF THE THREE DEPTH LAYERS: ~14, 16 AND 20 cm d^{-1}).	29
FIG. 12: SENSITIVITY ANALYSIS OF THE POROSITY DISTRIBUTION MEASURE n CONCERNING THE ABSOLUTE ERRORS OF THE WATER CONTENTS IN PLOT 2 (INITIAL n : 2).	30
FIG. 13: SENSITIVITY ANALYSIS OF THE AIR-ENTRY VALUE a CONCERNING THE ABSOLUTE ERRORS OF THE WATER CONTENTS IN PLOT 5 (INITIAL a : 0.0775 cm^{-1}).	30
FIG. 14: SENSITIVITY ANALYSIS OF THE SATURATED HYDRAULIC CONDUCTIVITY K_s CONCERNING THE ABSOLUTE ERRORS OF THE NO_3^- -N CONCENTRATION IN PLOT 5 (INITIAL K_s OF THE THREE DEPTH LAYERS: ~14, 16 AND 20 cm d^{-1}).	31
FIG. 15: SENSITIVITY ANALYSIS OF THE LONGITUDINAL DISPERSIVITY D_L CONCERNING THE ABSOLUTE ERRORS OF THE NO_3^- -N CONCENTRATION IN PLOT 2 (INITIAL D_L : 15 cm).	32
FIG. 16: SENSITIVITY ANALYSIS OF THE ZERO-ORDER RATE CONSTANT r_w (0–30 cm) CONCERNING THE ABSOLUTE ERRORS OF THE NO_3^- -N CONCENTRATION IN PLOT 5 (INITIAL r_w : 0.00025 $\text{mg cm}^{-3} \text{d}^{-1}$).	32
FIG. 17: SIMULATED WATER CONTENT CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 2 IN COMPARISON TO THE PUNCTUAL MEASURED WATER CONTENTS IN PLOT 2.....	33
FIG. 18: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 2 IN COMPARISON TO THE PUNCTUAL MEASURED NO_3^- -N CONCENTRATIONS IN PLOT 2.....	33
FIG. 19: SIMULATED WATER CONTENT CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 5 IN COMPARISON TO THE PUNCTUAL MEASURED WATER CONTENTS IN PLOT 5.....	34
FIG. 20: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 5 IN COMPARISON TO THE PUNCTUAL MEASURED NO_3^- -N CONCENTRATIONS IN PLOT 5.....	35
FIG. 21: SIMULATED WATER CONTENT CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 6 IN COMPARISON TO THE PUNCTUAL MEASURED WATER CONTENTS IN PLOT 6.....	35
FIG. 22: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 6 IN COMPARISON TO THE PUNCTUAL MEASURED NO_3^- -N CONCENTRATIONS IN PLOT 6.....	36
FIG. 23: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 2 WITH ADAPTED ZERO-ORDER RATE CONSTANT FOR THE TIME PERIOD FROM NOVEMBER TO APRIL IN COMPARISON TO THE PUNCTUAL MEASURED NO_3^- -N CONCENTRATIONS IN PLOT 2.....	37
FIG. 24: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 5 WITH ADAPTED ZERO-ORDER RATE CONSTANT FOR THE TIME PERIOD FROM NOVEMBER TO APRIL IN COMPARISON TO THE PUNCTUAL MEASURED NO_3^- -N CONCENTRATIONS IN PLOT 5.....	37

FIG. 25: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 6 WITH ADAPTED ZERO-ORDER RATE CONSTANT FOR THE TIME PERIOD FROM NOVEMBER TO APRIL IN COMPARISON TO THE PUNCTUAL MEASURED NO_3^- -N CONCENTRATIONS IN PLOT 6.	38
FIG. 26: SIMULATED WATER CONTENT CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 1 IN COMPARISON TO THE PUNCTUAL MEASURED WATER CONTENTS IN PLOT 1.	38
FIG. 27: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 1 IN COMPARISON TO THE PUNCTUAL MEASURED NO_3^- -N CONCENTRATIONS IN PLOT 1.	39
FIG. 28: SIMULATED WATER CONTENT CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 3 IN COMPARISON TO THE PUNCTUAL MEASURED WATER CONTENTS IN PLOT 3.	39
FIG. 29: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 3 IN COMPARISON TO THE PUNCTUAL MEASURED NO_3^- -N CONCENTRATIONS IN PLOT 3.	39
FIG. 30: SIMULATED WATER CONTENT CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 4 IN COMPARISON TO THE PUNCTUAL MEASURED WATER CONTENTS IN PLOT 4.	40
FIG. 31: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF PLOT 4 IN COMPARISON TO THE PUNCTUAL MEASURED NO_3^- -N CONCENTRATIONS IN PLOT 4.	40
FIG. 32: ACTUAL EVAPOTRANSPIRATION CALCULATED BY HYDRUS-1D FOR THE CALIBRATED PLOTS (P2, P5 AND P6), WITH THE WIND SPEED CURVE FROM THE ZAMG MEASUREMENT STATION.	41
FIG. 33: POTENTIAL EVAPORATION AND POTENTIAL TRANSPIRATION CALCULATED BY HYDRUS-1D FOR PLOT 5.	41
FIG. 34: SIMULATED WATER CONTENT CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF ALL PLOTS.	42
FIG. 35: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 16 TH OF APRIL 2018 OF ALL PLOTS.	42
FIG. 36: OBSERVED WATER CONTENTS (21 ST SEPTEMBER, 30 TH OCTOBER, 21 ST NOVEMBER, 16 TH APRIL) PLOTTED AGAINST THE SIMULATED WATER CONTENTS AT THAT EXACT DATES; WITH THE COLOURS DIVIDING INTO DIFFERENT PLOTS AND PLOT DEPTHS, AND AN R^2 VALUE OF 0.659.	43
FIG. 37: OBSERVED NO_3^- -N CONCENTRATIONS (21 ST SEPTEMBER, 30 TH OCTOBER, 21 ST NOVEMBER, 16 TH APRIL) PLOTTED AGAINST THE SIMULATED NO_3^- -N CONCENTRATIONS AT THAT EXACT DATES; WITH THE COLOURS DIVIDING INTO DIFFERENT PLOTS AND PLOT DEPTHS, AND AN R^2 VALUE OF 0.624.	44
FIG. 38: OBSERVED WATER CONTENTS (21 ST SEPTEMBER, 30 TH OCTOBER, 21 ST NOVEMBER, 16 TH APRIL) PLOTTED AGAINST THE SIMULATED WATER CONTENTS AT THAT EXACT DATES, WITH THE COLOURS DIVIDING INTO THE MONTHS WHEN THE MEASUREMENTS WERE CONDUCTED.	44
FIG. 39: OBSERVED NO_3^- -N CONCENTRATIONS (21 ST SEPTEMBER, 30 TH OCTOBER, 21 ST NOVEMBER, 16 TH APRIL) PLOTTED AGAINST THE SIMULATED NO_3^- -N CONCENTRATIONS AT THAT EXACT DATES, WITH THE COLOURS DIVIDING INTO THE MONTHS WHEN THE MEASUREMENTS WERE CONDUCTED.	45
FIG. 40: ACTUAL ROOT NO_3^- -N UPTAKE OF THE WINTER COVER CROPS IN ALL PLOTS, WITH THE TEMPERATURE CURVE FROM THE ZAMG MEASUREMENT STATION.	46
FIG. 41: LEFT: PRECIPITATION CURVE FROM THE ZAMG MEASUREMENT STATION IN WAGNA, WITH THE HEAVY PRECIPITATION EVENT ON THE 16 TH OF APRIL 2018; RIGHT: INUNDATED FIELD SITE ON THE 17 TH OF APRIL 2018 (©PHILIPP HOLLER).	46
FIG. 42: SIMULATED WATER CONTENT CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 2 ND OF MAY 2018 OF PLOT 5 IN COMPARISON TO THE PUNCTUAL MEASURED WATER CONTENTS IN PLOT 5; WITH THE HEAVY PRECIPITATION EVENT ON THE 16 TH OF APRIL 2017 AFTER THE FOURTH SOIL SAMPLING.	47
FIG. 43: SIMULATED NO_3^- -N CONCENTRATION CURVES FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 2 ND OF MAY 2018 OF PLOT 5 IN COMPARISON TO THE PUNCTUAL MEASURED NO_3^- -N CONCENTRATIONS IN PLOT 5; WITH THE HEAVY PRECIPITATION EVENT ON THE 16 TH OF APRIL 2017 AFTER THE FOURTH SOIL SAMPLING.	47
FIG. 44: OBSERVED WATER CONTENTS (2 ND MAY 2018) PLOTTED AGAINST THE SIMULATED WATER CONTENTS AT THAT EXACT DATE.	48
FIG. 45: OBSERVED NO_3^- -N CONCENTRATIONS (2 ND MAY 2018) PLOTTED AGAINST THE SIMULATED NO_3^- -N CONCENTRATIONS AT THAT EXACT DATE.	48
FIG. 46: BOTTOM NO_3^- -N FLUX [MG NO_3^- -N $\text{CM}^{-2} \text{D}^{-1}$] IN ALL PLOTS CALCULATED BY HYDRUS-1D FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 2 ND OF MAY 2018.	49

FIG. 47: CUMULATIVE BOTTOM NO_3^- -N FLUX [KG NO_3^- -N HA^{-1}] IN ALL PLOTS CALCULATED BY HYDRUS-1D FOR THE TIME PERIOD FROM 3 RD OF AUGUST 2017 TO 2 ND OF MAY 2018.	49
--	----

LIST OF TABLES

TABLE 1: SCHEDULE OF ACTIONS TAKEN AT THE FIELD SITE.....	10
TABLE 2: WATER RETENTION PARAMETERS AS DESCRIBED IN THE VAN GENUCHTEN WATER RETENTION FUNCTION (1.5) AND THE SOIL AND SOLUTE SPECIFIC PARAMETERS OF THE THREE DEPTH LAYERS.	24
TABLE 3: ROOT WATER UPTAKE PARAMETERS OF THE SIX PLOTS, COVERED BY DIFFERENT WINTER COVER CROP MIXTURES.	25
TABLE 4: LEACHED NO_3^- -N AMOUNTS [KG/HA] BELOW 1 M, CALCULATED EXEMPLARILY FOR TWO TIME PERIODS.	50
TABLE 5: COMPARISON OF THE MONTHLY ETP SUMS OF THE HYDRUS-MODEL AND THE <i>WASSERVERSORGUNGSPLAN STEIERMARK</i> (AMT DER STEIERMÄRKISCHEN LANDESREGIERUNG 2015)	54

APPENDIX

Table 6: Measured water contents from Philipp Holler (2019).

WATER CONTENTS [%]		August	September	October	November	April	May
Plot 1	0-30cm	17.46	25.49	20.07	25.87	23.76	23.47
	30-60cm	15.45	20.46	17.59	20.69	20.35	19.19
	60-90cm	13.13	18.08	16.70	18.99	18.79	17.90
Plot 2	0-30cm	18.07	26.99	20.45	26.19	23.33	23.77
	30-60cm	17.04	20.73	17.19	20.74	19.68	19.03
	60-90cm	18.60	18.24	16.12	17.89	18.06	17.21
Plot 3	0-30cm	15.35	26.17	20.49	26.30	23.36	24.84
	30-60cm	13.10	21.72	16.36	21.41	20.26	19.24
	60-90cm	11.30	18.08	17.50	18.87	18.42	21.23
Plot 4	0-30cm	18.82	26.13	19.58	26.73	23.36	23.74
	30-60cm	15.29	21.34	16.24	21.10	19.37	18.33
	60-90cm	13.66	17.17	16.86	18.82	17.93	17.87
Plot 5	0-30cm	17.78	25.80	23.33	26.44	24.42	24.28
	30-60cm	15.50	21.69	19.04	21.25	20.44	19.60
	60-90cm	13.55	18.12	17.80	20.04	18.45	18.43
Plot 6	0-30cm	18.13	27.36	22.64	27.39	23.84	23.28
	30-60cm	14.87	21.42	19.48	21.85	19.78	18.93
	60-90cm	12.54	17.46	17.22	18.52	18.61	21.65

Table 7: NO₃⁻-N concentrations, calculated from the measured values from Philipp Holler (2019).

NO ₃ -N [mg/kg]		August	September	October	November	April	May
Plot 1	0-30cm	13.487	2.843	3.309	1.480	12.338	9.615
	30-60cm	6.127	1.238	1.401	0.613	5.801	4.904
	60-90cm	3.045	1.292	0.761	1.690	3.242	1.946
Plot 2	0-30cm	25.899	3.775	2.875	2.912	13.382	5.895
	30-60cm	6.543	2.086	0.831	0.928	5.963	3.749
	60-90cm	3.195	1.668	0.616	1.180	3.585	1.827
Plot 3	0-30cm	9.268	4.067	2.691	2.106	11.640	7.609
	30-60cm	7.742	2.835	1.678	1.058	5.044	4.314
	60-90cm	3.496	3.681	2.167	2.022	2.724	4.483
Plot 4	0-30cm	12.452	3.298	3.336	3.144	11.738	7.618
	30-60cm	6.987	2.309	0.921	1.275	5.467	4.358
	60-90cm	3.447	2.349	1.008	1.113	3.701	2.711
Plot 5	0-30cm	16.445	2.403	5.720	6.387	13.571	8.777
	30-60cm	8.678	4.257	3.250	3.670	8.504	6.454
	60-90cm	7.550	3.475	4.085	3.944	5.506	4.483
Plot 6	0-30cm	14.878	4.187	4.110	3.762	12.681	7.151
	30-60cm	8.013	6.603	3.501	1.453	5.543	4.174
	60-90cm	4.133	3.783	3.074	2.107	4.091	4.444

Table 8: Meteorological input parameters for HYDRUS, measured by the ZAMG station in Wagna.

	Precipitation [cm/d]	Solar radiation [MJ/m ² d]	Temperature [°C]	Humidity [%]	Wind speed [km/d]
03.08.2017	0	24.81	26.58	65.58	100.08
04.08.2017	0	23.81	28.55	60.71	110.88
05.08.2017	0.01	24.18	26.34	62.96	142.2
06.08.2017	1.95	7.46	21.29	78.96	113.04
07.08.2017	0.47	22.12	19.93	70.21	83.52
08.08.2017	0	20.27	20.52	78.25	124.2
09.08.2017	0.01	17.83	23.40	77.75	84.24
10.08.2017	0.92	16.69	24.10	77.46	111.96
11.08.2017	0.02	19.1	21.69	70.71	87.12
12.08.2017	0.05	8.95	18.26	69.25	101.52
13.08.2017	0	22.86	20.02	62.79	127.08
14.08.2017	0	24.85	20.31	66.21	74.52
15.08.2017	0	24.36	21.35	64.58	117
16.08.2017	0	18.05	21.03	73.13	97.56
17.08.2017	0	23.31	22.58	69.67	78.12
18.08.2017	0	22.1	24.38	63.04	135.36
19.08.2017	0.97	6.6	18.96	80.75	109.8
20.08.2017	0.57	16.51	17.17	70.33	130.32
21.08.2017	0	19.15	15.40	66.08	102.24
22.08.2017	0	22.6	15.67	66.42	83.52
23.08.2017	0	22.65	17.85	68.38	93.24
24.08.2017	0	22.08	20.91	70.08	92.16
25.08.2017	0	19.03	22.77	72.04	64.08
26.08.2017	0	21.13	24.38	67.33	53.64
27.08.2017	1.05	20.62	24.86	63.46	84.24
28.08.2017	0.08	11.34	19.23	83.21	83.88
29.08.2017	0	21.45	18.69	70.38	68.04
30.08.2017	0	21.76	19.30	69.13	79.92
31.08.2017	0	20.59	22.48	69.88	83.52
01.09.2017	1.91	8.37	19.17	86.21	78.84
02.09.2017	0.76	5.59	15.11	86.00	92.88
03.09.2017	0.01	10.01	14.50	70.96	168.12
04.09.2017	0	19.69	14.40	68.79	61.56
05.09.2017	0	21.16	15.99	66.83	59.04
06.09.2017	0	11.09	17.47	77.50	42.12
07.09.2017	0.22	4.11	14.87	89.21	103.32
08.09.2017	0	11.79	14.16	84.00	79.56
09.09.2017	0	15.3	16.08	77.17	110.88
10.09.2017	0.04	7.42	18.71	77.79	70.2
11.09.2017	1.06	3.04	16.24	92.67	59.04
12.09.2017	0.4	5.11	13.73	84.29	99.72
13.09.2017	0	18.74	14.97	71.25	95.4
14.09.2017	1.84	19.43	16.60	72.50	217.44
15.09.2017	0.89	5.12	11.67	85.33	55.44

16.09.2017	1.33	3.32	12.40	92.13	36.36
17.09.2017	2.37	5.08	12.64	92.13	48.6
18.09.2017	0.76	10.9	12.04	86.17	80.28
19.09.2017	4.84	1.35	9.71	91.58	194.4
20.09.2017	0	4.34	11.05	76.46	137.52
21.09.2017	0	14.77	12.46	65.04	180
22.09.2017	0	16.05	12.61	69.46	55.44
23.09.2017	0	16.88	13.35	75.83	71.28
24.09.2017	0	6.96	13.69	83.21	56.52
25.09.2017	0.01	4.38	12.29	89.79	26.64
26.09.2017	0.31	3.27	12.53	92.58	81.36
27.09.2017	0.19	4.48	13.90	93.00	40.68
28.09.2017	0	15.95	14.28	80.13	79.56
29.09.2017	0	13.97	11.24	82.21	74.88
30.09.2017	0	14.48	10.46	75.88	82.8
01.10.2017	0	16.07	9.11	75.29	54
02.10.2017	0	15.45	9.57	78.25	58.32
03.10.2017	0.37	3.74	10.82	91.83	41.04
04.10.2017	0	13.57	12.57	83.83	76.68
05.10.2017	0	14.88	16.50	69.96	119.16
06.10.2017	0.27	11.63	11.45	76.83	120.24
07.10.2017	0	13.53	9.20	71.79	74.52
08.10.2017	0	10.57	9.66	79.29	33.84
09.10.2017	0	8.38	9.65	82.50	72.36
10.10.2017	0	13.45	12.10	79.50	81.36
11.10.2017	0	13.23	12.09	80.96	57.6
12.10.2017	0	13	12.95	78.46	68.04
13.10.2017	0	12.91	13.20	79.75	41.04
14.10.2017	0	12.16	12.60	83.71	41.4
15.10.2017	0	12.49	13.27	82.96	43.56
16.10.2017	0	12.07	13.28	83.71	37.08
17.10.2017	0	12.24	13.54	81.00	39.6
18.10.2017	0	9.25	12.03	85.83	29.52
19.10.2017	0	11.94	11.34	82.21	40.32
20.10.2017	0	10.48	10.55	84.04	27.36
21.10.2017	0	7.77	11.49	82.67	41.76
22.10.2017	1.82	2.16	8.93	91.38	93.24
23.10.2017	0	2.81	9.28	73.04	146.16
24.10.2017	0	10.21	10.71	63.25	168.48
25.10.2017	0	8.99	7.71	81.04	24.84
26.10.2017	0	9.42	10.38	82.50	50.04
27.10.2017	0	2.27	9.61	78.25	107.28
28.10.2017	0	5.52	7.70	69.67	64.8
29.10.2017	0	7.6	11.47	51.79	350.64
30.10.2017	0	9.88	7.33	53.38	142.92
31.10.2017	0	8.18	5.06	66.13	63.72

01.11.2017	0	6.7	5.58	76.13	42.48
02.11.2017	0	4.78	5.48	85.17	32.4
03.11.2017	0.04	4.56	7.85	86.71	60.48
04.11.2017	0	3.44	5.78	94.92	56.88
05.11.2017	0	5.06	9.42	82.00	90
06.11.2017	2.53	0.57	7.67	93.17	81.36
07.11.2017	0.6825	0.54	7.45	91.29	87.66
08.11.2017	1.62	1.51	6.89	93.13	68.04
09.11.2017	0	5.05	6.27	89.54	65.16
10.11.2017	0	5.07	7.13	88.58	50.4
11.11.2017	0	5.02	4.93	92.25	41.76
12.11.2017	0	2.42	4.82	93.33	38.52
13.11.2017	2.48	1.37	3.83	87.92	92.88
14.11.2017	0	6.95	2.40	79.92	61.2
15.11.2017	0	3.82	1.07	83.88	30.96
16.11.2017	0	2.27	4.79	78.88	39.96
17.11.2017	0.11	0.95	4.69	89.00	38.88
18.11.2017	0.01	5.47	4.55	89.71	57.24
19.11.2017	0	3.3	0.32	96.08	43.92
20.11.2017	0	7.98	2.22	76.46	52.92
21.11.2017	0	4.49	3.15	79.33	27
22.11.2017	0	5.6	4.46	85.08	27.72
23.11.2017	0	7.33	3.81	80.83	73.8
24.11.2017	0	6.85	3.40	86.29	52.56
25.11.2017	0.82	2.81	5.87	86.58	122.4
26.11.2017	1.22	2.62	3.98	76.08	115.2
27.11.2017	0	7.13	0.66	72.25	64.44
28.11.2017	0	4.87	-1.10	86.04	23.76
29.11.2017	0.62	1.37	0.70	89.67	44.28
30.11.2017	1.04	3.44	1.03	95.75	53.64
01.12.2017	0.09	1.9	0.49	95.67	43.56
02.12.2017	0	5.69	0.14	86.96	59.4
03.12.2017	0	6.1	-2.32	82.96	33.12
04.12.2017	0	2.85	-4.45	87.33	28.8
05.12.2017	0	6.46	-0.61	83.46	28.44
06.12.2017	0	6.44	-0.35	85.63	29.52
07.12.2017	0	6.7	-2.38	87.00	27.72
08.12.2017	1.55	1.91	1.49	82.29	106.56
09.12.2017	0.11	2.94	-0.99	88.92	83.52
10.12.2017	0	4.86	-2.70	76.54	192.6
11.12.2017	0.18	0.94	8.59	74.96	309.24
12.12.2017	0.59	3.42	12.03	63.79	248.4
13.12.2017	0.65	2.42	0.72	94.17	48.6
14.12.2017	0	2.61	1.15	94.46	47.88
15.12.2017	0.7	0.79	1.13	96.46	71.28

16.12.2017	0	3.93	2.40	88.71	51.12
17.12.2017	0.01	5.89	0.48	83.50	59.76
18.12.2017	0	6.12	-1.83	81.92	46.08
19.12.2017	0	6.34	-2.80	80.83	43.92
20.12.2017	0	3.77	-0.98	84.04	52.56
21.12.2017	0	3.92	-2.20	85.83	35.64
22.12.2017	0	3.06	0.70	85.38	52.56
23.12.2017	0	5.61	0.58	85.00	43.2
24.12.2017	0	5.07	2.35	82.96	47.16
25.12.2017	0	6.16	-0.17	86.67	35.28
26.12.2017	0	5.29	1.71	82.38	95.76
27.12.2017	0.12	2.47	4.63	84.75	126
28.12.2017	0.68	1.22	3.99	87.25	85.32
29.12.2017	0	5.41	0.71	78.42	59.04
30.12.2017	0	4.43	-0.93	82.54	28.08
31.12.2017	0	5.64	1.64	82.92	32.4
01.01.2018	0.62	0.71	3.38	85.58	107.28
02.01.2018	0.24	2.82	0.68	96.17	51.84
03.01.2018	0	3.4	2.67	73.83	176.04
04.01.2018	0	2.93	1.71	79.38	72.36
05.01.2018	0	4.02	3.96	82.38	64.8
06.01.2018	0	6.41	7.66	73.75	135.72
07.01.2018	0	5.74	4.72	85.71	80.64
08.01.2018	0.04	1.88	5.16	96.79	48.24
09.01.2018	0.28	0.67	5.51	97.83	68.76
10.01.2018	0.57	1.46	5.62	94.96	57.96
11.01.2018	0	0.89	4.63	89.38	76.32
12.01.2018	0.09	2.2	4.12	87.21	52.92
13.01.2018	0.28	0.56	2.64	92.71	69.12
14.01.2018	0.52	1.09	-0.25	93.04	117.72
15.01.2018	0.1	2.72	-1.93	89.67	68.76
16.01.2018	0	4.67	-0.87	91.58	61.2
17.01.2018	0	2.99	1.81	77.71	98.64
18.01.2018	0	5.77	3.28	59.63	162
19.01.2018	0.03	1.37	0.93	84.04	36.36
20.01.2018	0	6.62	0.08	87.29	52.56
21.01.2018	0	6.36	2.49	71.50	110.52
22.01.2018	0	4.18	0.23	77.92	79.56
23.01.2018	0	6.29	1.76	80.46	43.92
24.01.2018	0	6.31	0.31	84.83	37.08
25.01.2018	0	7.16	0.19	85.63	35.28
26.01.2018	0	5.22	0.65	88.63	38.52
27.01.2018	0	1.25	0.33	94.67	73.44
28.01.2018	0	2.47	0.60	93.17	80.28
29.01.2018	0	8.49	4.02	84.04	37.8
30.01.2018	0	8.1	4.73	79.13	83.88
31.01.2018	0	3.68	1.72	92.38	36.36

01.02.2018	0.27	3.84	4.92	82.63	147.24
02.02.2018	2	1.27	0.65	95.29	51.84
03.02.2018	0.14	2.72	0.70	94.38	47.52
04.02.2018	0	8.89	0.42	87.33	67.32
05.02.2018	0	6.94	0.28	84.25	77.04
06.02.2018	0.02	2.32	0.23	83.00	123.48
07.02.2018	1.29	3.27	0.00	90.96	72
08.02.2018	0.19	3.5	1.95	91.17	55.44
09.02.2018	0	3.13	0.95	89.79	92.52
10.02.2018	0	1.46	0.50	87.17	52.92
11.02.2018	0	8.9	0.20	81.50	66.24
12.02.2018	1.64	2.66	0.35	92.33	50.04
13.02.2018	0	5.28	-0.01	86.79	72
14.02.2018	0	10.37	-0.83	76.04	71.64
15.02.2018	0	11.43	-1.86	74.75	87.84
16.02.2018	0	5.84	0.16	77.17	40.68
17.02.2018	0.01	11.09	2.08	81.29	110.88
18.02.2018	0.24	5.56	1.67	78.63	96.12
19.02.2018	0.14	5.04	0.24	72.67	57.96
20.02.2018	0.6	3.3	-0.33	91.46	122.04
21.02.2018	0.25	3.85	0.37	88.71	52.56
22.02.2018	1.23	2.92	-0.95	89.71	82.44
23.02.2018	0.69	4.05	-0.57	90.75	81
24.02.2018	0.05	5.25	-2.06	74.92	114.84
25.02.2018	0	12.44	-8.28	64.50	82.08
26.02.2018	0.21	5.39	-10.52	79.13	112.68
27.02.2018	0.27	7.66	-8.91	81.71	131.76
28.02.2018	0	12.89	-12.38	74.79	73.08
01.03.2018	0	8.99	-11.38	78.25	57.96
02.03.2018	0.06	4.23	-6.50	81.79	54
03.03.2018	0.06	8.21	-2.76	78.46	61.92
04.03.2018	0	6.95	-0.32	77.33	114.12
05.03.2018	0.34	4.34	-0.55	89.54	92.16
06.03.2018	0.79	10.58	0.44	85.71	62.64
07.03.2018	0	11.27	2.63	82.42	43.92
08.03.2018	0.05	14.41	5.08	75.21	104.76
09.03.2018	0	11.17	3.22	74.71	36.36
10.03.2018	0	11.72	7.15	66.00	151.2
11.03.2018	0	9.46	8.16	75.88	112.68
12.03.2018	0	9.83	11.03	65.75	188.64
13.03.2018	0.59	10.86	8.63	73.67	93.96
14.03.2018	0	11.11	8.05	77.13	83.52
15.03.2018	0	10.25	9.14	67.71	148.68

16.03.2018	0.53	5.95	7.92	81.71	81.72
17.03.2018	2.43	1.5	3.30	93.29	96.12
18.03.2018	0.21	2.66	-0.94	89.83	153.36
19.03.2018	0.21	5.22	-2.29	84.58	117.72
20.03.2018	0	8.96	-1.15	78.00	87.84
21.03.2018	0	10.48	0.46	67.21	143.64
22.03.2018	0	18.2	-0.19	66.83	74.88
23.03.2018	0	7.7	1.69	70.08	84.96
24.03.2018	0	6.79	4.36	65.83	108.72
25.03.2018	0	4.77	2.45	76.96	75.96
26.03.2018	0	13.18	5.22	68.54	52.92
27.03.2018	0.37	10.62	5.56	76.92	73.08
28.03.2018	0	12.82	6.44	85.63	96.84
29.03.2018	0.65	2.74	8.57	86.63	81.72
30.03.2018	0.41	3.05	6.72	90.92	70.2
31.03.2018	0.37	4.25	7.22	93.71	90.72
01.04.2018	0.02	12.21	10.00	54.21	230.04
02.04.2018	0	19.29	9.74	49.88	113.04
03.04.2018	0	15.37	15.23	39.67	354.96
04.04.2018	0	7.44	14.61	50.00	404.28
05.04.2018	0.4	9.24	12.29	71.25	147.6
06.04.2018	0	21.31	9.43	64.46	119.88
07.04.2018	0	17.2	8.74	70.29	142.2
08.04.2018	0	16.49	11.87	66.79	85.32
09.04.2018	0	18.29	14.82	54.00	165.96
10.04.2018	0	13.17	14.01	58.92	211.32
11.04.2018	0	14.47	15.68	54.75	140.76
12.04.2018	0.54	10.56	13.15	79.58	82.08
13.04.2018	0.07	18.45	14.48	61.46	187.56
14.04.2018	0	21.79	13.61	59.96	81.72
15.04.2018	0.04	12.17	15.65	69.33	88.2
16.04.2018	8.29	11.5	16.40	79.25	119.88
17.04.2018	0.58	13.64	15.23	74.46	108
18.04.2018	0	21.49	14.80	64.42	82.08
19.04.2018	0	22.26	16.59	58.00	79.2
20.04.2018	0	22.01	17.49	59.63	77.04
21.04.2018	0	22.61	17.80	58.25	134.64
22.04.2018	0	21.27	19.51	59.33	118.8
23.04.2018	0.44	18.86	19.60	63.92	108
24.04.2018	0	14.75	18.01	68.25	109.8
25.04.2018	0	21.82	18.78	64.54	84.24
26.04.2018	1.07	7.34	13.95	76.75	154.44
27.04.2018	0.01	19.05	13.24	75.83	100.44
28.04.2018	0	22.43	17.07	68.50	110.88
29.04.2018	0	22.05	19.33	63.17	151.2
30.04.2018	0	23.33	18.61	51.54	250.92
01.05.2018	0	18.84	17.51	46.54	139.68
02.05.2018	0.7	22.54	19.01	58.79	104.4

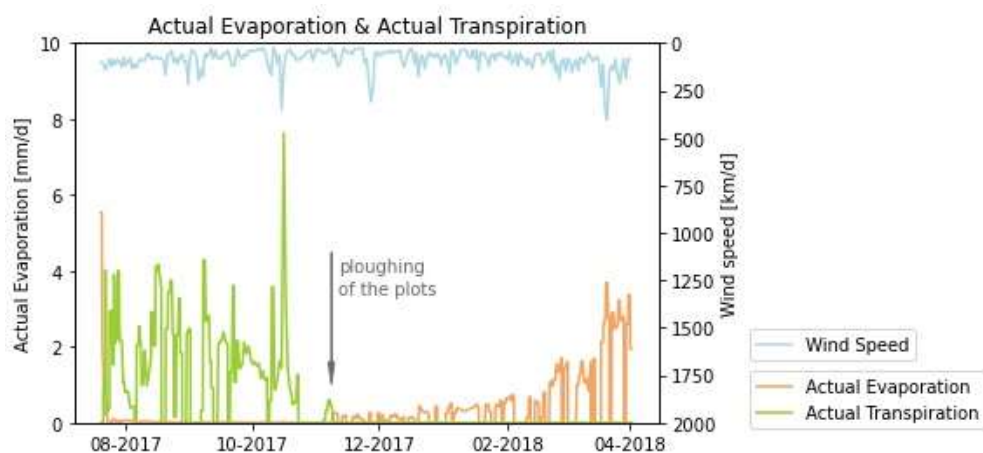


Fig. 48: Actual evaporation and actual transpiration of plot 5.

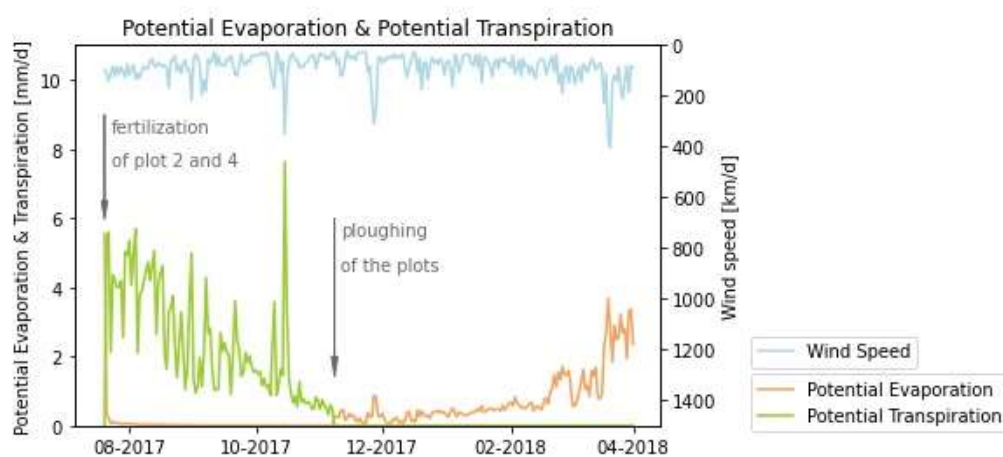


Fig. 49: Potential evaporation and potential transpiration of plot 5.

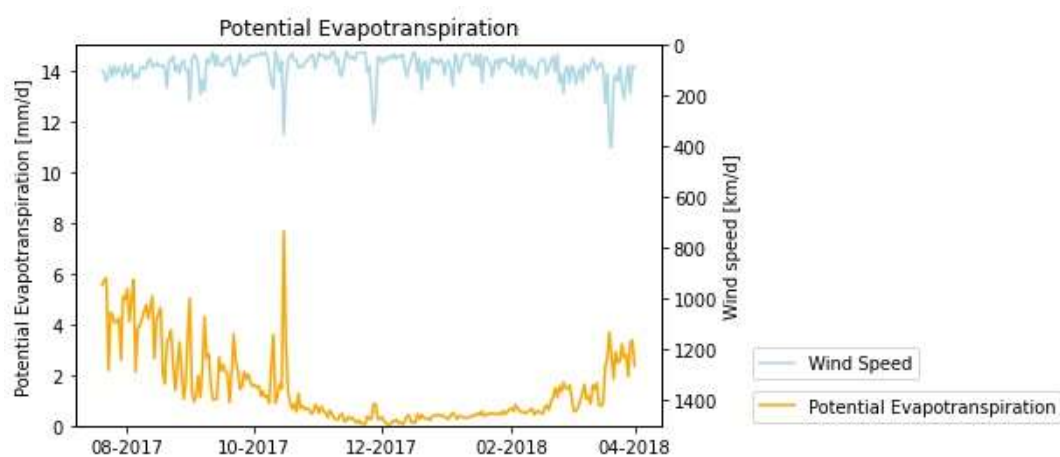


Fig. 50: Potential evapotranspiration of plot 5.